Biochemistry



Section K
Lipid Metabolism

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1g葡萄糖释放约16KJ的能量
1g脂肪可以放出约39KJ的能量。

K1 Structures and roles of fatty acids

K2 Fatty Acid Breakdown

K3 Fatty Acid Synthesis

K4 Metabolism of Triacylglycerols



K1

Structures and roles of fatty acids



1. Lipid

Biological lipids are a chemically diverse group of compounds, the common and defining feature of which is their insolubility in water.

Fats and oils (storage lipids)

Phospholipids and sterol (major elements of membranes)



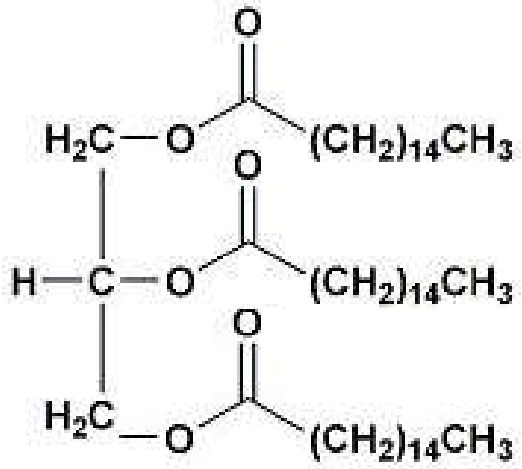
生物体内的脂类(lipid)

（脂质，类脂）：脂肪酸和醇组成或其衍生物

- 皂化脂类
  - 单纯脂类
    - 三酰甘油
    - 蜡
  - 复合脂类
    - 磷脂
    - 糖脂
    - 硫脂
- 非皂化脂类
  - 萜　类
  - 类固醇

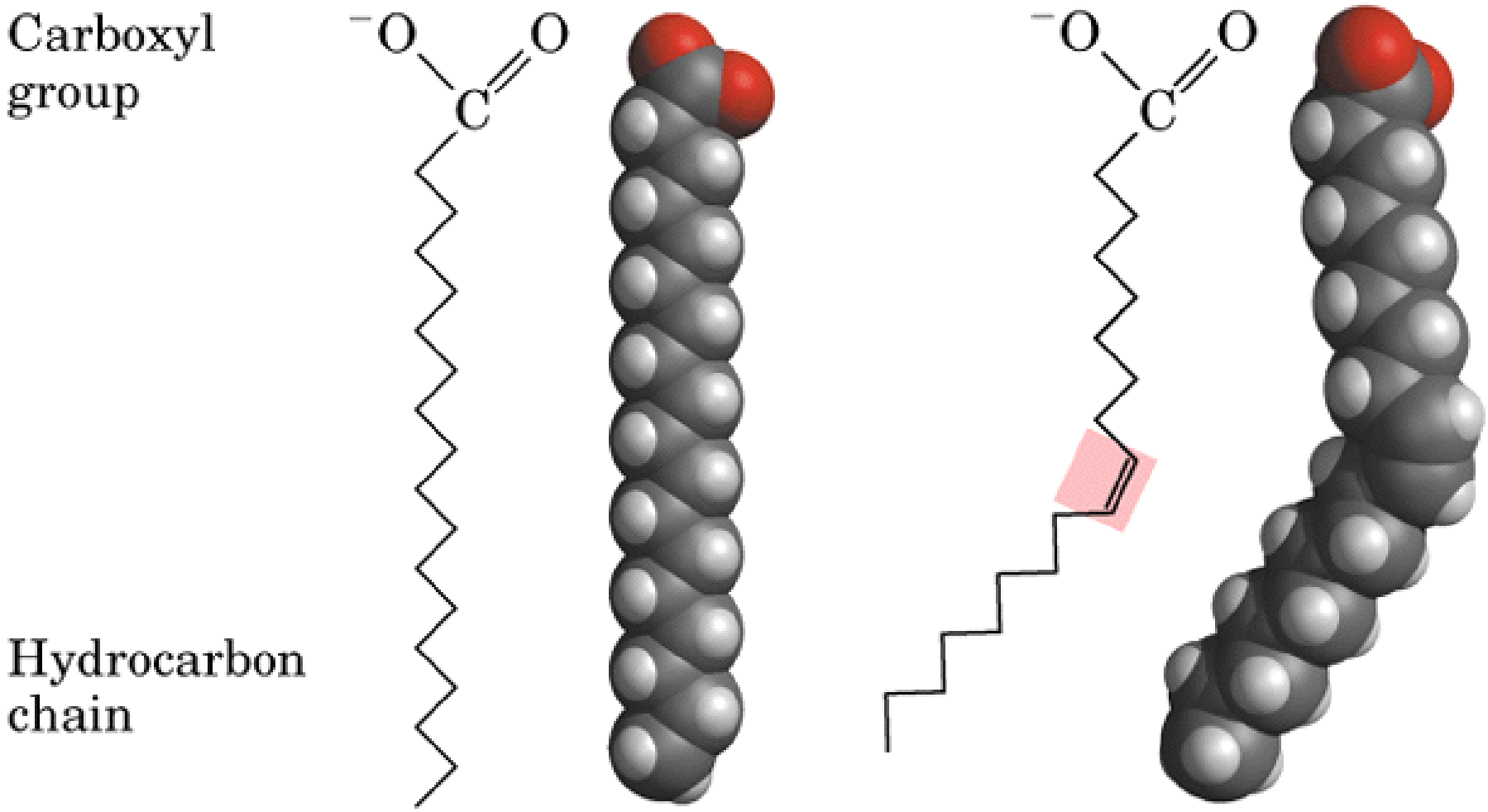
The fats and oils used almost universally as stored forms of energy in living organisms are derivatives of fatty acids.

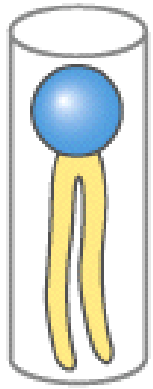
Typical type of fatty acid-containing compounds are triacylglycerols.



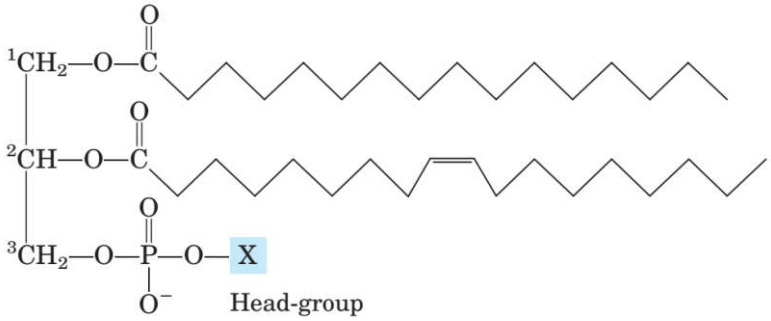
(C4 to C36)

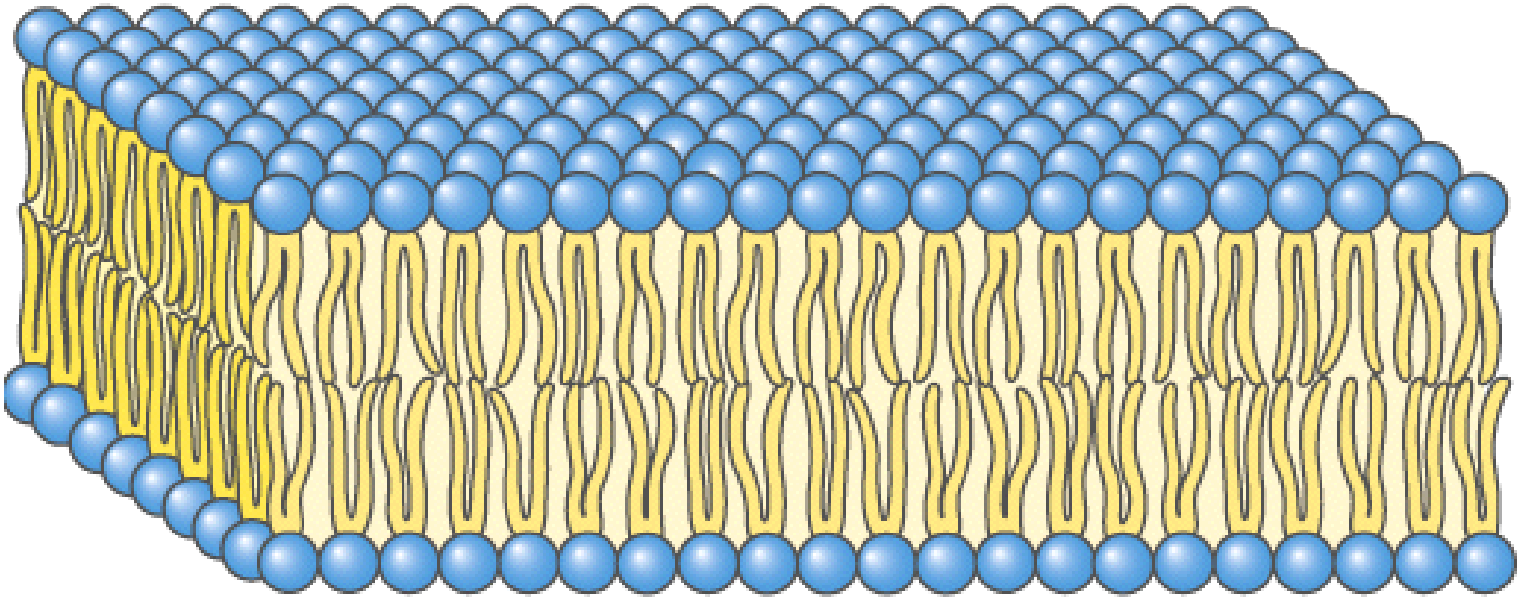
2. Structure and properties of Fatty acids





Individual units are cylindrical (cross-section of head equals that of side chain)

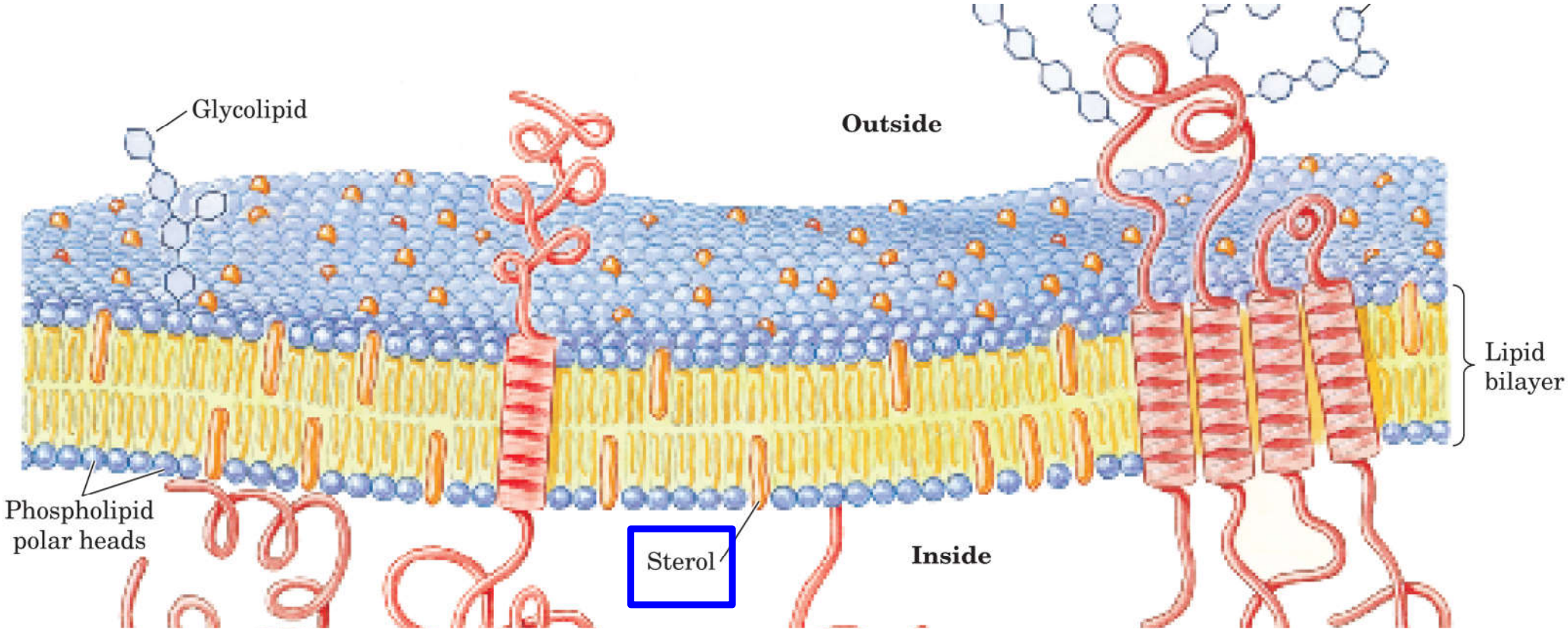




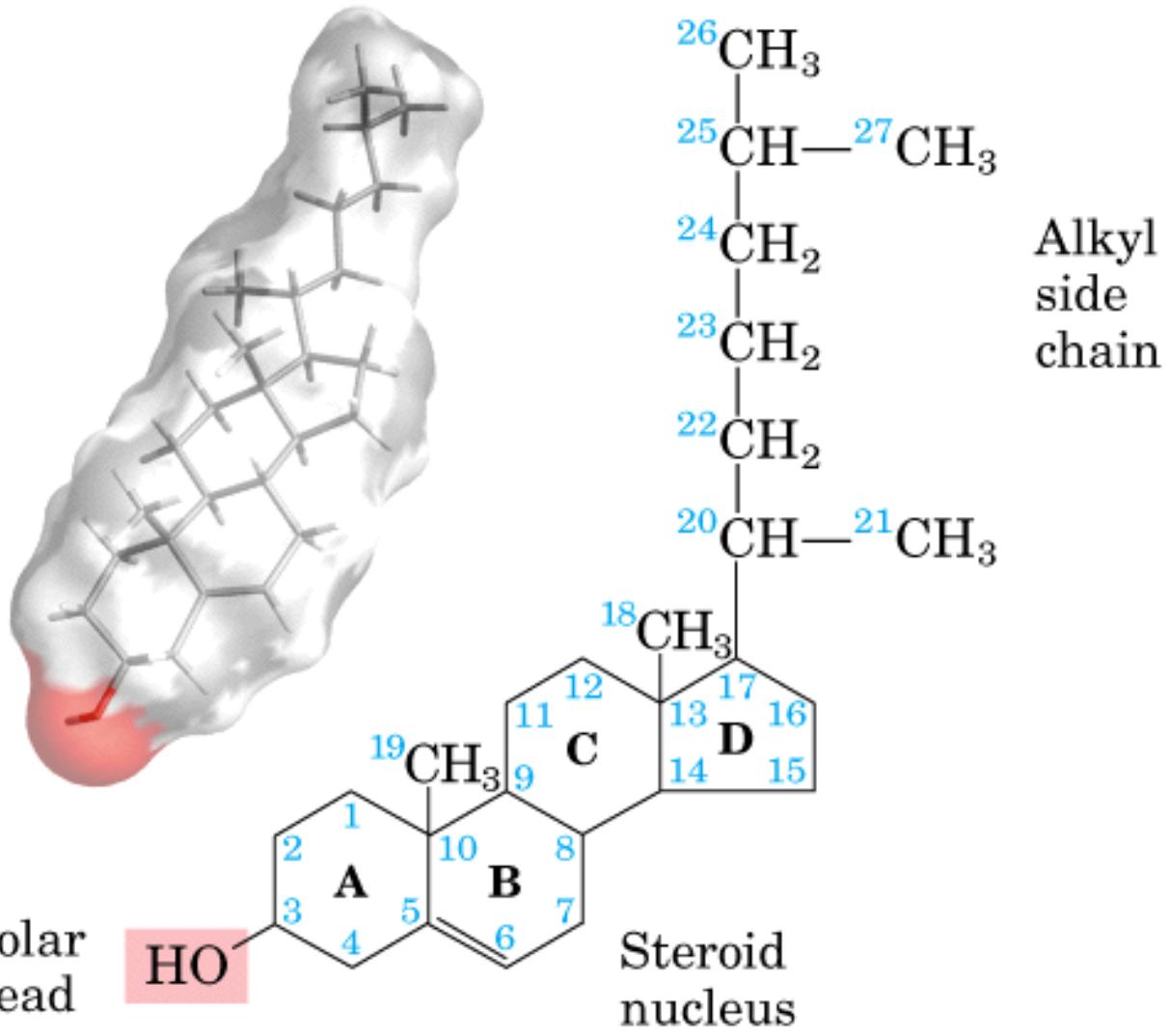
Bilayer

Cell membrane

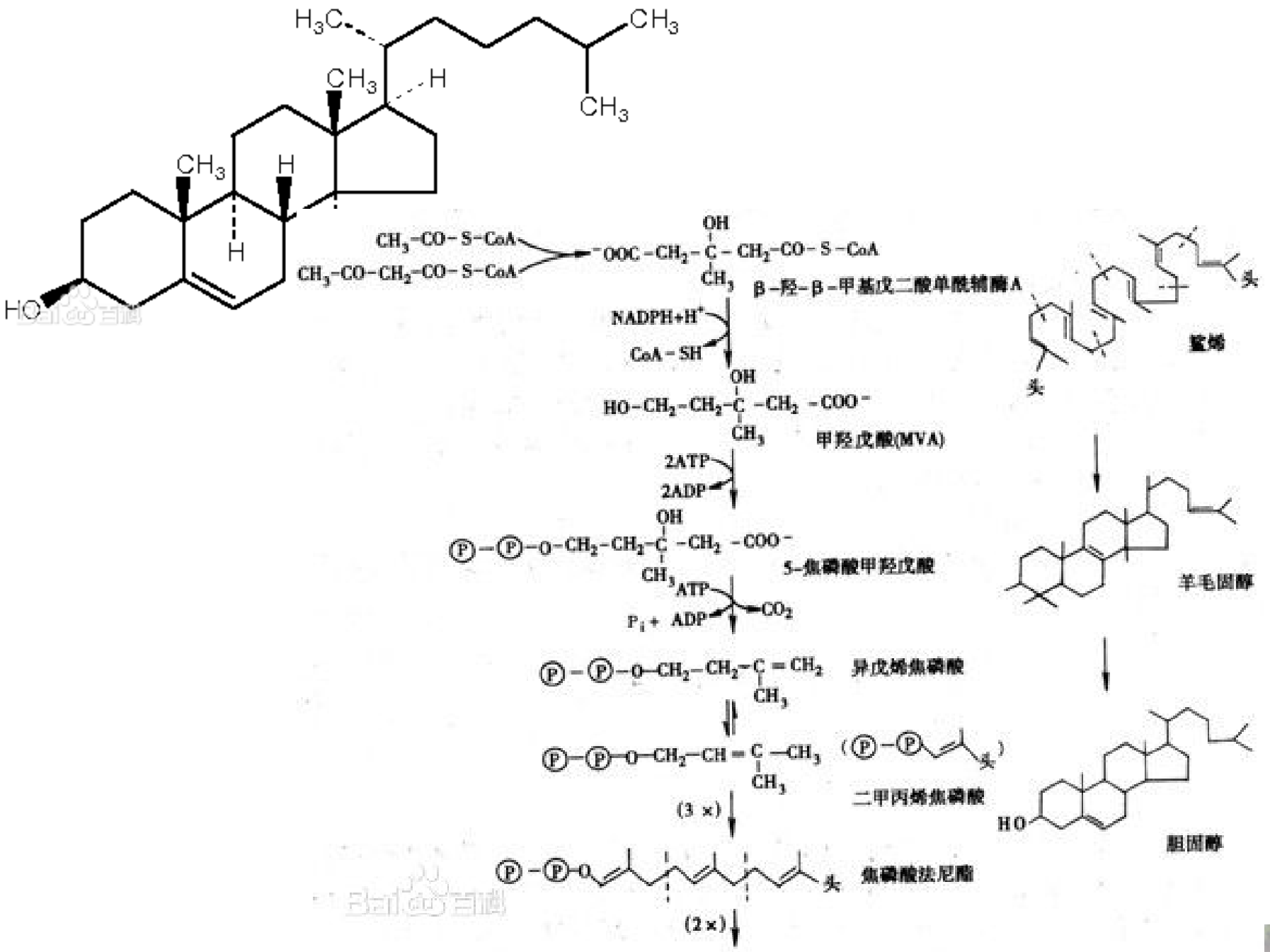



Glycolipid
Outside
Lipid bilayer
Phospholipid polar heads
Sterol
Inside

胆固醇

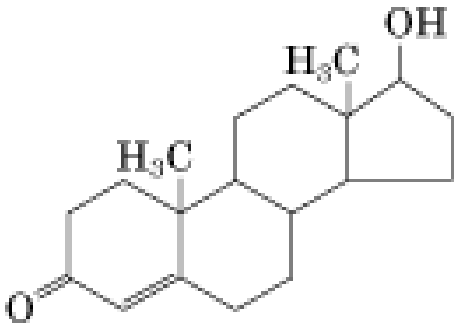


Sterols Have Four Fused Carbon Rings



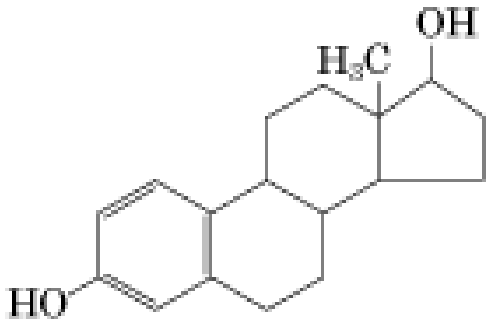
胆固醇

Biochemistry

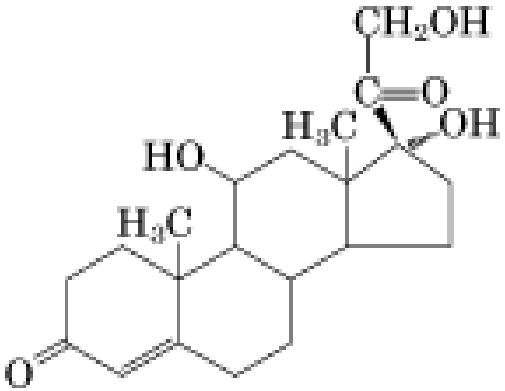


Testosterone

睾丸素

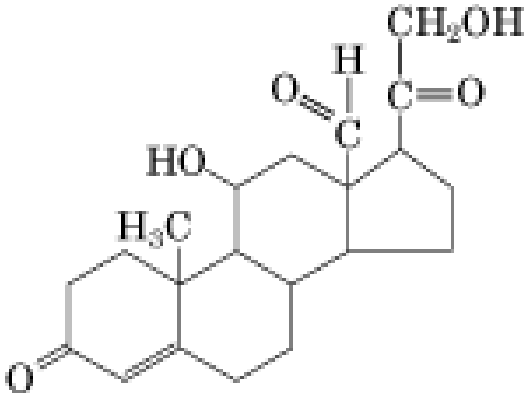


Estradiol 雌二醇

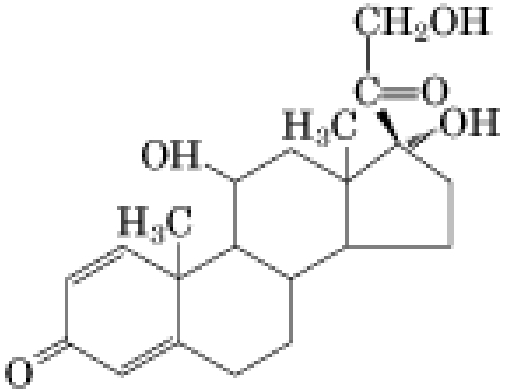


Cortisol

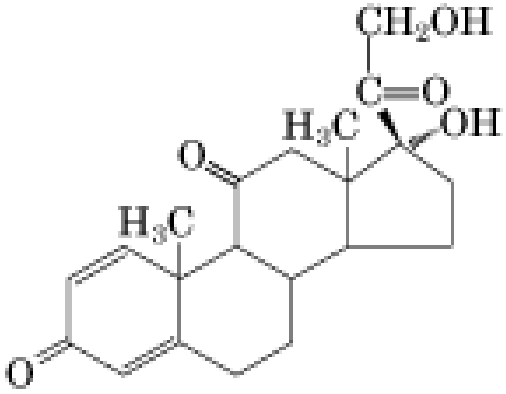
皮质（甾）醇



Aldosterone 醛甾酮



Prednisolone



Prednisone



3. Nomenclature

Fatty acids are named according to the number of carbon atoms in the chain and the number and position of any double bonds.

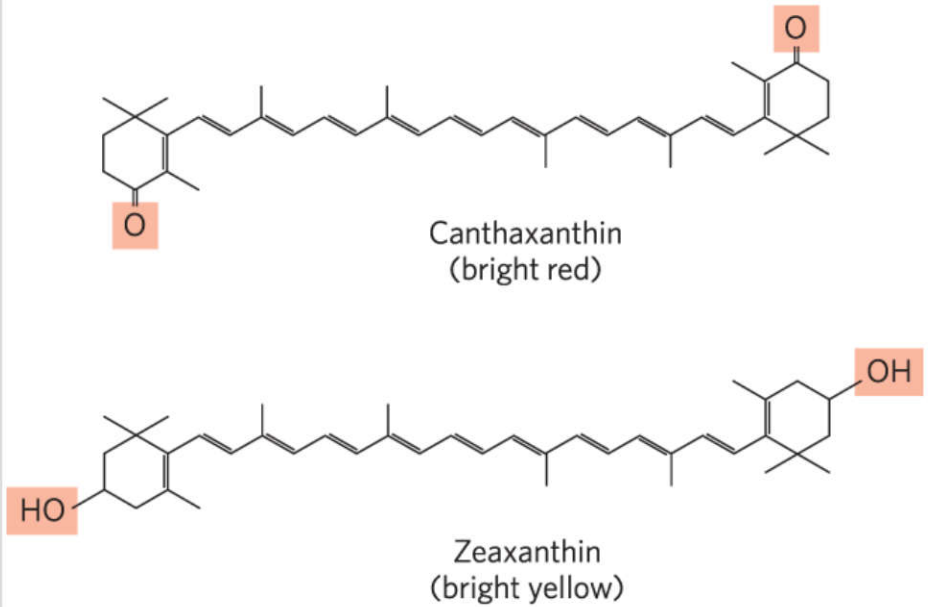
Palmitate	**(C16:0)**	**棕榈酸（软脂酸）**
Stearate	**(C18:0)**	**硬脂酸**
Oleate	**(C18:1（Δ^{9}）)**	**油酸**
Linoleate	**(C18:2（$\Delta^{9,12}$）)**	**亚油酸**
Linolenate	**(C18:3 ($\Delta^{9,12,15}$))**	**亚麻酸**
Arachidonate	**(C20:4 ($\Delta^{5,8,11,14}$))**	**花生四稀酸**



4. Roles

- Energy stores (triacylglycerols) and fuel molecule
- Components of membranes (glycerophospholipids and sphingolipids)
- As hormones , intracellular second messengers (DAG, diacylglycerol), and pigment
- Covalently joined with some proteins







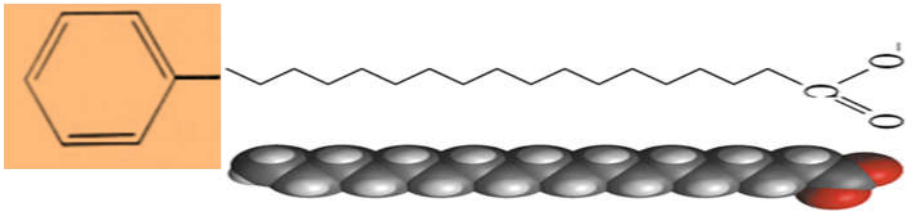
Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e,
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Fatty Acid Breakdown (K2)

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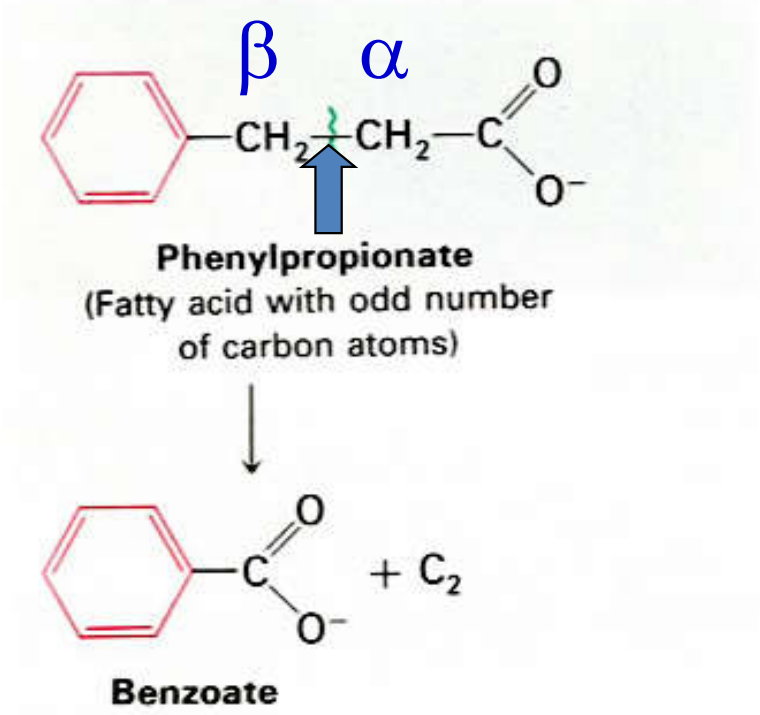


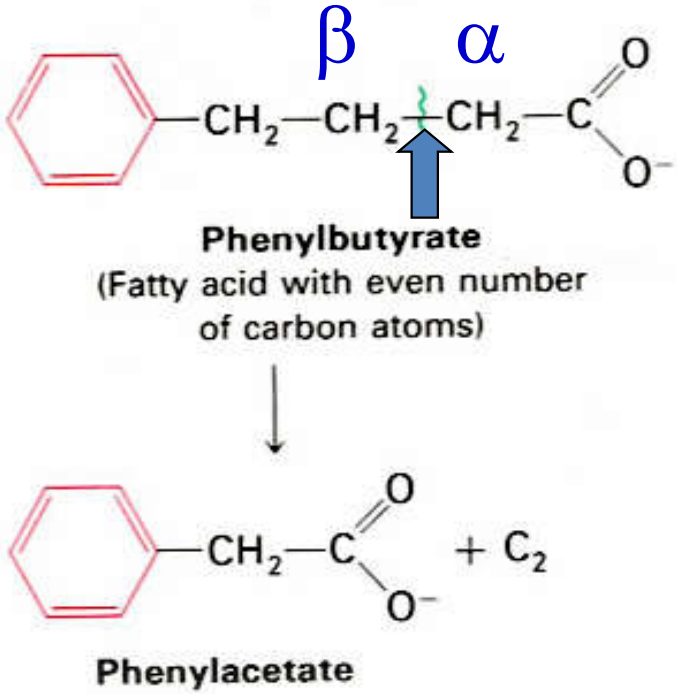
1. Early labeling experiments (1904): fatty acids are degraded by sequential removal of two-carbon units



- **When dogs were fed with odd-numbered fatty acids attached to a phenyl group, benzoate was excreted; and when fed with even-numbered, phenylacetate was excreted.**
- **Hypothesis: the β-carbon is oxidized,with two-carbon units released by each round of oxidation.**







Franz Knoop's labeling Experiments (1904):

fatty acids are degraded by oxidation at the β carbon, i.e., **β oxidation**.



•**These experiments are a landmark in biochemistry, in using synthetic label (the phenyl group here) to elucidate reaction mechanism, and was done long before radioisotopes was used in biochemistry!**



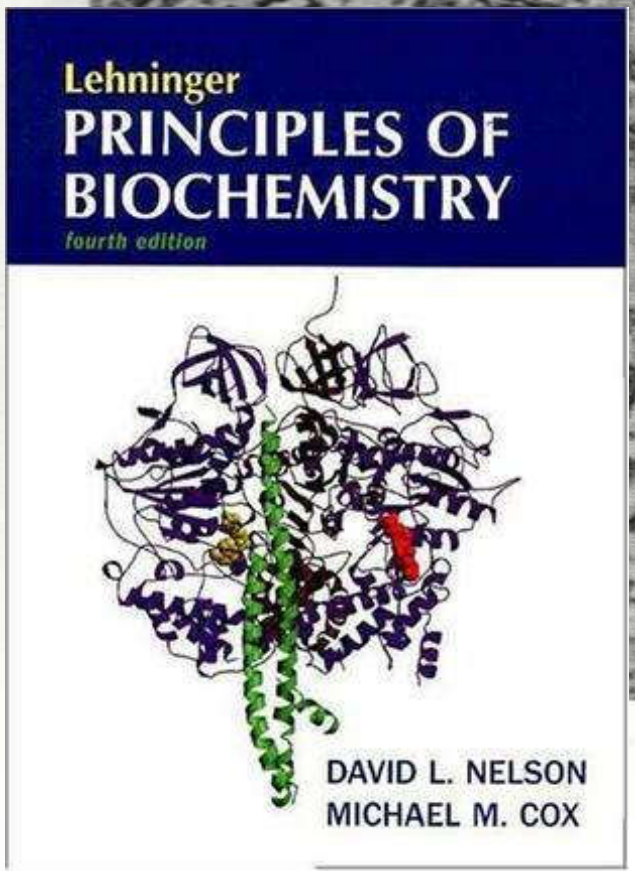
2. Fatty acid oxidation was found to occur in mitochondria







Albert L. Lehninger
1917–1986





Lehninger Principle of biochemistry

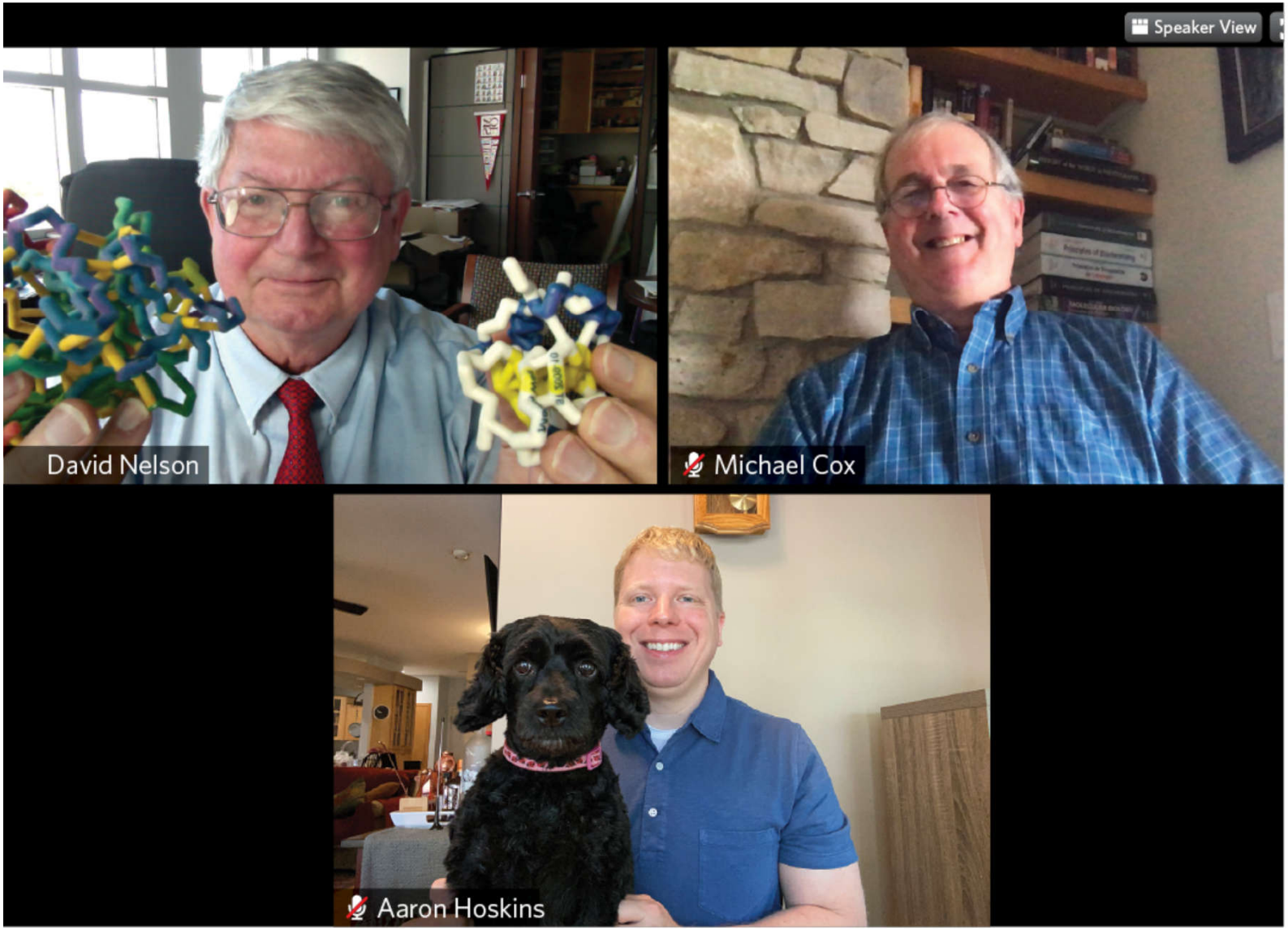
1982年第1版，**A. Lehninger**
1993年第2版，Albert Lehninger, David Nelson, and Michael Cox；
2000年第3版， David Nelson, and Michael Cox；
2005年第4版， David Nelson, and Michael Cox；
2008年第5版， David Nelson, and Michael Cox；
2012年第6版， David Nelson, and Michael Cox；
2016年第7版， David Nelson, and Michael Cox；
2021年第8班， David Nelson, Michael Cox , and Aaron A. Hoskins





David L. Nelson and Michael M. Cox.

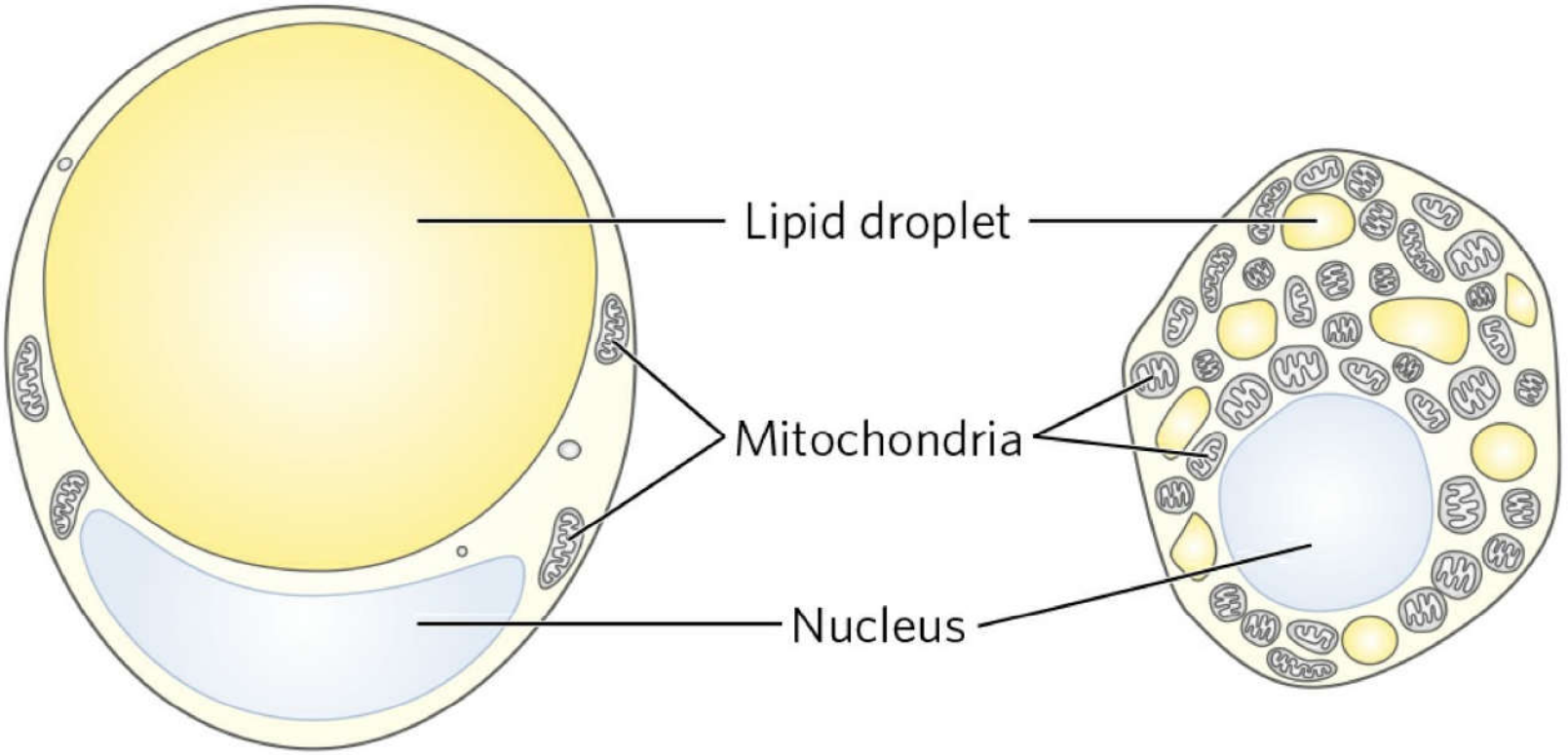
Biochemistry



From top left: David L. Nelson, Michael M. Cox, and Aaron A. Hoskins (Photos by David L. Nelson, Michael M. Cox, and Aaron A. Hoskins)

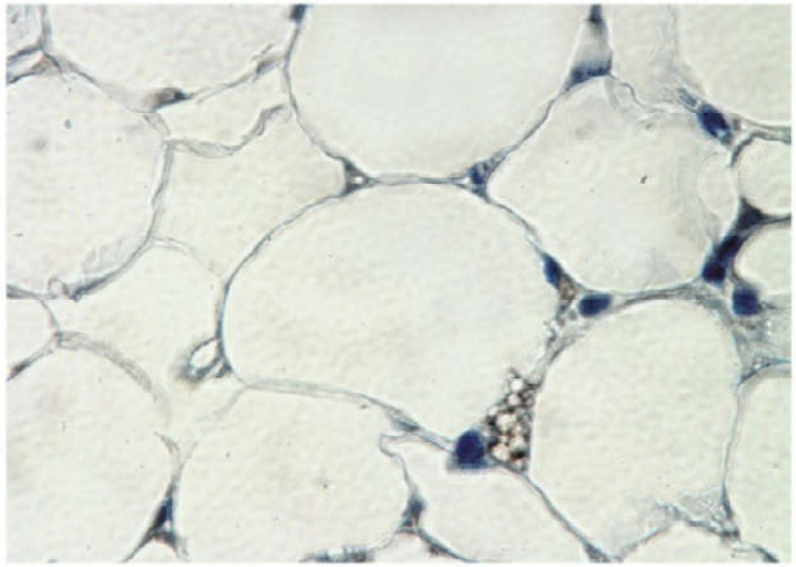
Aaron A. Hoskins

Biochemistry

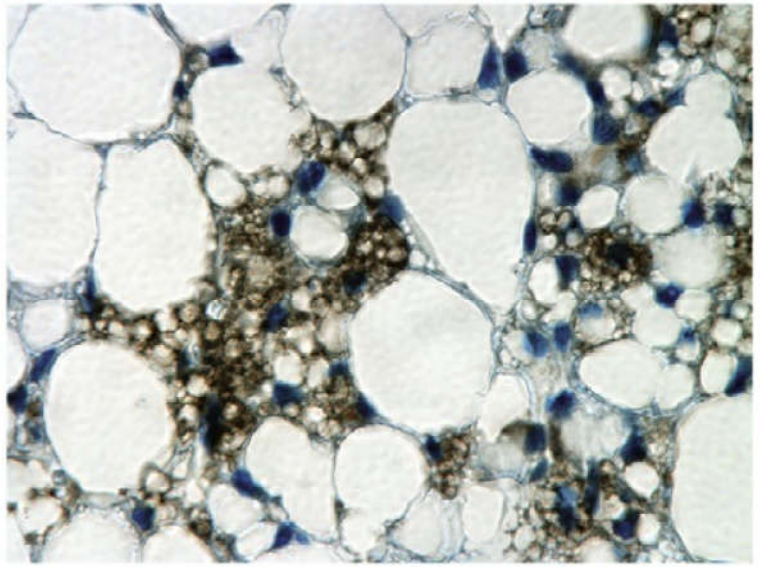


(a) White adipocyte

(b) Brown adipocyte

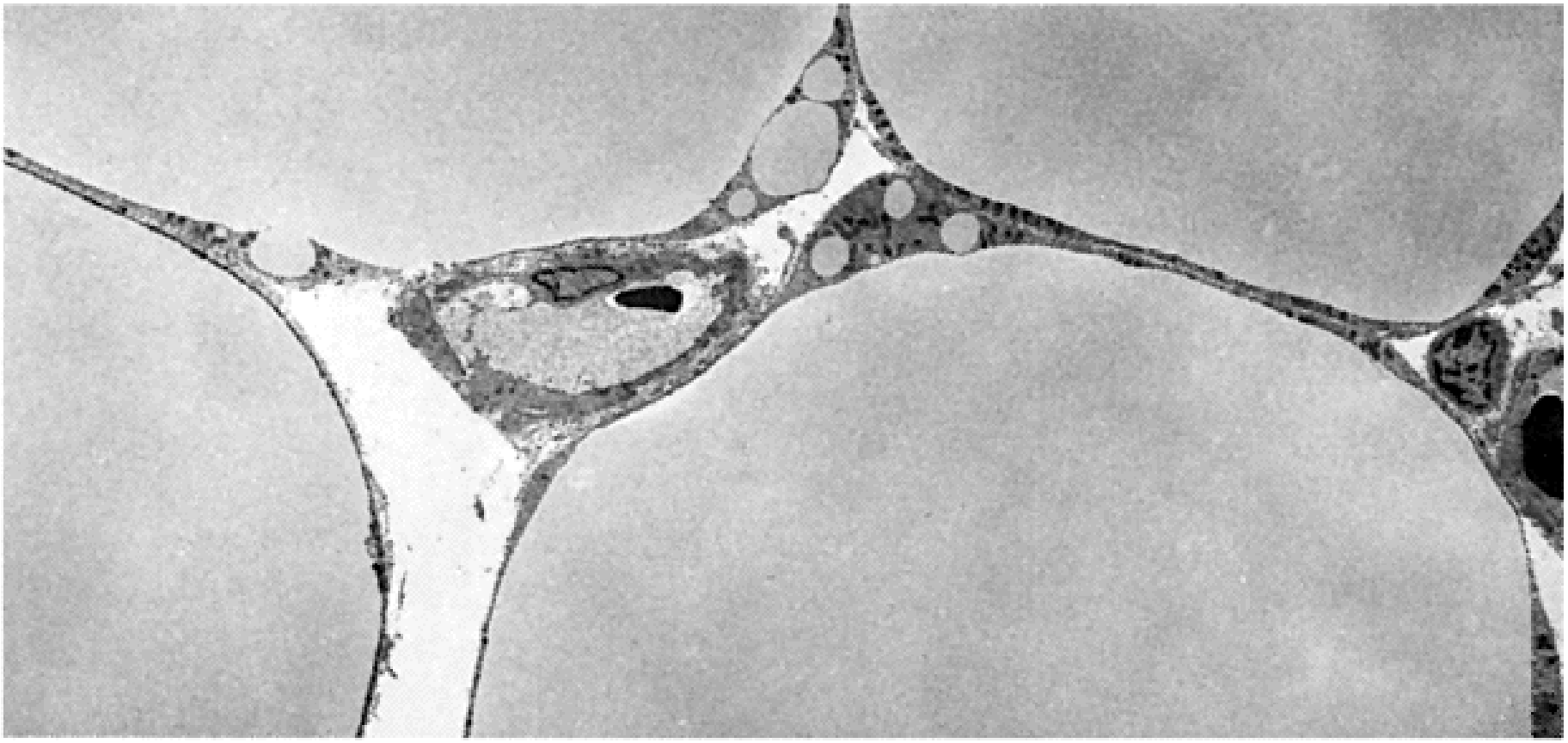


(c)



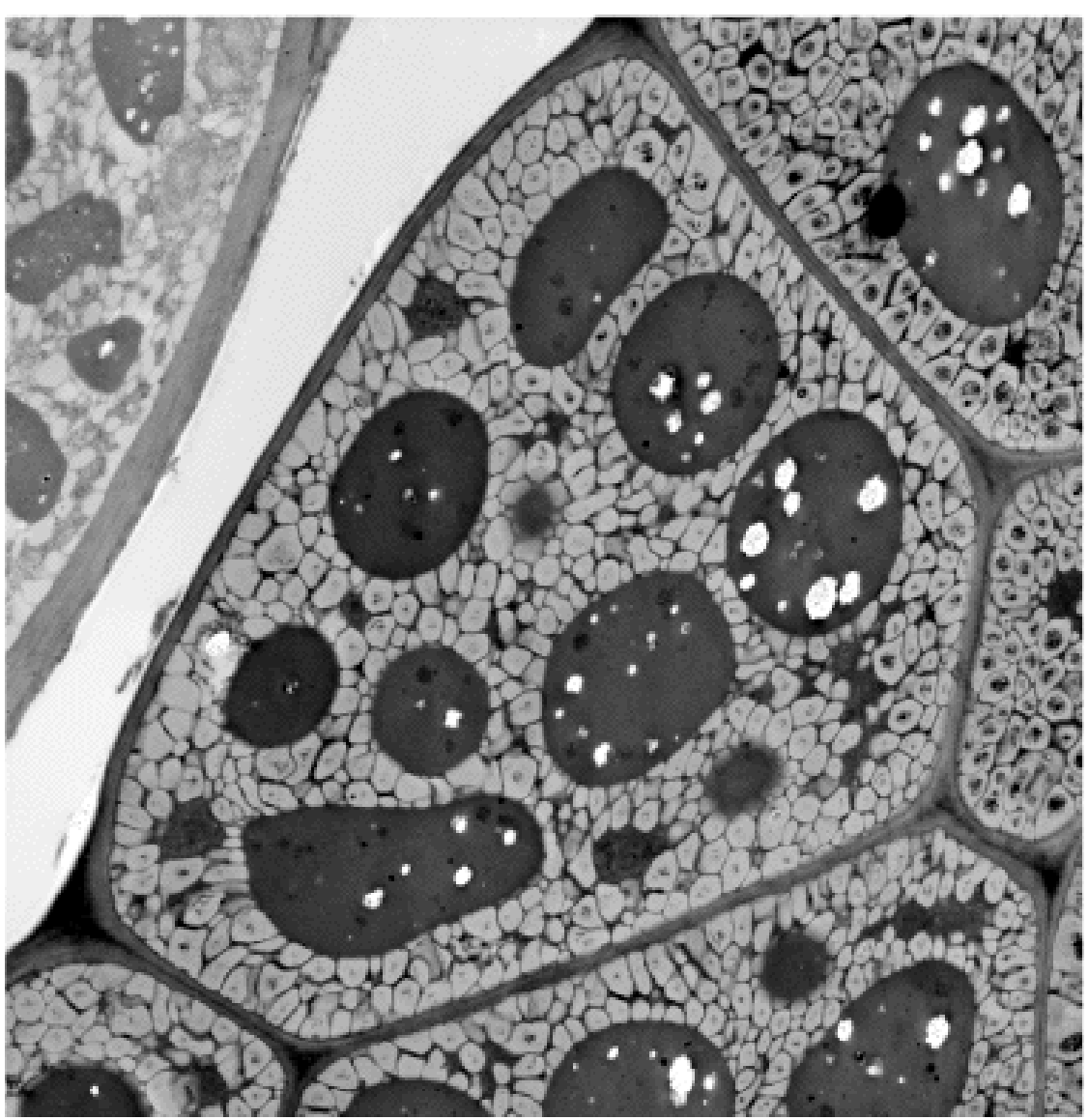
(d)

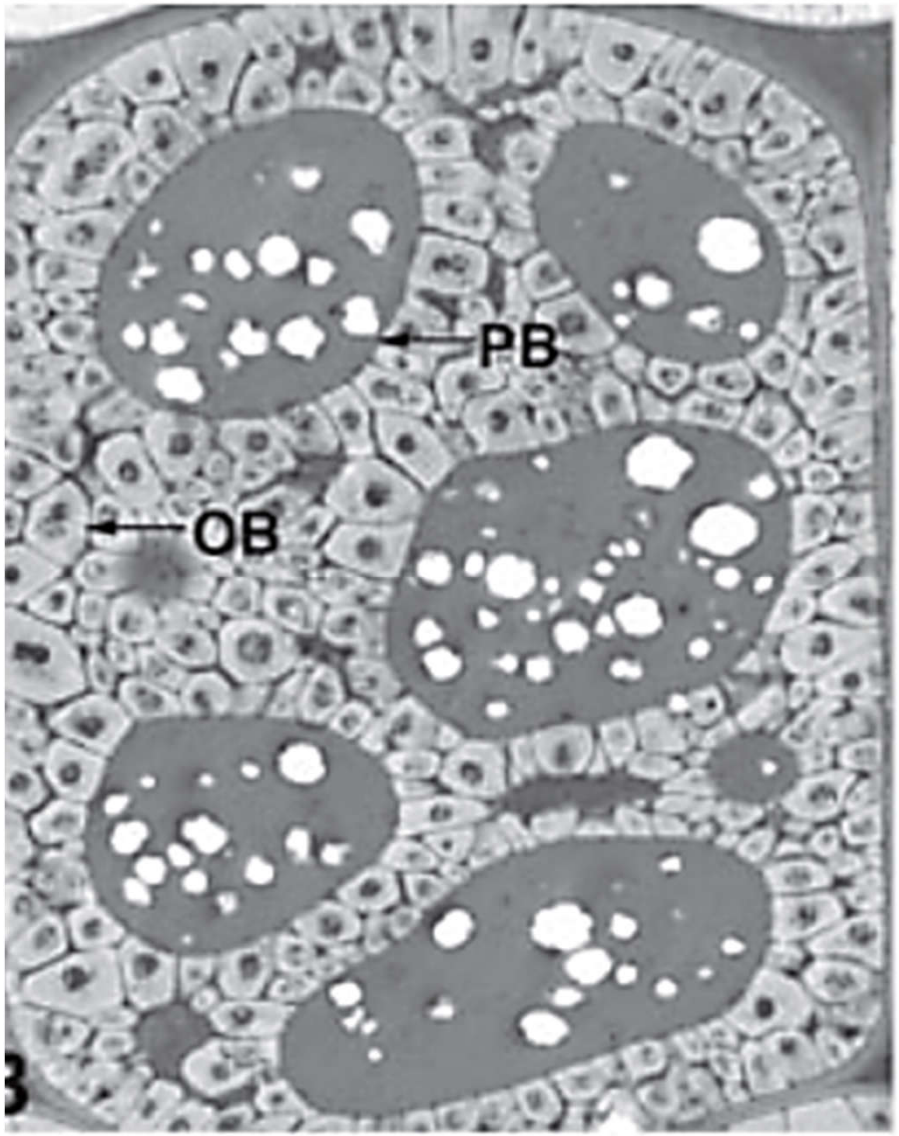




8 μm

(a)



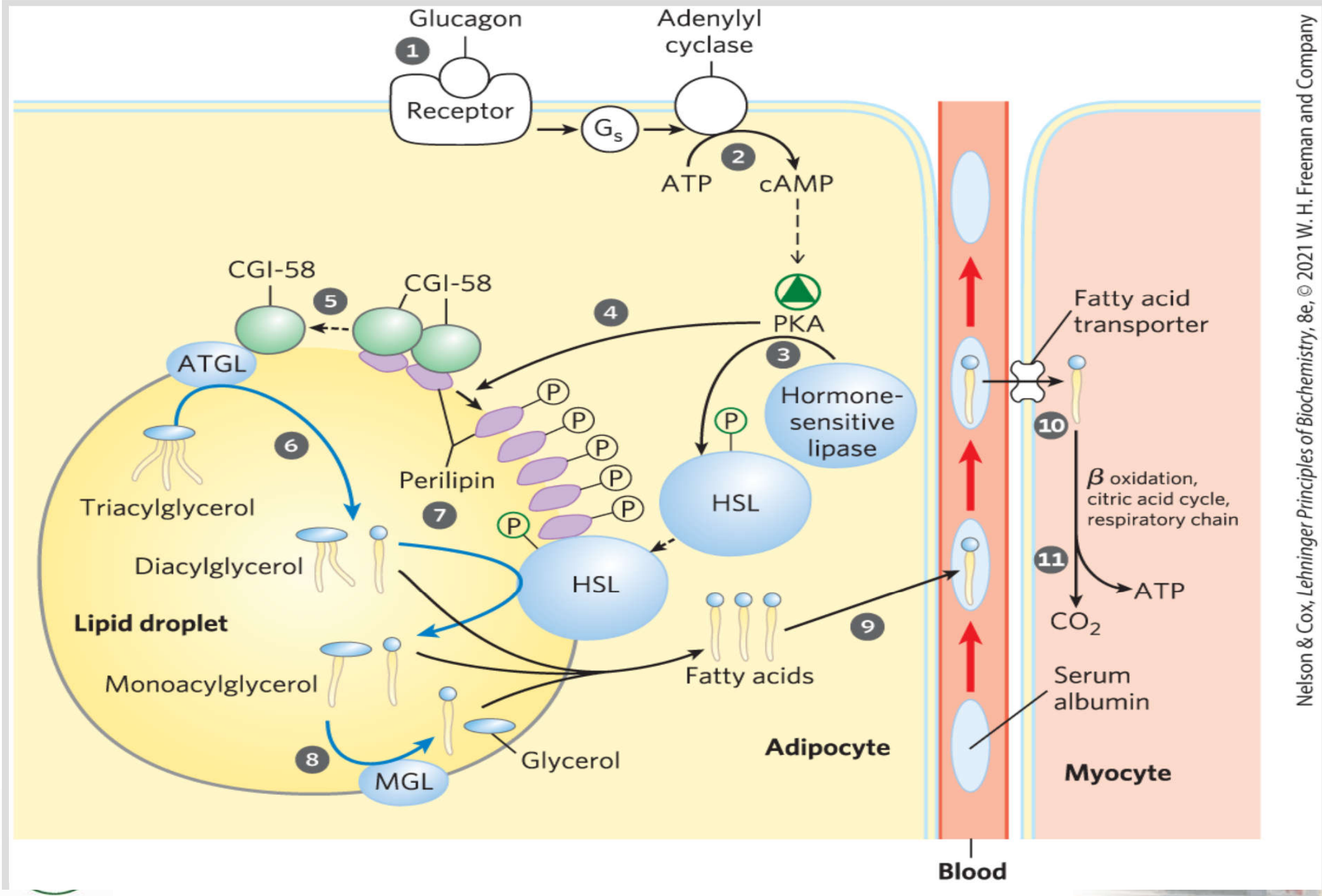


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OB: oil body；PB：protein body



Mobilization of triacylglycerols stored in adipose tissue



Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2021 W. H. Freeman and Company

1. Activation of Fatty acid (acyl-COA)
2. Transport of fatty acyl-COA (acyl-carnitine)
3. β-oxidation：(mitochondria matrix)

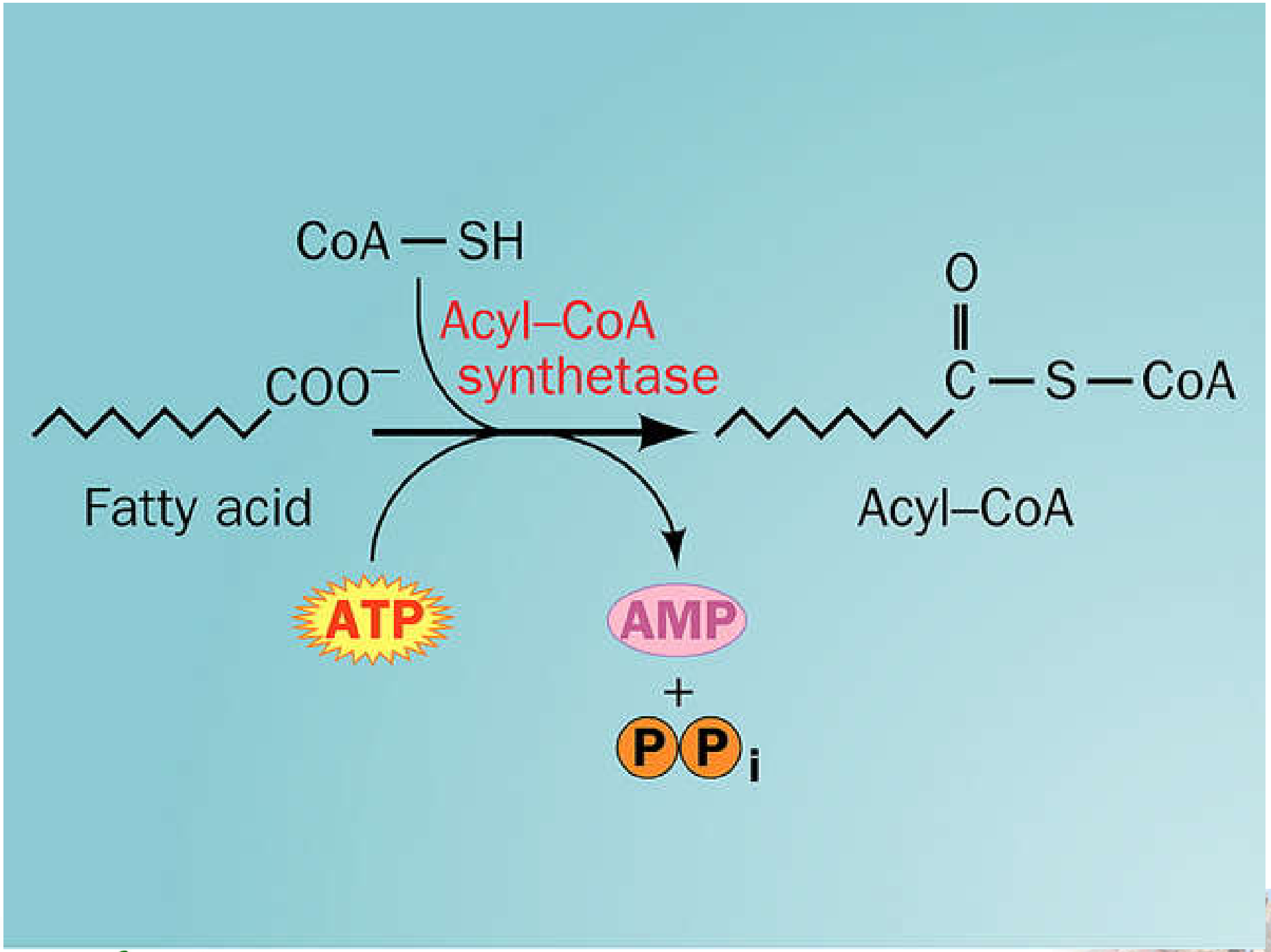
   Oxidation by FAD
   Hydration
   Oxidation by NAD^+
   Thiolysis by CoA.



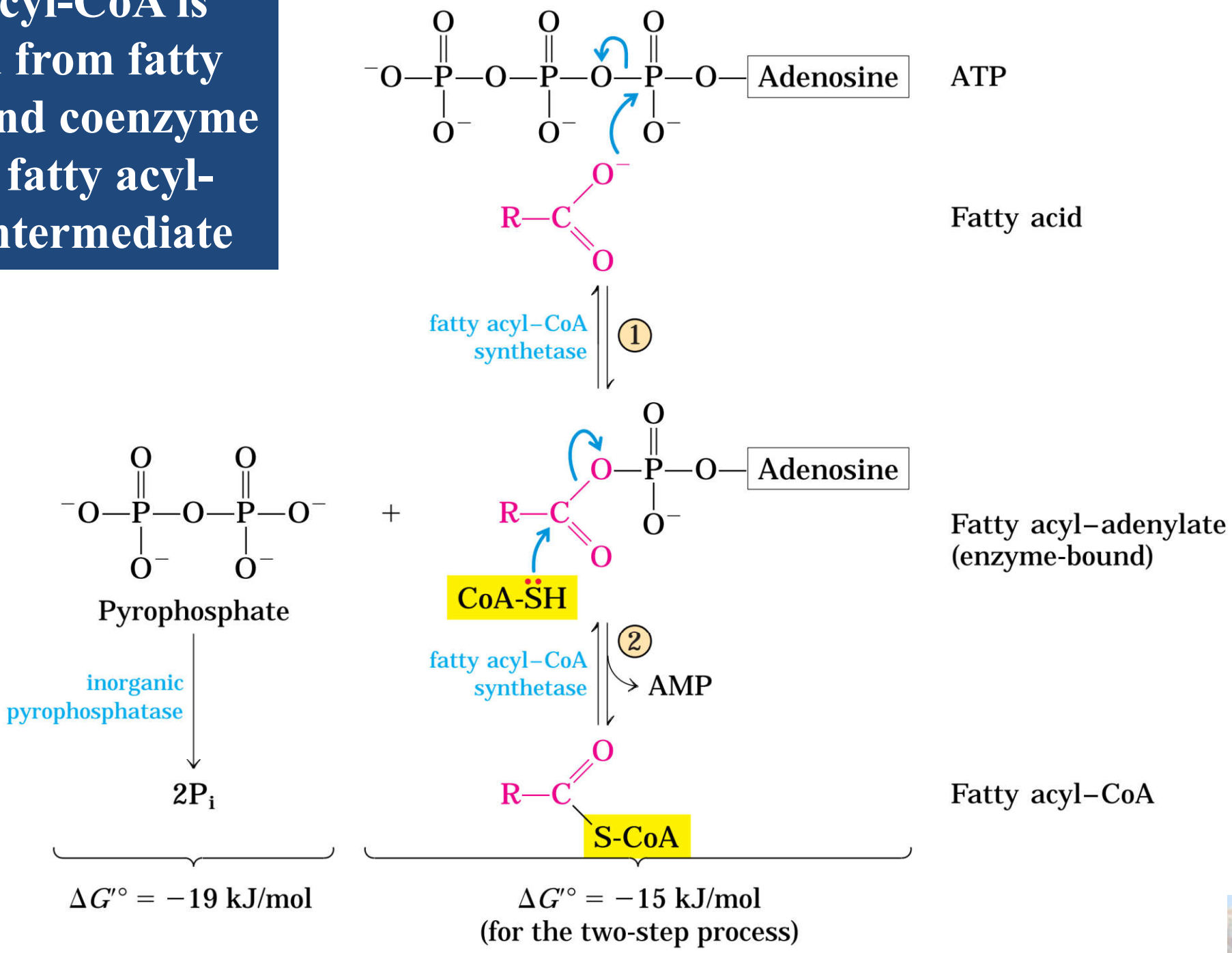
3. Fatty acids are activated on the outer membrane of mitochondria

- **Fatty acids are converted to fatty acyl-CoA (a high energy compound) via a fatty-acyl-adenylate intermediate (enzyme-bound) by the action of fatty acyl-CoA synthetases (also called fatty acid thiokinase，硫激酶).**

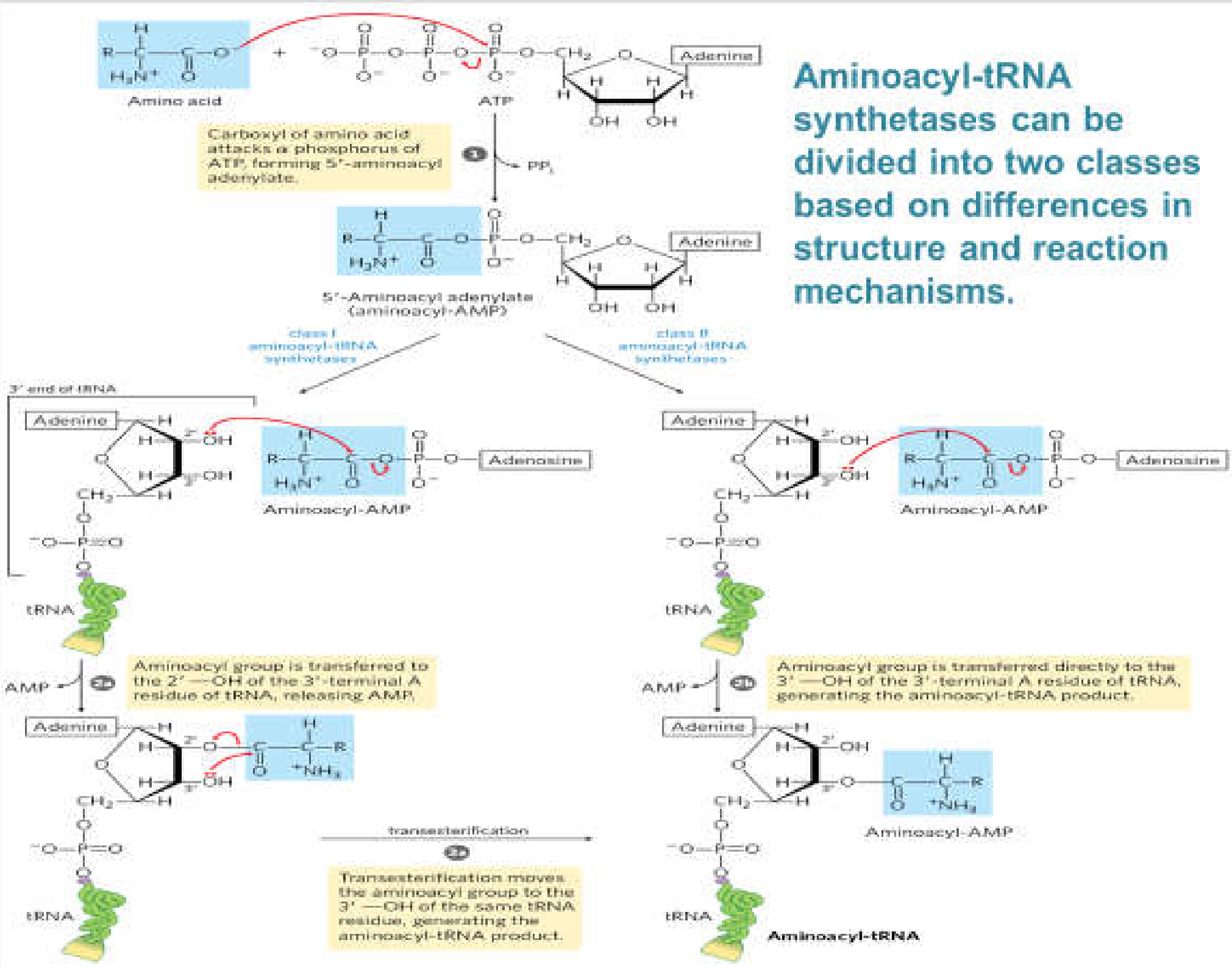



CoA—SH
Acyl–CoA
synthetase
COO⁻
O
C—S—CoA
Fatty acid
Acyl–CoA
ATP
AMP
+
PPi

Fatty acyl-CoA is formed from fatty acids and coenzyme A via a fatty acyl-AMP intermediate

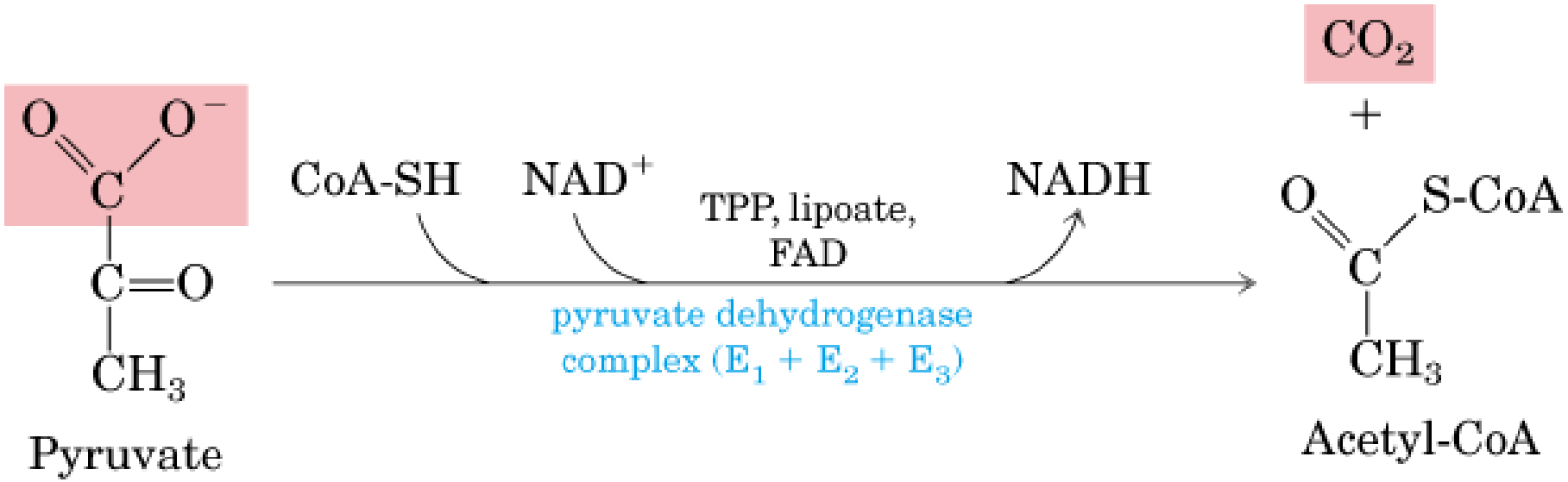


Biochemistry



Aminoacyl-tRNA synthetases can be divided into two classes based on differences in structure and reaction mechanisms.

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$\Delta G'^{\circ} = -33.4$ kJ/mol

The oxidative decarboxylation of pyruvate in mitochondria: producing acetyl-CoA and CO_2.



$$\text{Fatty acid} + \text{CoA} + \text{ATP} \rightleftharpoons \text{fatty acyl–CoA} + \text{AMP} + \text{PP}_i$$

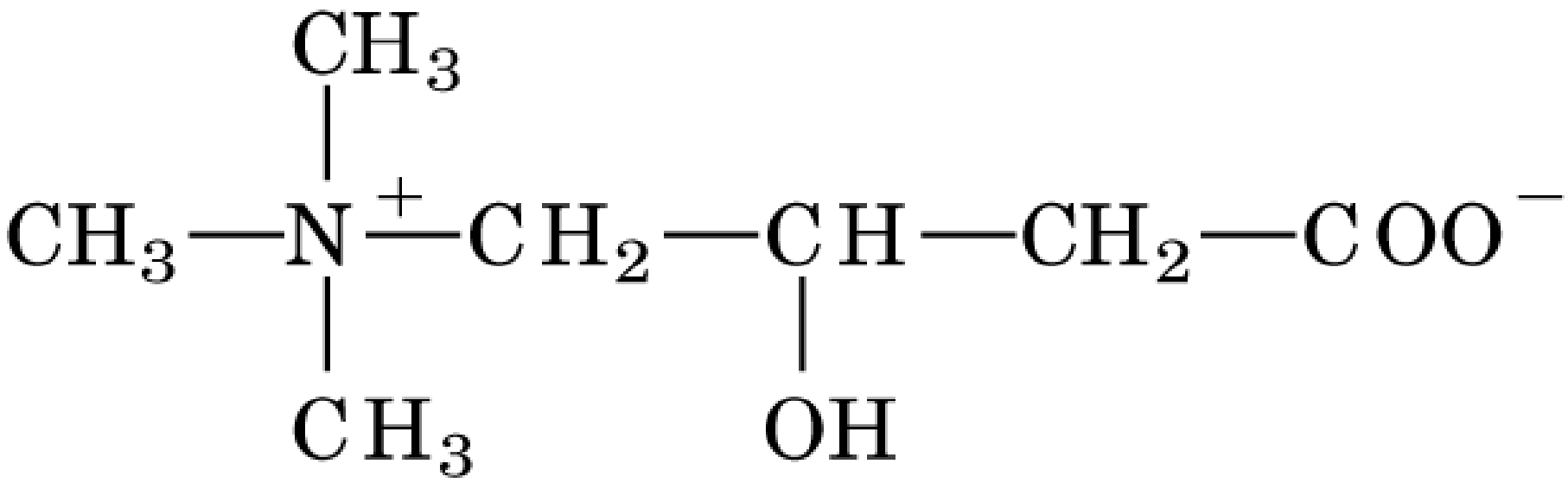
Fatty acid + CoA + ATP → fatty acyl-CoA + AMP + 2Pi

The fatty acid is activated by forming a thioester link with CoA before entering the mitochondria.

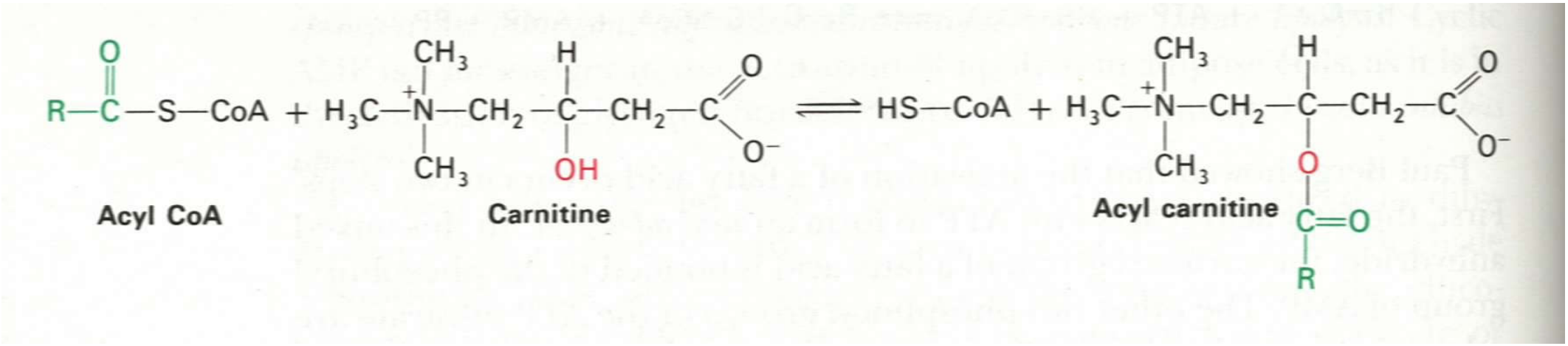
The inner mitochondrial membrane is not permeable to long-chain acyl CoA derivatives.

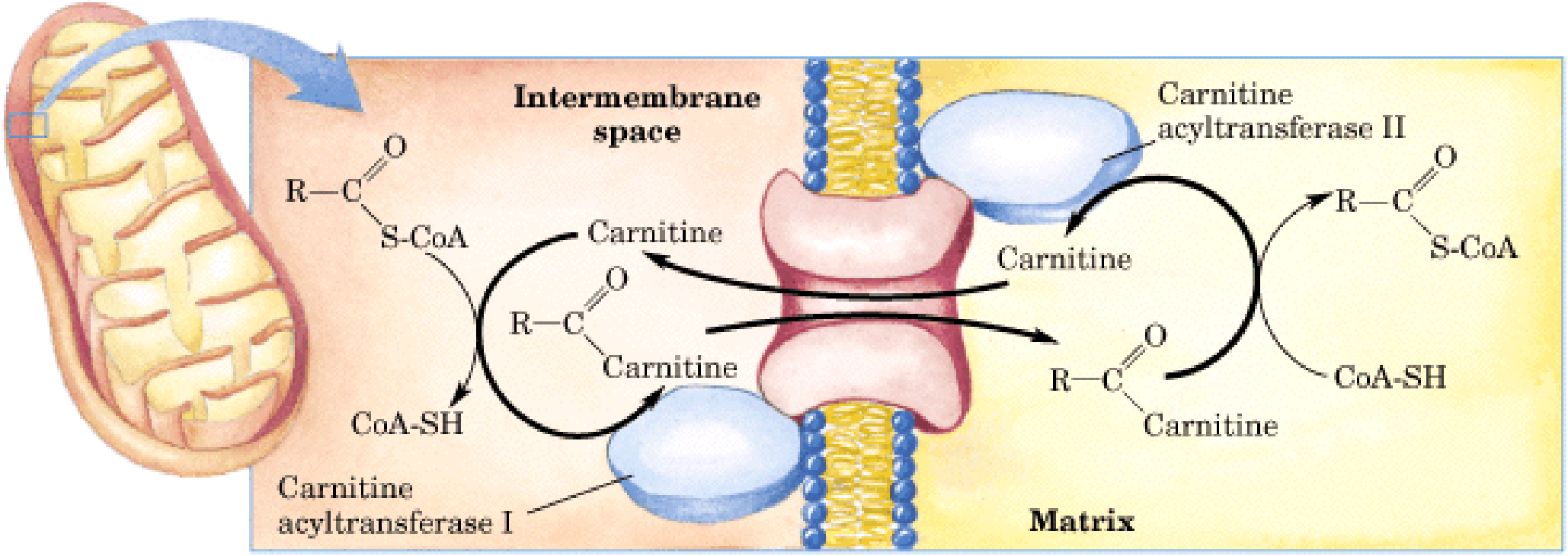


4. Activated (long chain) fatty acids are carried into the matrix by carnitine



Carnitine 肉碱（肉毒碱）


Acyl CoA
Carnitine
Acyl carnitine


Intermembrane space
Carnitine acyltransferase II
Carnitine
Carnitine
CoA-SH
Carnitine
acyltransferase I
Matrix
CoA-SH

- The fatty acyl group is attached to **carnitine** (肉碱) by the action of **carnitine acyltransferase I** located on the outer face of the inner membrane, forming **fatty acyl-carnitine**, leaving the CoA in the cytosol.
- The **acyl-carnitine/carnitine transporter** moves acyl-carnitine across the inner membrane of mitochondria.
- The acyl group is then transferred back to CoA to form fatty acyl-CoA by the action of **carnitine acyltransferase II** located on the inner face of the inner membrane.

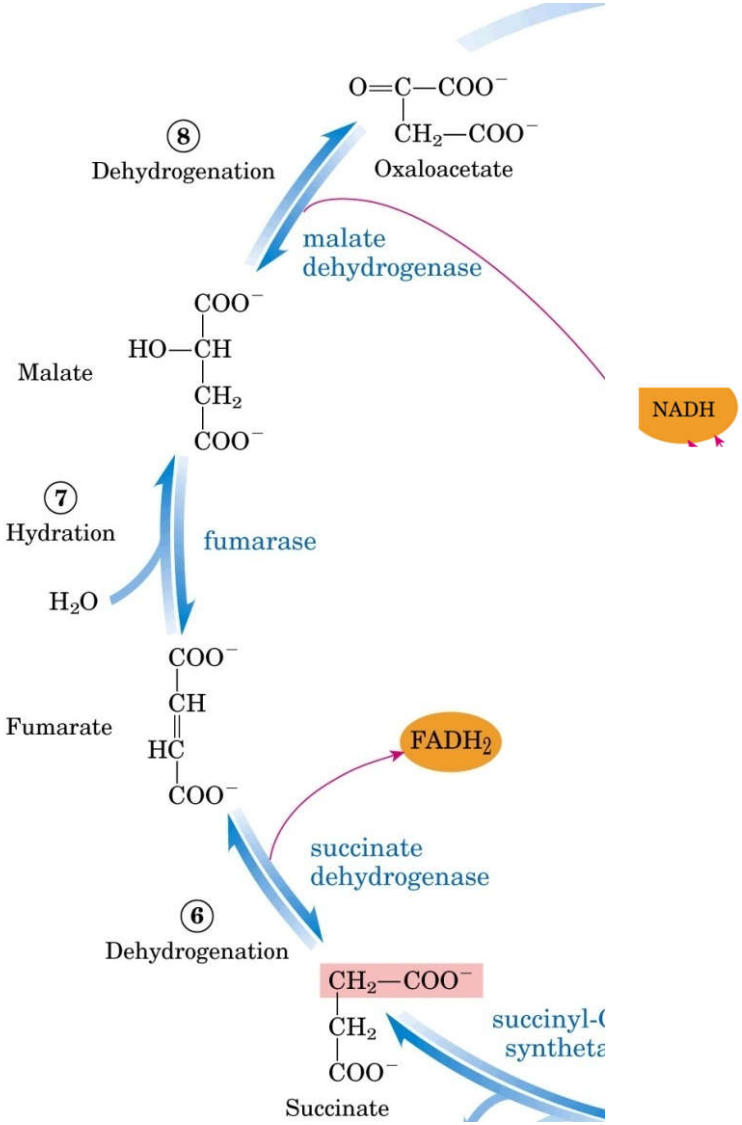


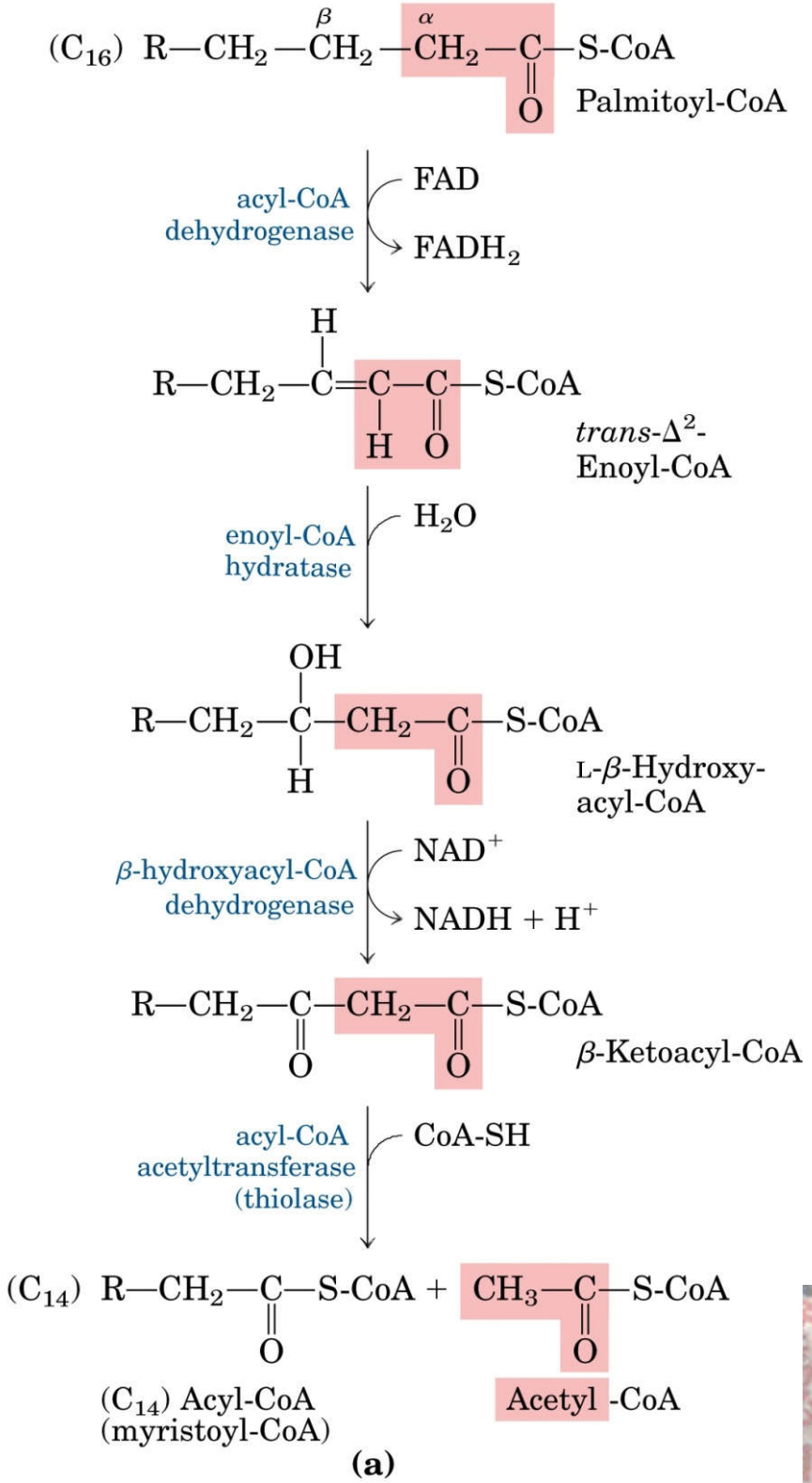
5. Fatty acyl-CoA is oxidized to acetyl-CoA via multiple rounds of β oxidation

The β oxidation consists of four reactions:

- Oxidation by FAD
- Hydration
- Oxidation by NAD^+
- Thiolysis by CoA.




⑧
Dehydrogenation
Oxaloacetate
malate
dehydrogenase
NADH
Malate
⑦
Hydration
fumarase
H_2O
Fumarate
$FADH_2$
succinate
dehydrogenase
⑥
Dehydrogenation
Succinate


(C_{16}) R—CH_2—CH_2—CH_2—C—S-CoA
Palmitoyl-CoA
acyl-CoA dehydrogenase
FAD
$FADH_2$
R—CH_2—C=C—C—S-CoA
trans-Δ^2-Enoyl-CoA
enoyl-CoA hydratase
H_2O
R—CH_2—C(OH)H—CH_2—C—S-CoA
L-β-Hydroxyacyl-CoA
β-hydroxyacyl-CoA dehydrogenase
NAD^+
$NADH + H^+$
R—CH_2—C—CH_2—C—S-CoA
β-Ketoacyl-CoA
acyl-CoA acetyltransferase (thiolase)
CoA-SH
(C_{14}) R—CH_2—C—S-CoA + CH_3—C—S-CoA
(C_{14}) Acyl-CoA (myristoyl-CoA)
Acetyl -CoA
(a)

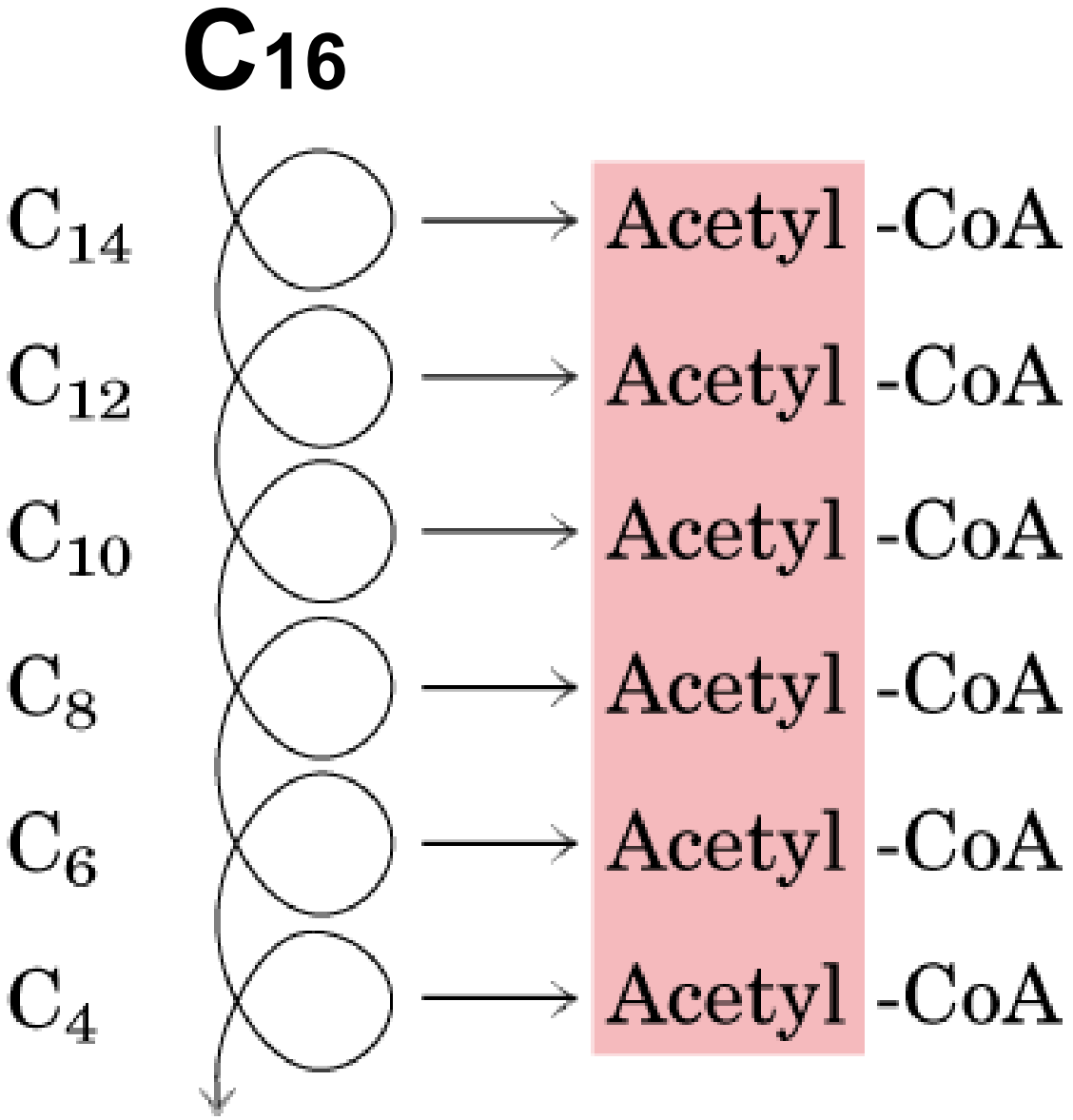
- 1. The 1st oxidation is catalyzed by the membrane-bound acyl-CoA dehydrogenase, converting acyl-CoA to *trans*-Δ^2-enoyl-CoA with electrons collected by FAD.
- 2. The hydration step, catalyzed by enoyl-CoA hydratase, converts the *trans*-Δ^2-enoyl-CoA to L-β-hydroxyacyl-CoA.
- 3. The second oxidation is catalyzed by L-β-hydroxyacyl-CoA dehydrogenase, converting L-β-hydroxylacyl-CoA to β-ketoacyl-CoA, with electrons collected by NAD^+.

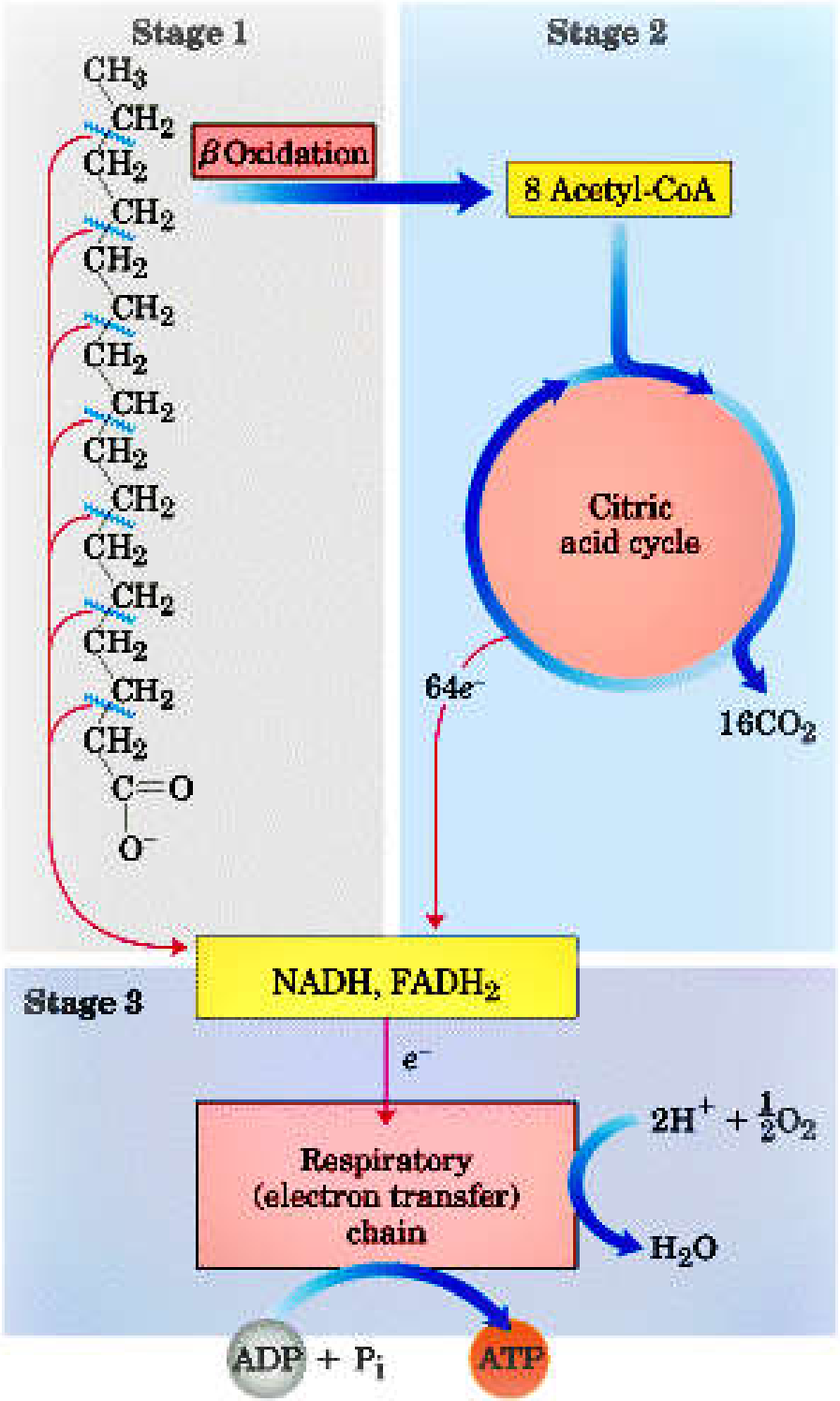


- 4. The acyl-CoA acetyltransferase (or commoly called thiolase) catalyzes the attack of CoA, cleaving β-ketoacyl-CoA between the α and β carbon (thiolysis), generating two acyl-CoA molecules with one entering the citric acid cycle and the other reenter the β oxidation pathway.



- Emergy yield




Stage 1
Stage 2
β Oxidation
8 Acetyl-CoA
Citric acid cycle
64e⁻
16CO2
NADH, FADH2
Stage 3
e⁻
Respiratory (electron transfer) chain
2H⁺ + ½O2
H2O
ADP + Pi
ATP

Biochemist

$$\text{Palmitoyl-CoA} + 7\text{CoA} + 7\text{FAD} + 7\text{NAD}^+ + 7\text{H}_2\text{O} \longrightarrow 8\ \text{acetyl-CoA} + 7\text{FADH}_2 + 7\text{NADH} + 7\text{H}^+$$

•Emergy yield

•The complete oxidization of each 16-carbon palmitate (to H_2O and CO_2) yields 106 ATP

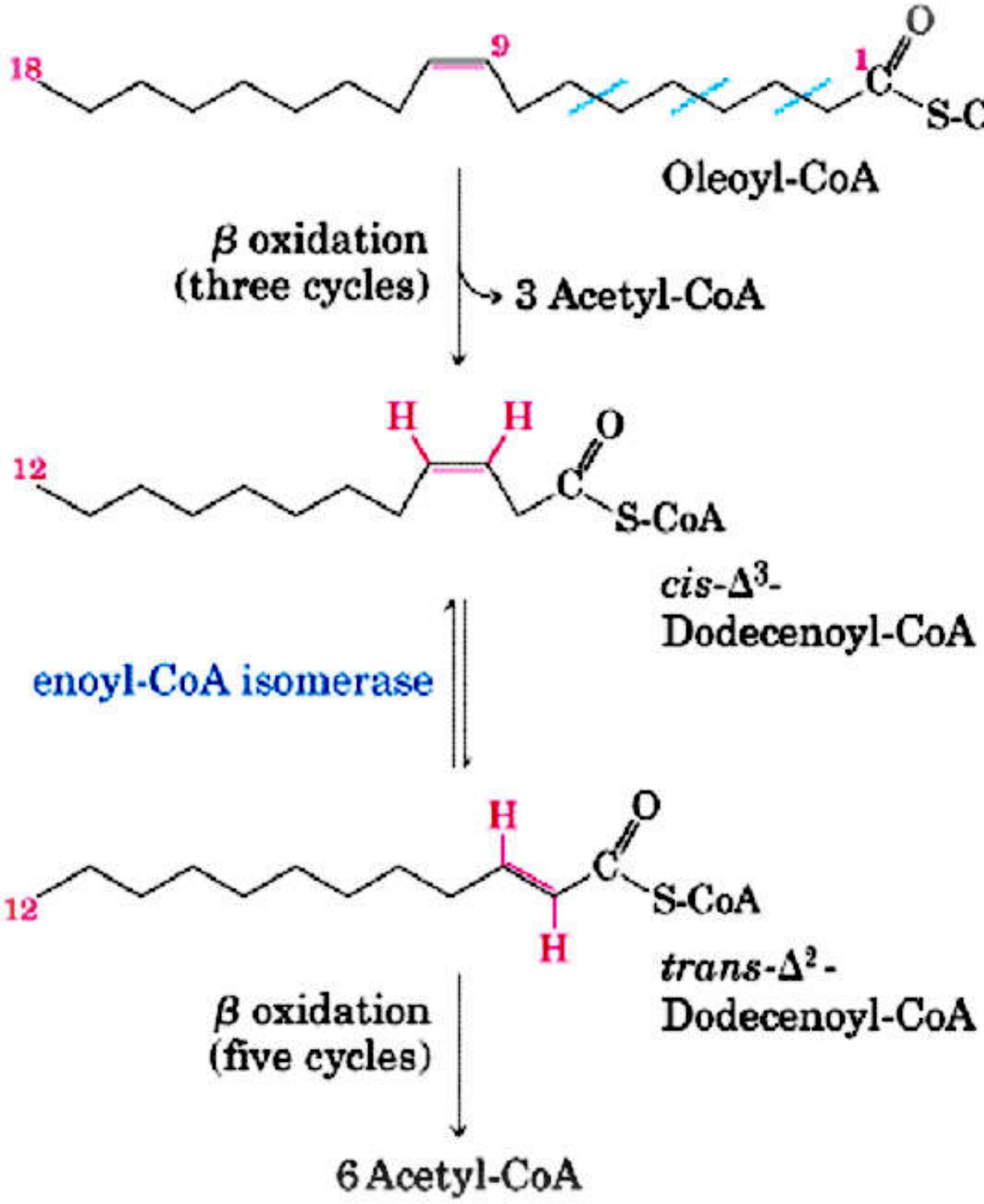
•(32 ATP per glucose, both having about 60% of actual energy recovery).



6. Oxidation of unsaturated fatty acids requires one or two auxiliary enzymes, an isomerase and a reductase

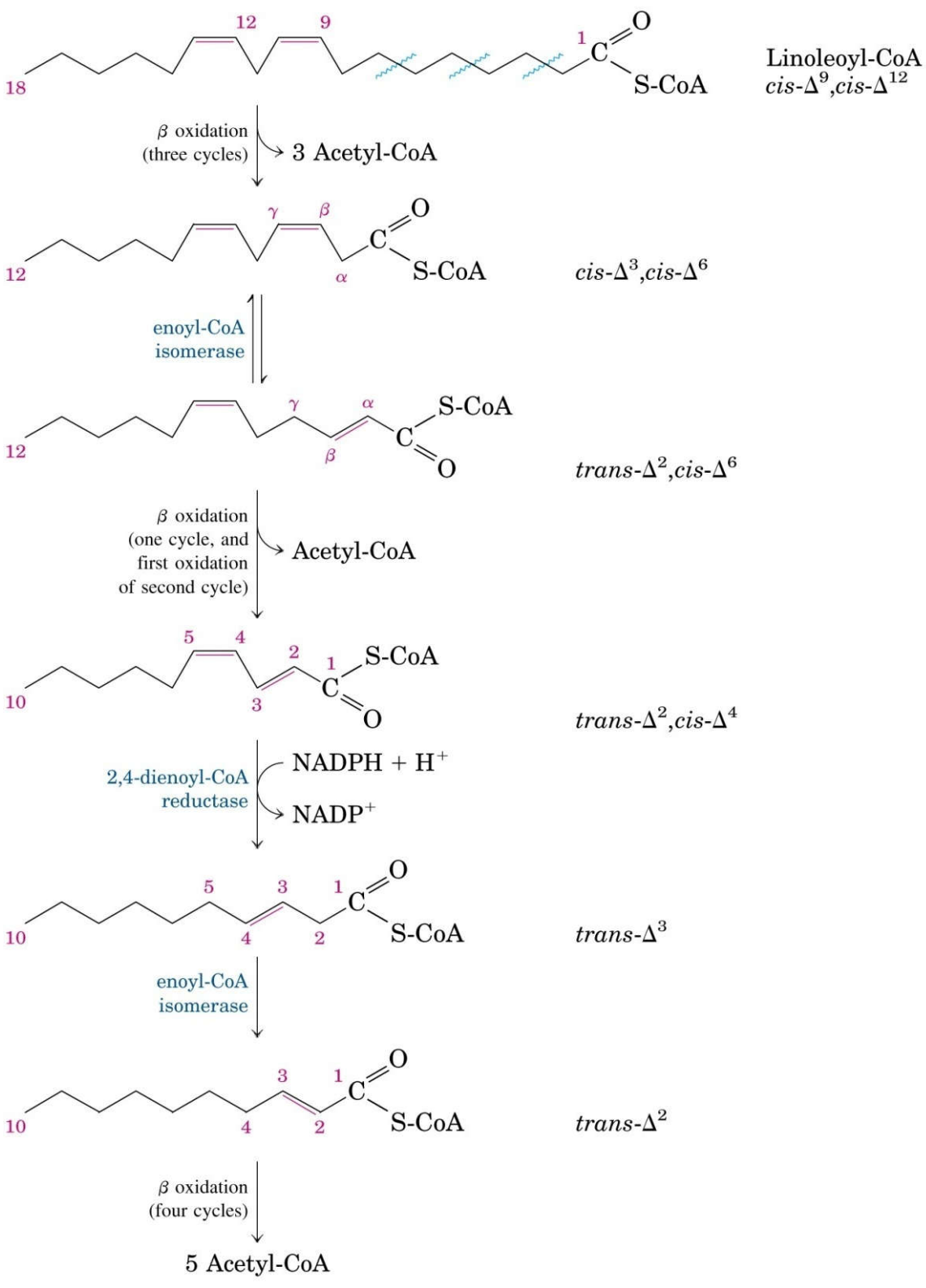
- The isomerase converts a *cis*-Δ^3 double bond to a *trans*-Δ^2 double bond.
- The reductase (2,4-dienoyl-CoA reductase) converts a *trans*-Δ^2, cis-Δ^4 structure to a *trans*-Δ^3 structure, which will be further converted to a *trans*-Δ^2 structure by the isomerase.
- NADPH is needed for the reduction (from two double bonds to one).





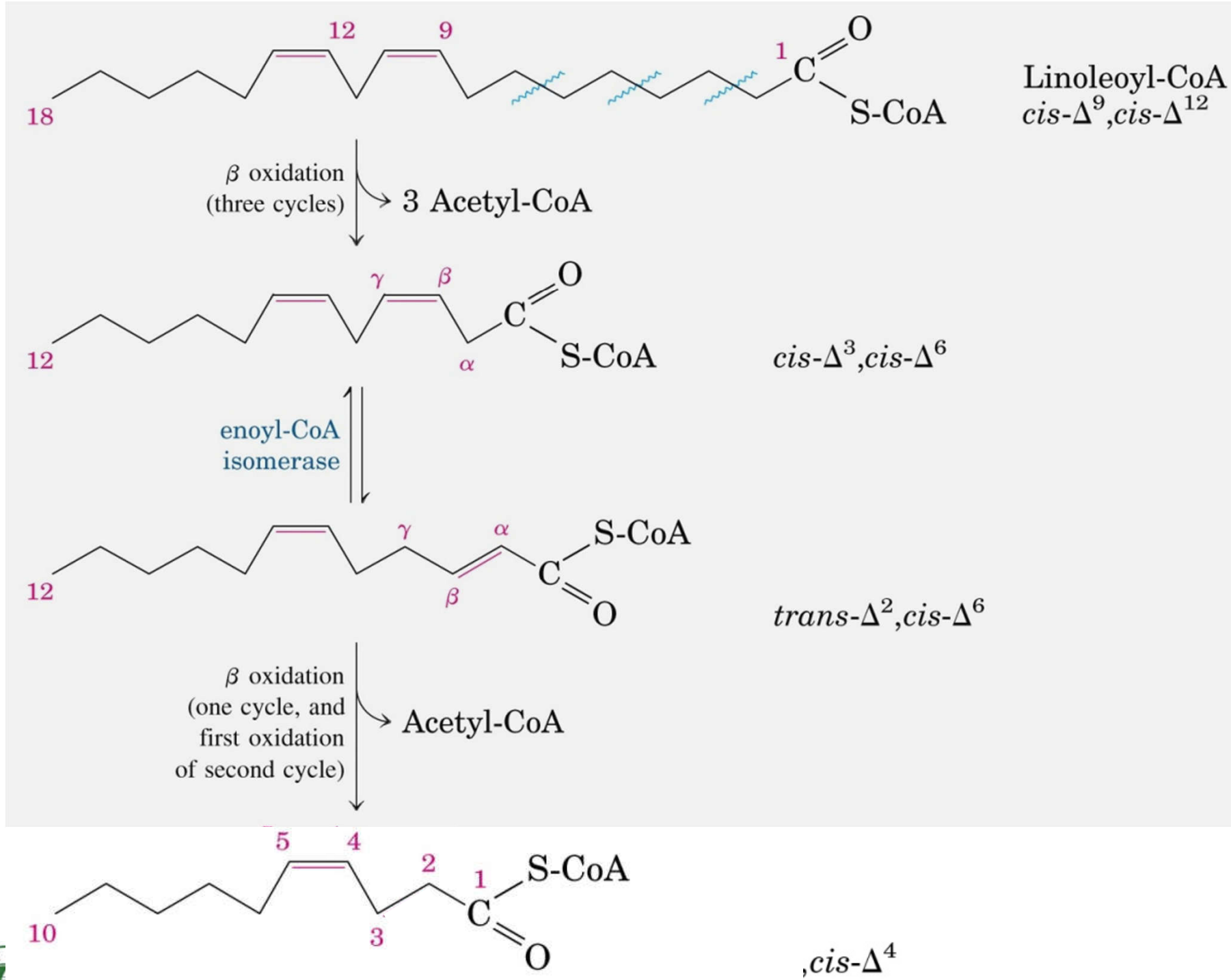
Oxidation of a monounsaturated fatty acid: the enoyl-CoA isomerase helps to reposition the double bond

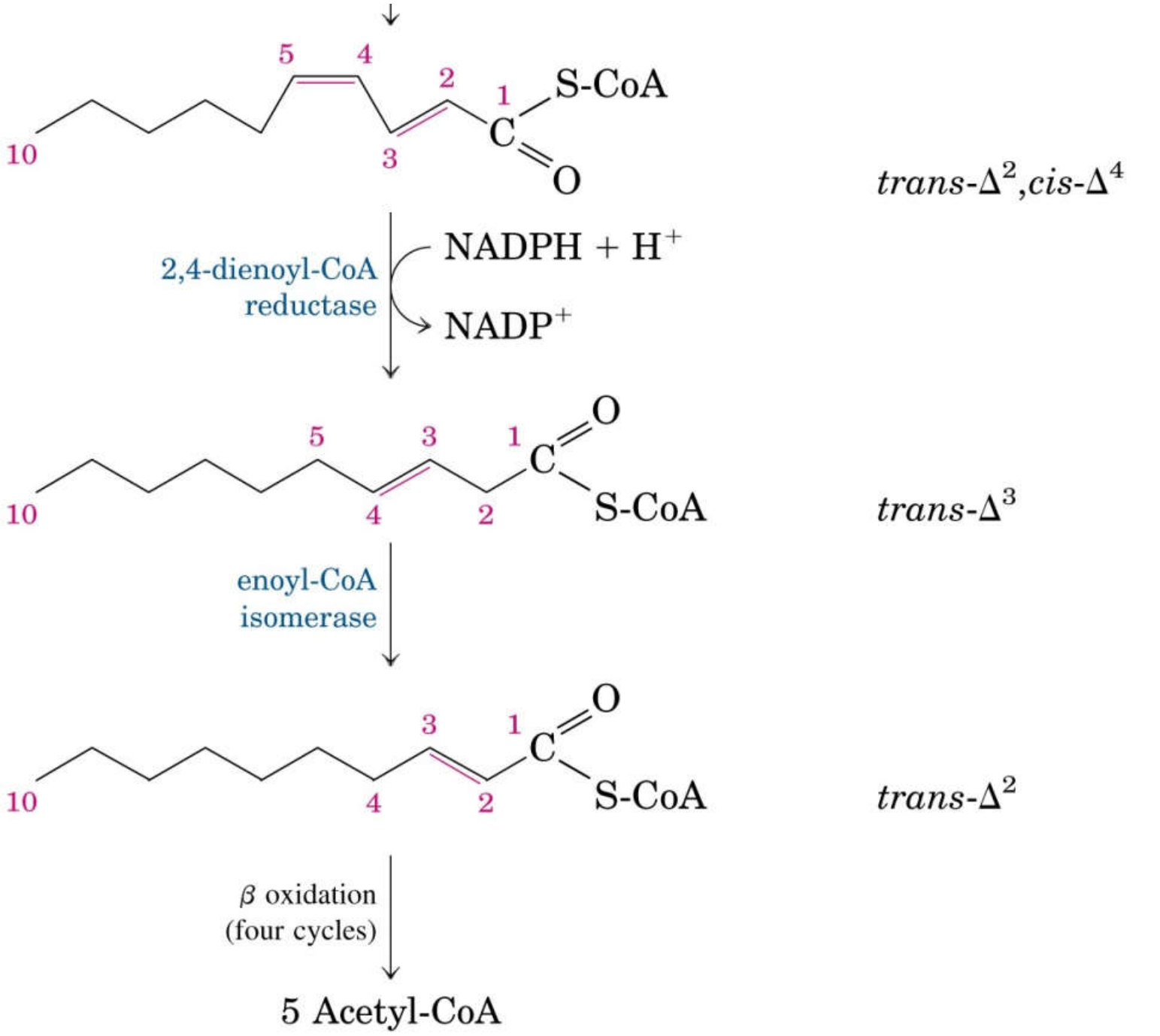




Both an isomerase and a reductase are needed for oxidizing polyunsaturated fatty acids.




12
9
1
O
C
18
S-CoA
Linoleoyl-CoA
cis-Δ9,cis-Δ12
β oxidation
(three cycles)
3 Acetyl-CoA
γ
β
O
C
12
α
S-CoA
cis-Δ3,cis-Δ6
enoyl-CoA
isomerase
γ
α
S-CoA
C
12
β
O
trans-Δ2,cis-Δ6
β oxidation
(one cycle, and
first oxidation
of second cycle)
Acetyl-CoA
5
4
2
1
S-CoA
10
C
3
O
,cis-Δ4


5
4
2
1
S-CoA
10
3
C
O
trans-Δ2,cis-Δ4
2,4-dienoyl-CoA reductase
NADPH + H+
NADP+
5
3
1
O
C
10
4
2
S-CoA
trans-Δ3
enoyl-CoA isomerase
3
1
O
C
10
4
2
S-CoA
trans-Δ2
β oxidation (four cycles)
5 Acetyl-CoA

7. The oxidation of odd-chain fatty acids forms C3 propionyl-CoA

- Fatty acids having an odd number of carbon atoms are also degraded by the β–oxidation pathway in the same way as those with an even number of carbon atoms.
- The only difference is that in the final round the five carbon acyl CoA intermediate is cleaved into C3 **propionyl CoA** and acetyl CoA.



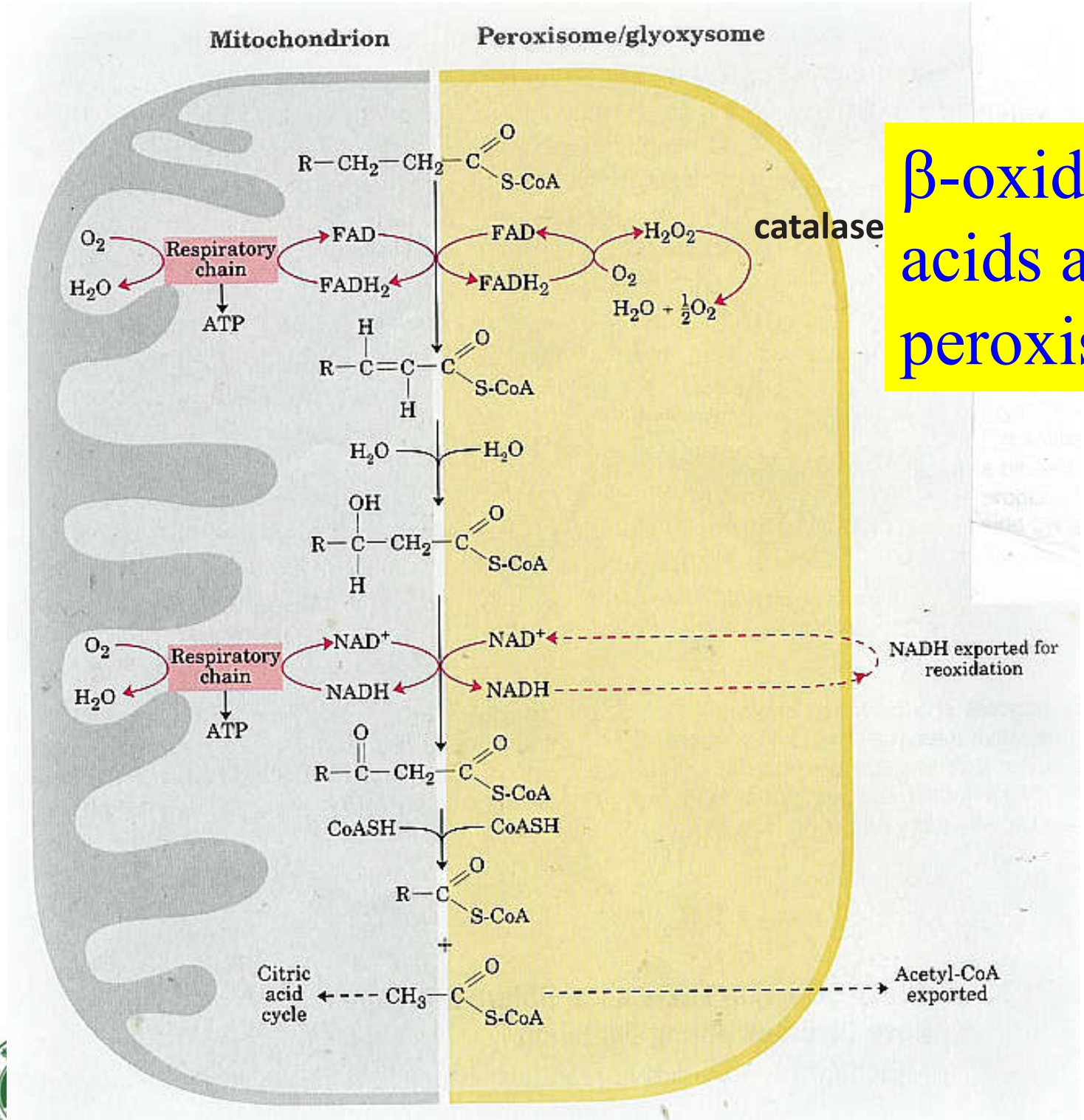
8. Fatty acid oxidation also occurs in peroxisomes (glyoxysomes)

- Peroxisome is now recognized as the principal organelle in which fatty acids are oxidized in most cell types.
- The acetyl-CoA produced in animal peroxisomes is transported into cytosol, where it is used in the synthesis of cholesterol and other metabolites.



- The acetyl-CoA produced in plant peroxisomes/ glyoxysomes (especially in germinating seeds) is converted to succinate via the glyoxylate cycle, and then to glucose via gluconeogenesis.

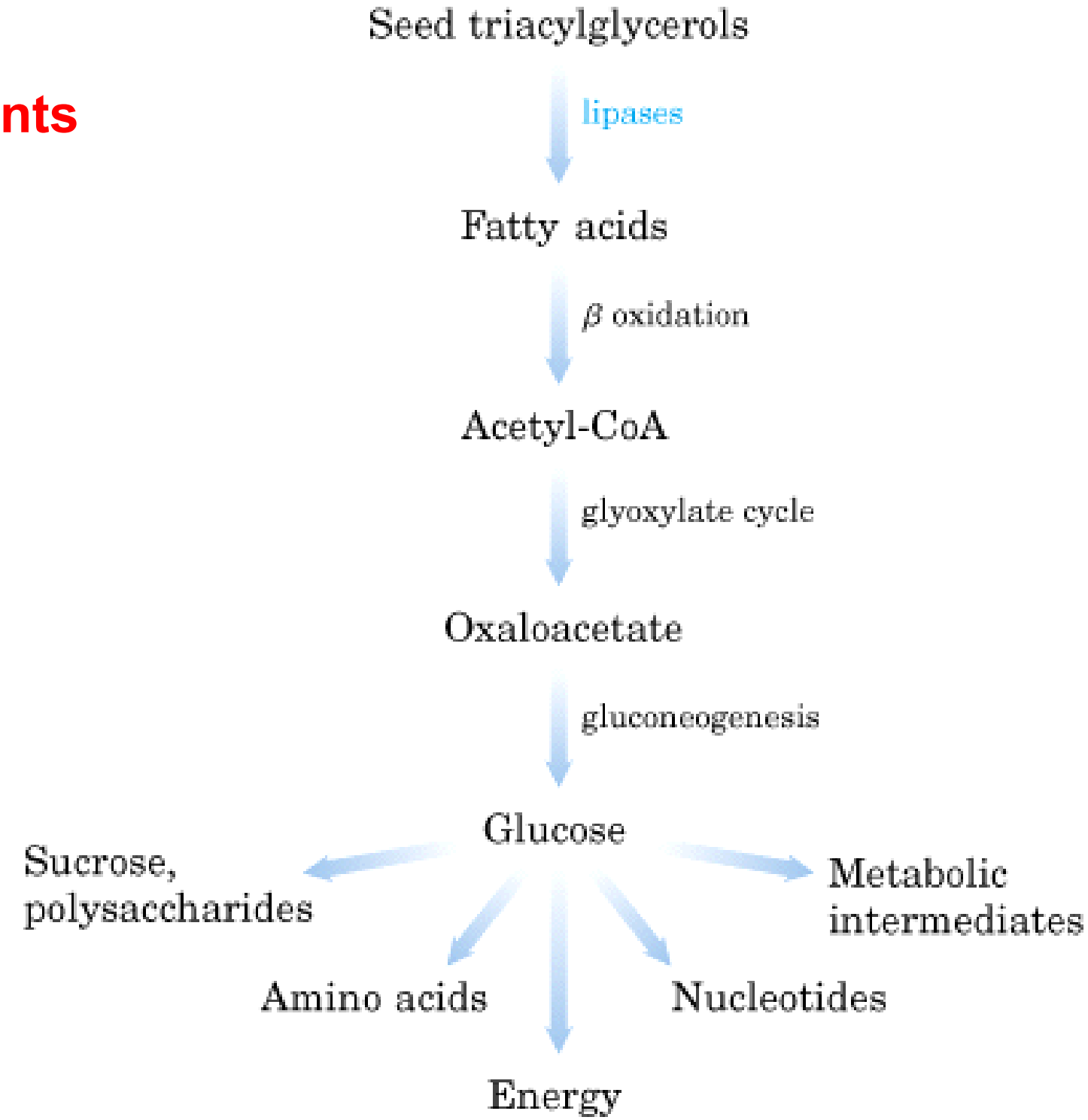




β-oxidation of fatty acids also occurs in peroxisomes



In Plants


Seed triacylglycerols
lipases
Fatty acids
β oxidation
Acetyl-CoA
glyoxylate cycle
Oxaloacetate
gluconeogenesis
Glucose
Sucrose, polysaccharides
Metabolic intermediates
Amino acids
Nucleotides
Energy

9. Acetyl-CoA in liver can be converted to ketone bodies when carbohydrate supply is not optimal

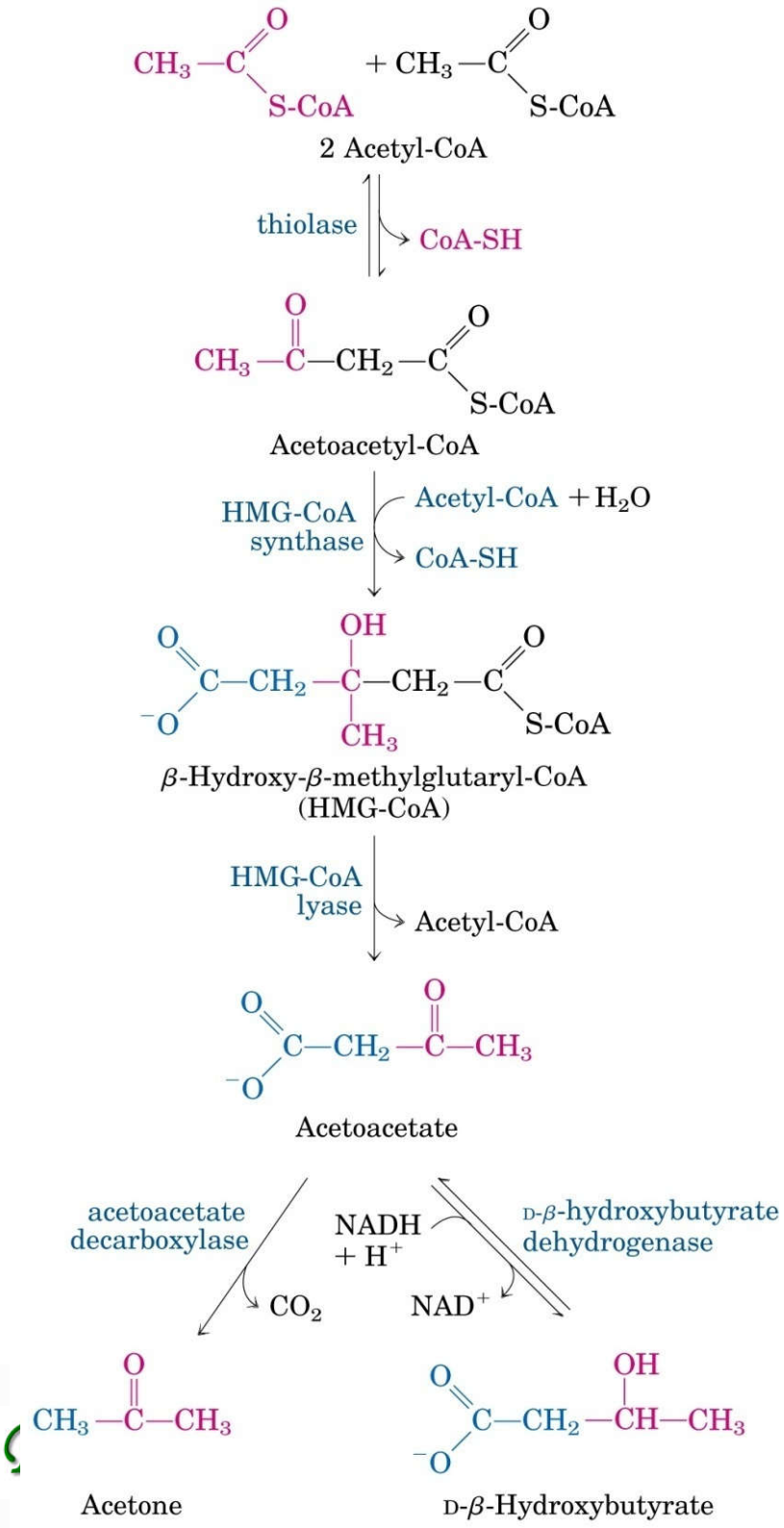


•In liver, The acetyl-CoA generated from active fatty acid oxidation can not be oxidized via the citric acid cycle and will be converted to acetoacetate（乙酰乙酸）, β-hydroxybutyrate（羟基丁酸）, and acetone（丙酮）(i.e., the ketone bodies) in mitochondria for export to other tissues.



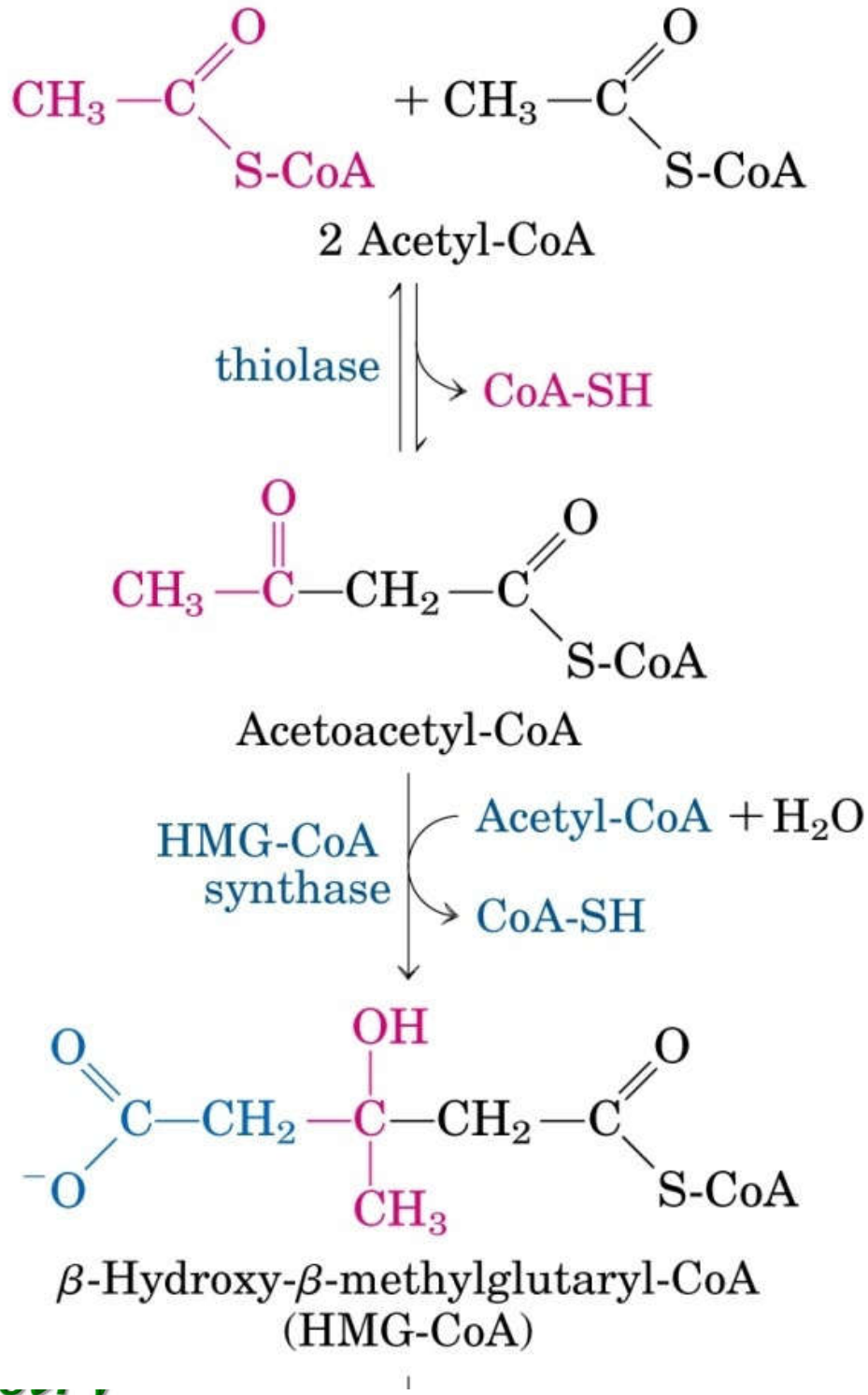
- For ketone body formation, first two acetyl-CoAs condense to form acetoacetyl-CoA (乙酰乙酰**CoA**) catalyzed by thiolase; then addition of another acetyl-CoA forms β-hydroxyl-β-methylglutaryl-CoA (β–羟－ β－甲基戊二酰**CoA**).
- Acetoacetate can be decarboxylated to form acetone (decarboxylase) or reduced to D-β-hydroxylbutyrate (β–羟丁酸) (dehydrogenase).

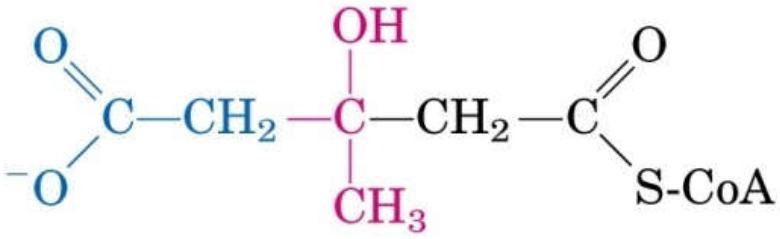




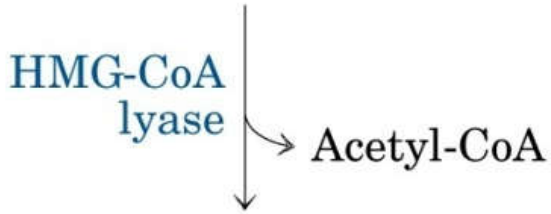
Acety-CoA can be converted to ketone bodies in liver.

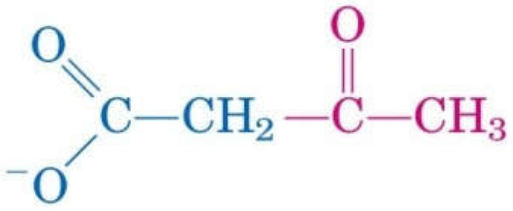



CH3—C(=O)—S-CoA + CH3—C(=O)—S-CoA
2 Acetyl-CoA
thiolase
CoA-SH
CH3—C(=O)—CH2—C(=O)—S-CoA
Acetoacetyl-CoA
HMG-CoA synthase
Acetyl-CoA + H2O
CoA-SH
β-Hydroxy-β-methylglutaryl-CoA
(HMG-CoA)

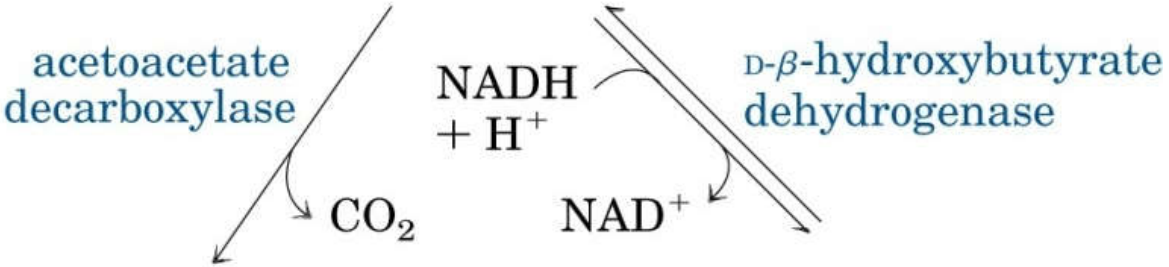


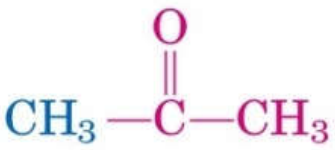
β-Hydroxy-β-methylglutaryl-CoA
(HMG-CoA)



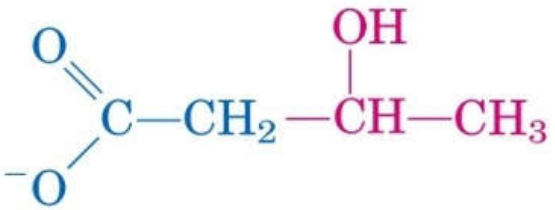


Acetoacetate





Acetone

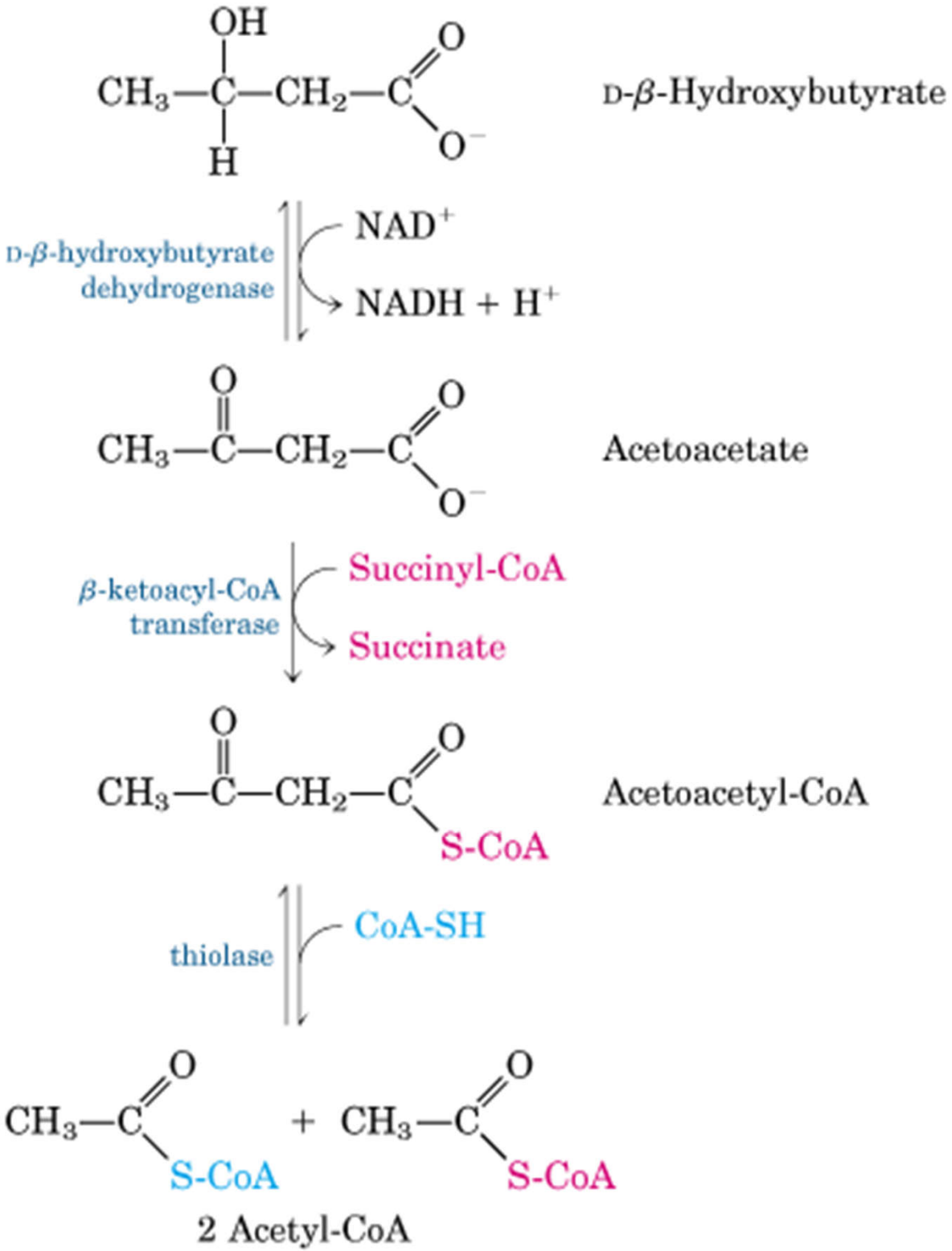


D-β-Hydroxybutyrate



10. Ketone bodies are converted back to acetyl-CoA in extrahepatic tissues





Ketone bodies are converted to acetyl-CoA in extrahepatic tissues.



Summary

- Fatty acids are activated to the acyl-CoA form and is then carried into mitochondria by carnitine with the help of two carnitine acyltranseferase isozymes (I and II) located on the outside and inside of the inner membrane.
- Acyl-CoA is converted to acetyl-CoA after going through multiple rounds of the four-step (dehydrogenation, hydration, dehydrogenation and thiolysis) β-oxidation pathway.



- Oxidative degradation of unsaturated fatty acids need two extra enzymes: an isomerase and a reductase.
- The rate of β-oxidation pathway is controlled by the rate at which acyl-CoA is transported into mitochondria.
- Excess acetyl-CoA (under conditions when glucose metabolism is not optimal) can be converted to ketone bodies (acetoacetate, β-hydroxylbutyrate and acetone) in the liver cells and reconverted into acetyl-CoA in extrahepatic cells.



Fatty Acid Synthesis

(K3)



1. Fatty acid synthesis takes a different pathway from its degradation

- **Occurs in the cytosol (chloroplasts in plants).**
- **Acetyl-CoA provides the first two carbons, which is elongated by sequential addition of two-carbon units donated from malonyl-CoA.**
- **Intermediates are attached to the -SH groups of an acyl carrier protein (ACP).**



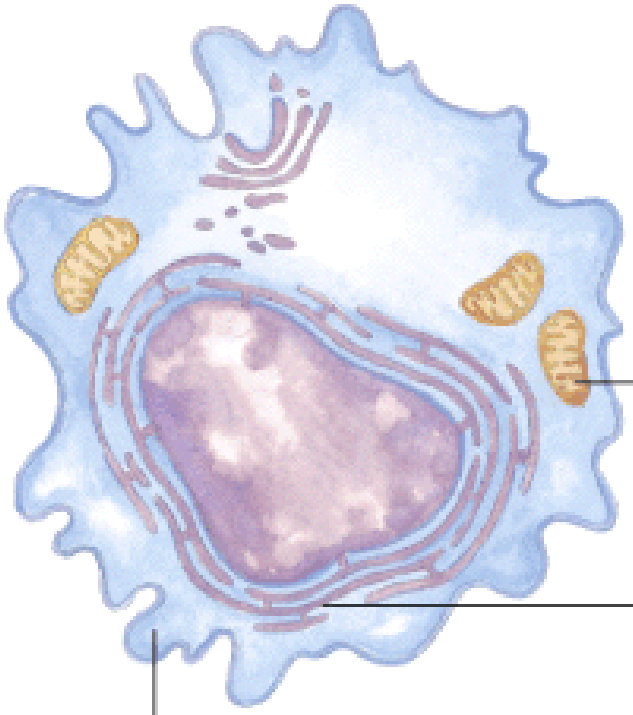
- **NADPH is the reductant.**
- **The enzymes are associated as a multi-enzyme complex or even being in one polypeptide chain in higher organisms (fatty acid synthase).**
- **Elongation by the fatty acid synthase complex stops upon formation of palmitate (C_{16}), further elongation and desaturation are carried out by other enzyme systems.**

unsaturated fatty acids
saturated fatty acids
desatureation



Animal cells, yeast cells

Plant cells



Mitochondria

- No fatty acid oxidation (plant cells)
- Fatty acid oxidation
- Acetyl-CoA production
- Ketone body synthesis
- Fatty acid elongation

Endoplasmic reticulum

- Phospholipid synthesis
- Sterol synthesis (late stages)
- Fatty acid elongation
- Fatty acid desaturation

Cytosol

- NADPH production (pentose phosphate pathway; malic enzyme)
- [NADPH]/[$NADP^+$] high
- Isoprenoid and sterol synthesis (early stages)
- Fatty acid synthesis

Chloroplasts

- NADPH, ATP production
- [NADPH]/[$NADP^+$] high
- Fatty acid synthesis

Peroxisomes

- Fatty acid oxidation ($\longrightarrow H_2O_2$)
- Catalase, peroxidase: $H_2O_2 \longrightarrow H_2O$

1. Acetyl-COA 转运（柠檬酸转运系统）

2. Acetyl-COA 羧化为malonyl-COA，作为脂肪酸合成的主要前体

3. Fatty acid synthesis：

Acetyl-COA进位KS巯基, malonyl-COA进位ACP巯基

Condensation

Reduction

Dehydration

Reduction

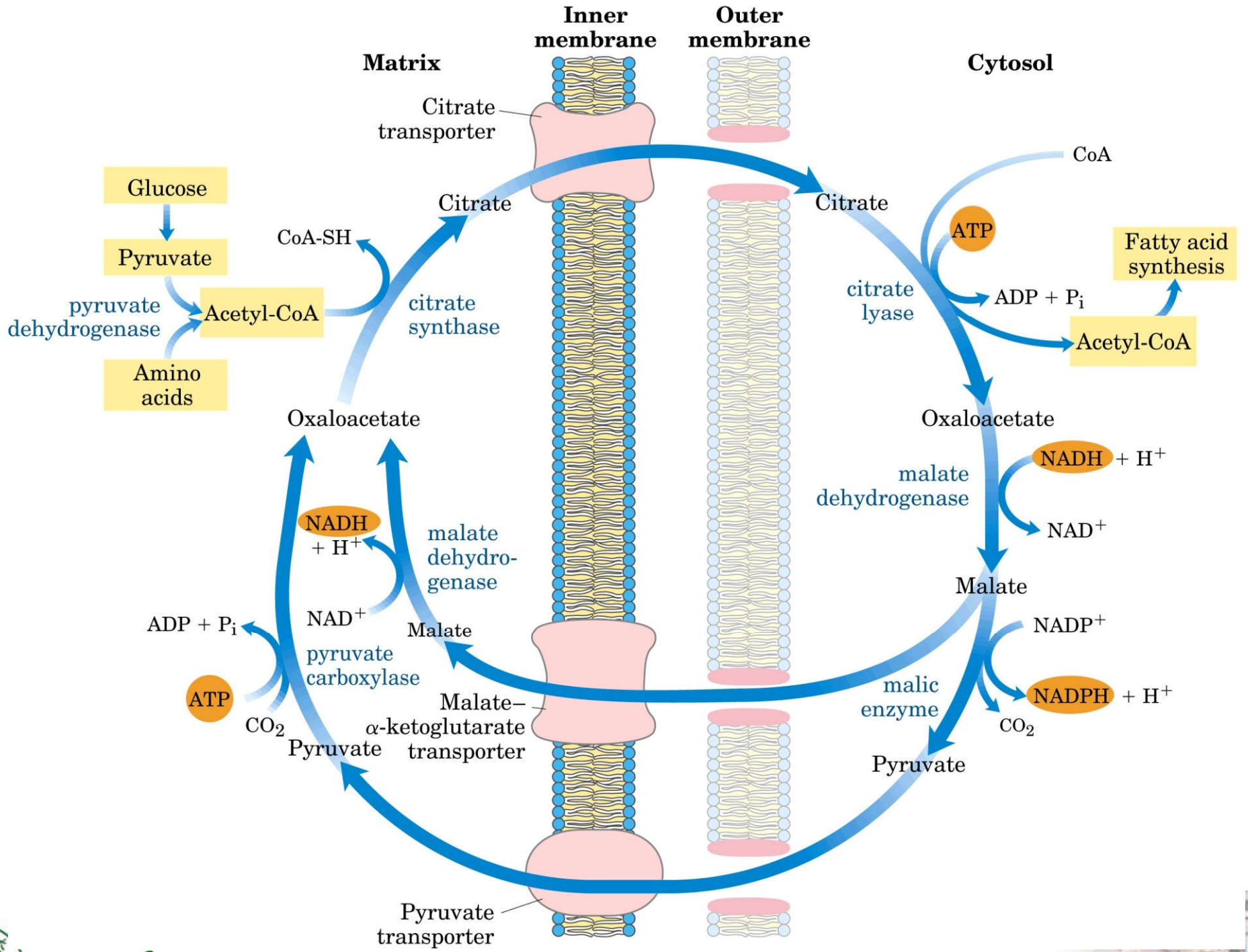
延长2C fatty acyl 移位至外周巯基



2. The acetyl groups of the mitochondrion are transported into the cytosol in the form of citrate (柠檬酸转运系统）

- The acetyl-CoA molecules are made from glucose and amino acids in mitochondria.
- They are shuttled into the cytosol in the form of citrate via the citrate transporter of the inner membrane.
- Acetyl-CoA is regenerated by the action of ATP-dependent citrate lyase in the cytosol.
- Oxaloacetate is shuttled back into the mitochondria as malate or pyruvate.



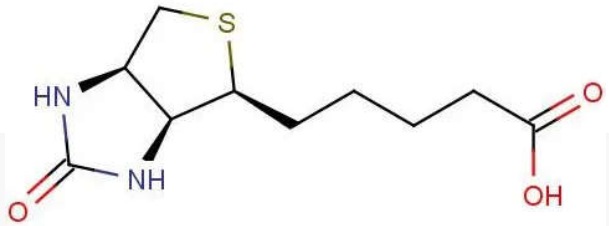


Biochemistry

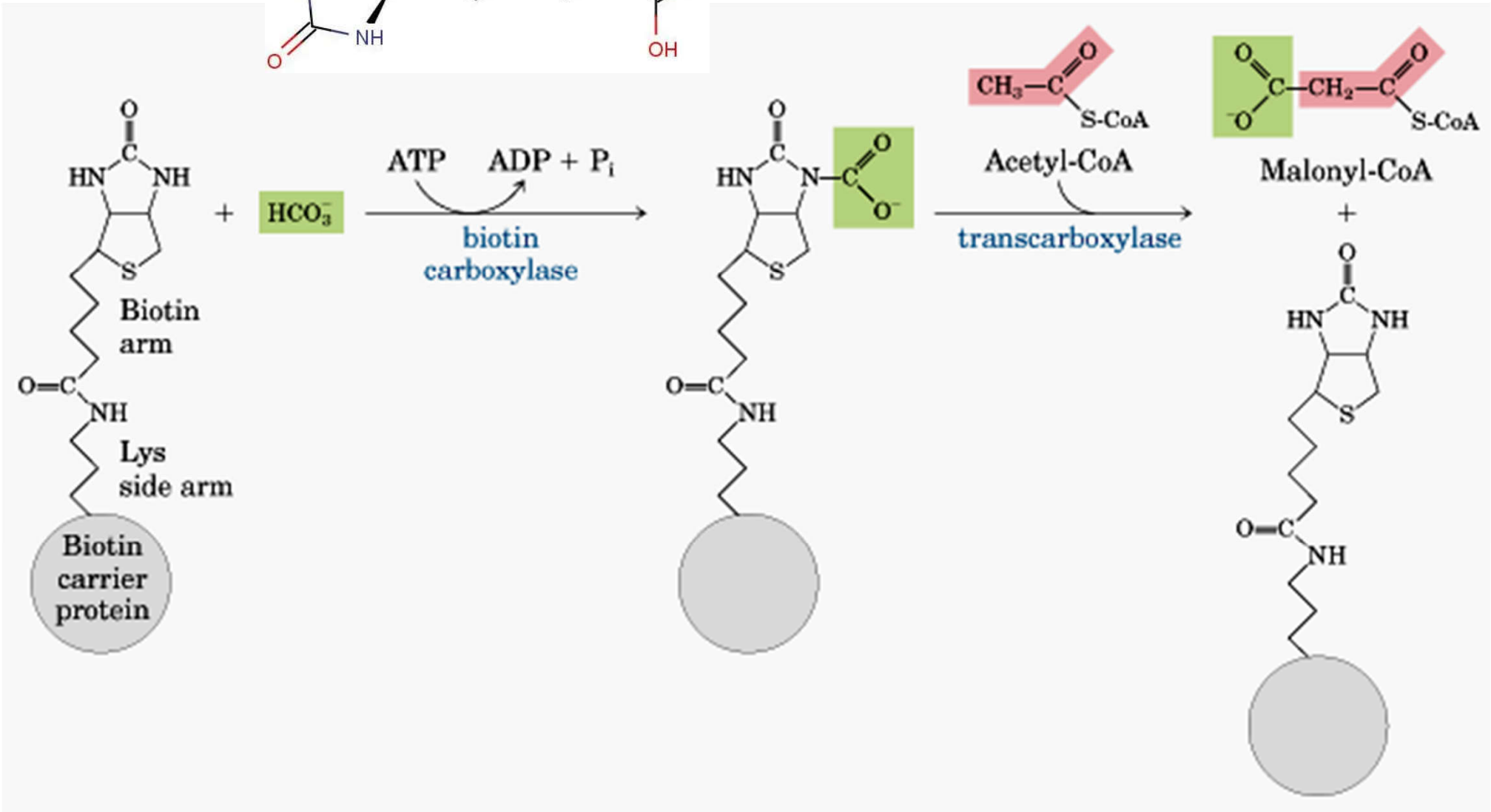
3. Malonyl-CoA is formed from acetyl-CoA and bicarbonate

- Salih Wakil discovered that HCO_3^- is required for fatty acid synthesis.
- Acetyl-CoA carboxylase catalyzes this carboxylation reaction.
- The enzyme has three functional parts: a biotin carrier protein; an ATP-dependent biotin carboxylase; and a transcarboxylase.





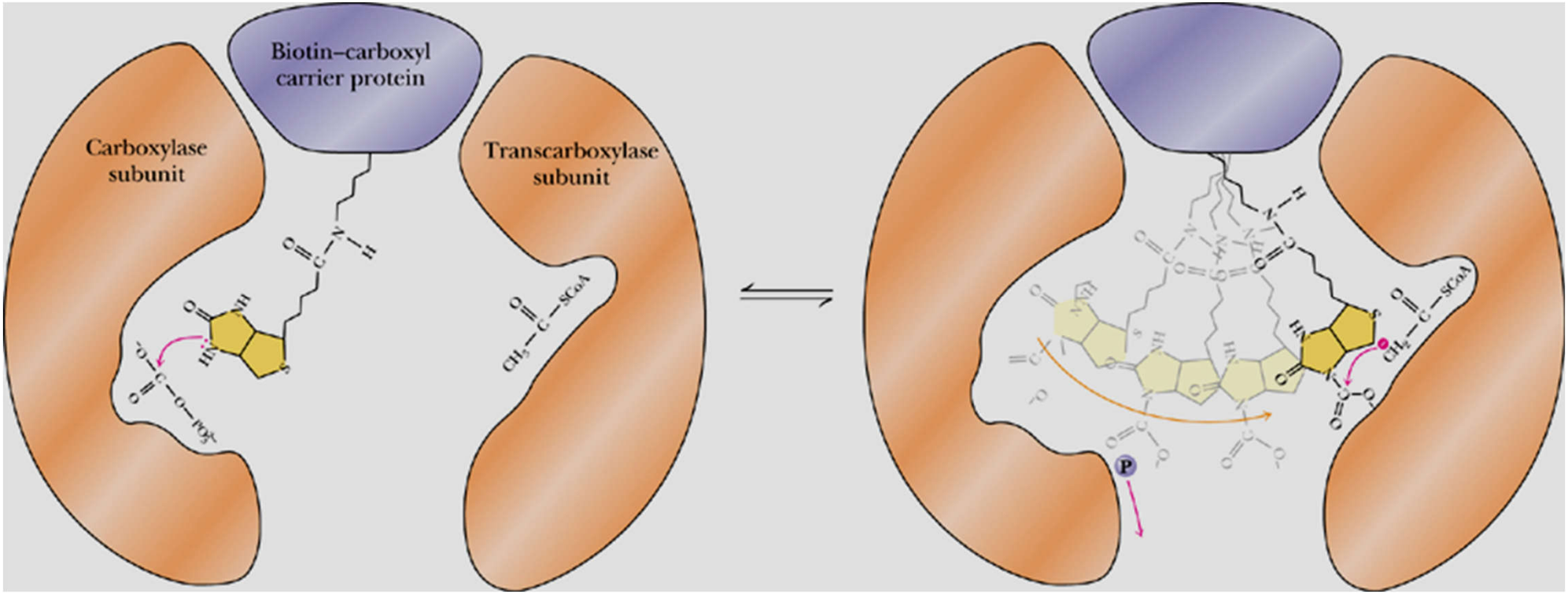
[维生素B7 生物素]



biotin carboxyl-carrier protein **BCCP**

Biochemistry




Biotin–carboxyl carrier protein
Carboxylase subunit
Transcarboxylase subunit
P

4. The acetyl and malonyl groups are first transferred to two –SH groups of the fatty acid synthase complex

- The acetyl group of acetyl-CoA is first transferred to the –SH group of a Cys residue on the β-ketoacyl-ACP synthase (KS) in a reaction catalyzed by acetyl-CoA-ACP transacetylase (AT).
- The malonyl group is transferred from malonyl-CoA to the –SH group of the acyl carrier protein (ACP) catalyzed by malonyl-CoA-ACP transacetylase (MT).

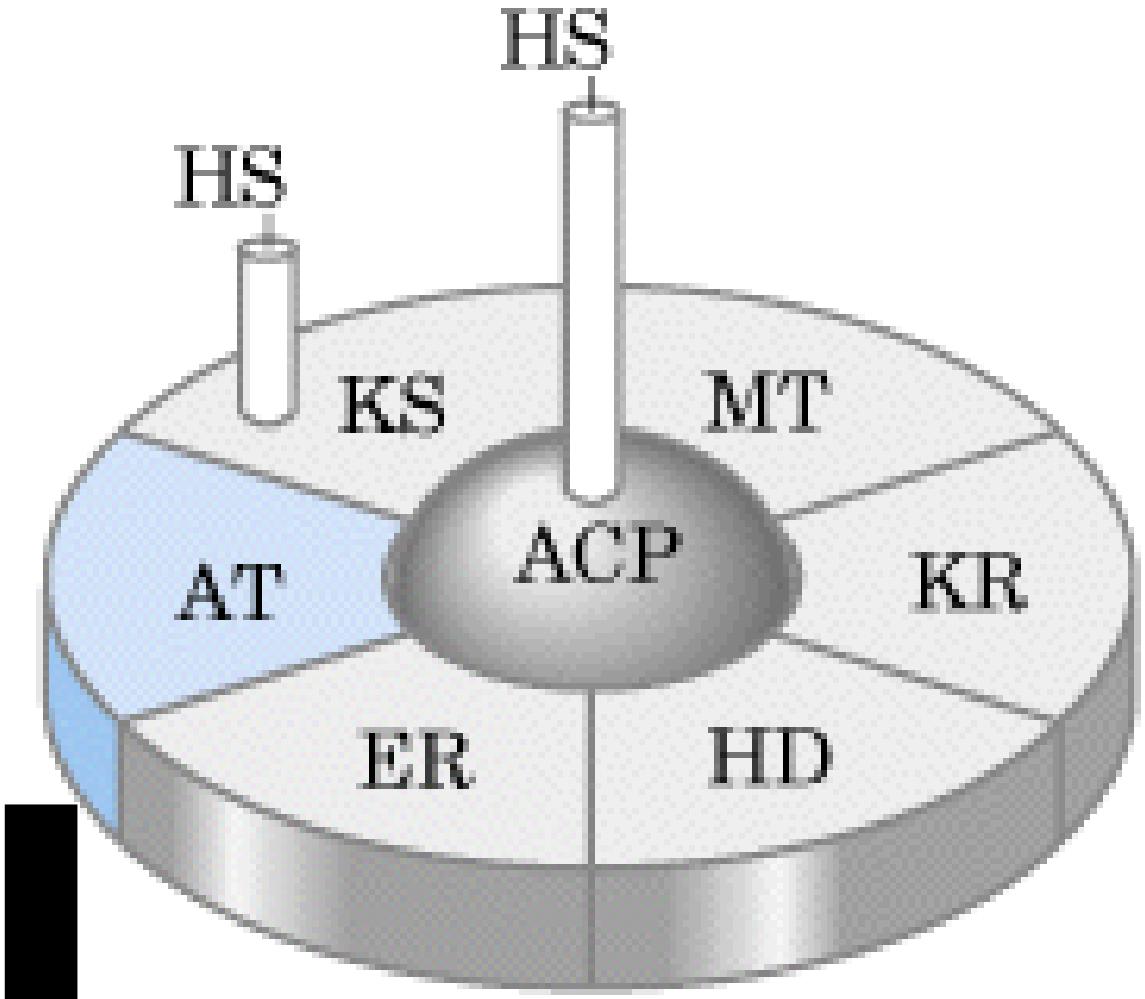


table 21–1

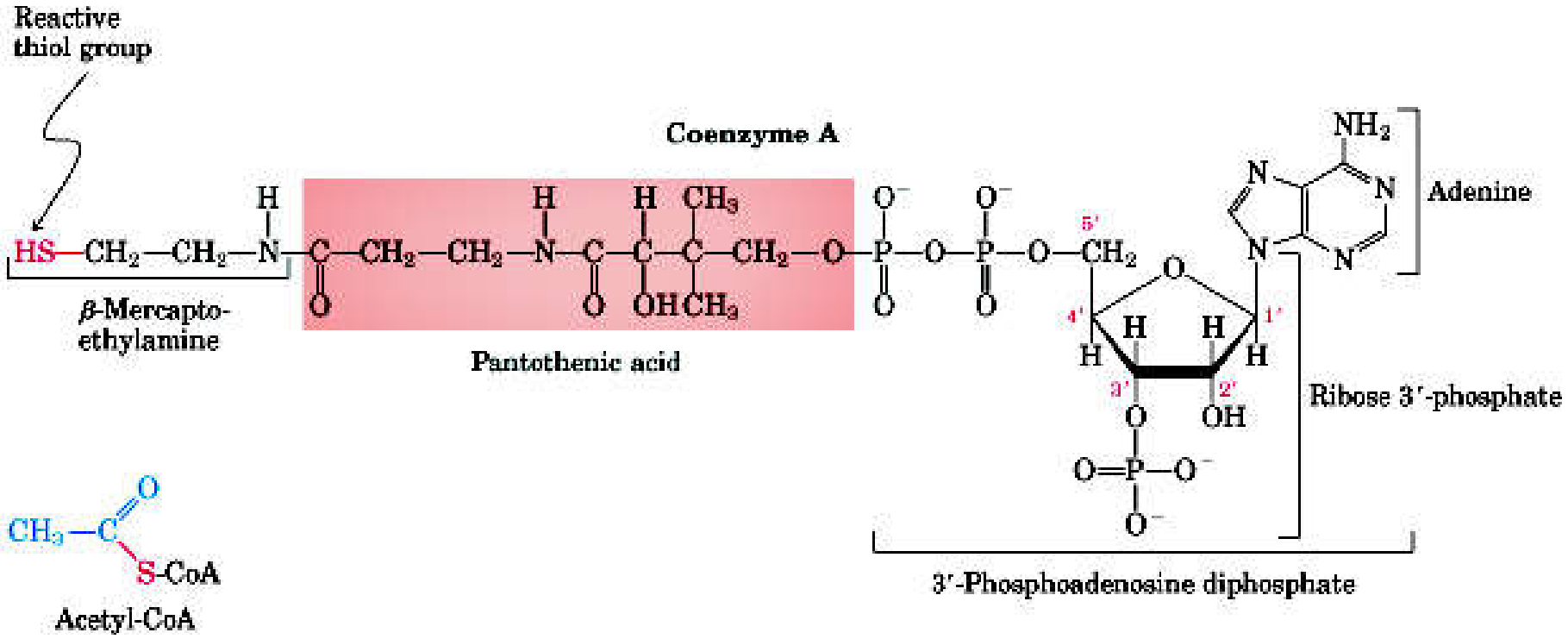
Proteins of the Fatty Acid Synthase Complex of *E. coli*

Protein	Role
Acyl carrier protein (ACP)	Carries acyl groups in thioester linkage
Acetyl-CoA–ACP transacetylase (AT)	Transfers acyl group from CoA to Cys residue of KS
β-Ketoacyl-ACP synthase (KS)	Condenses acyl and malonyl groups (there are at least three isozymes of KS)
Malonyl-CoA–ACP transferase (MT)	Transfers malonyl group from CoA to ACP
β-Ketoacyl-ACP reductase (KR)	Reduces β-keto group to β-hydroxy group
β-Hydroxyacyl-ACP dehydratase (HD)	Removes H_2O from β-hydroxyacyl-ACP, creating double bond
Enoyl-ACP reductase (ER)	Reduces double bond, forming saturated acyl-ACP

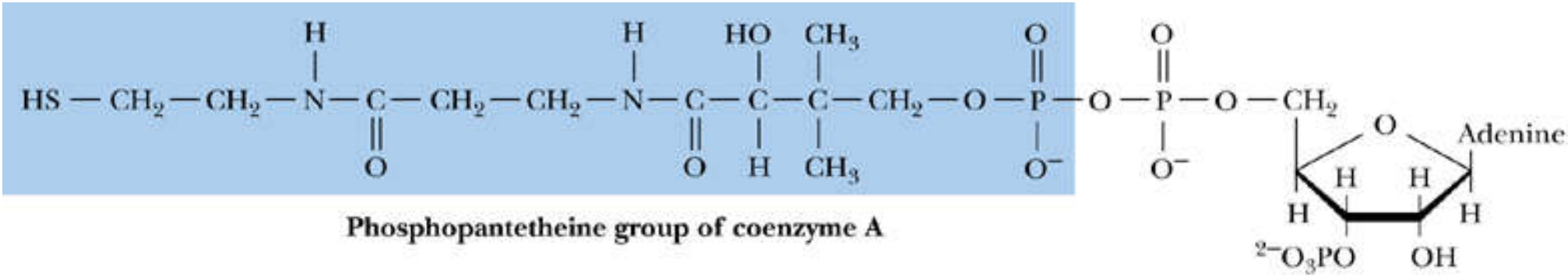



HS
HS
KS
MT
AT
ACP
KR
ER
HD

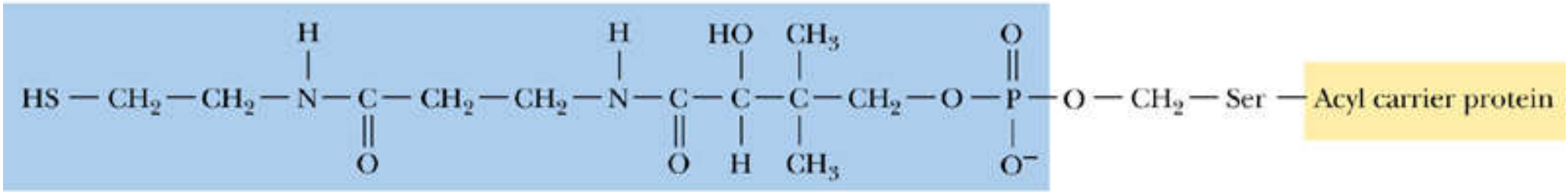
B5 泛酸(遍多酸) 自然界中广泛



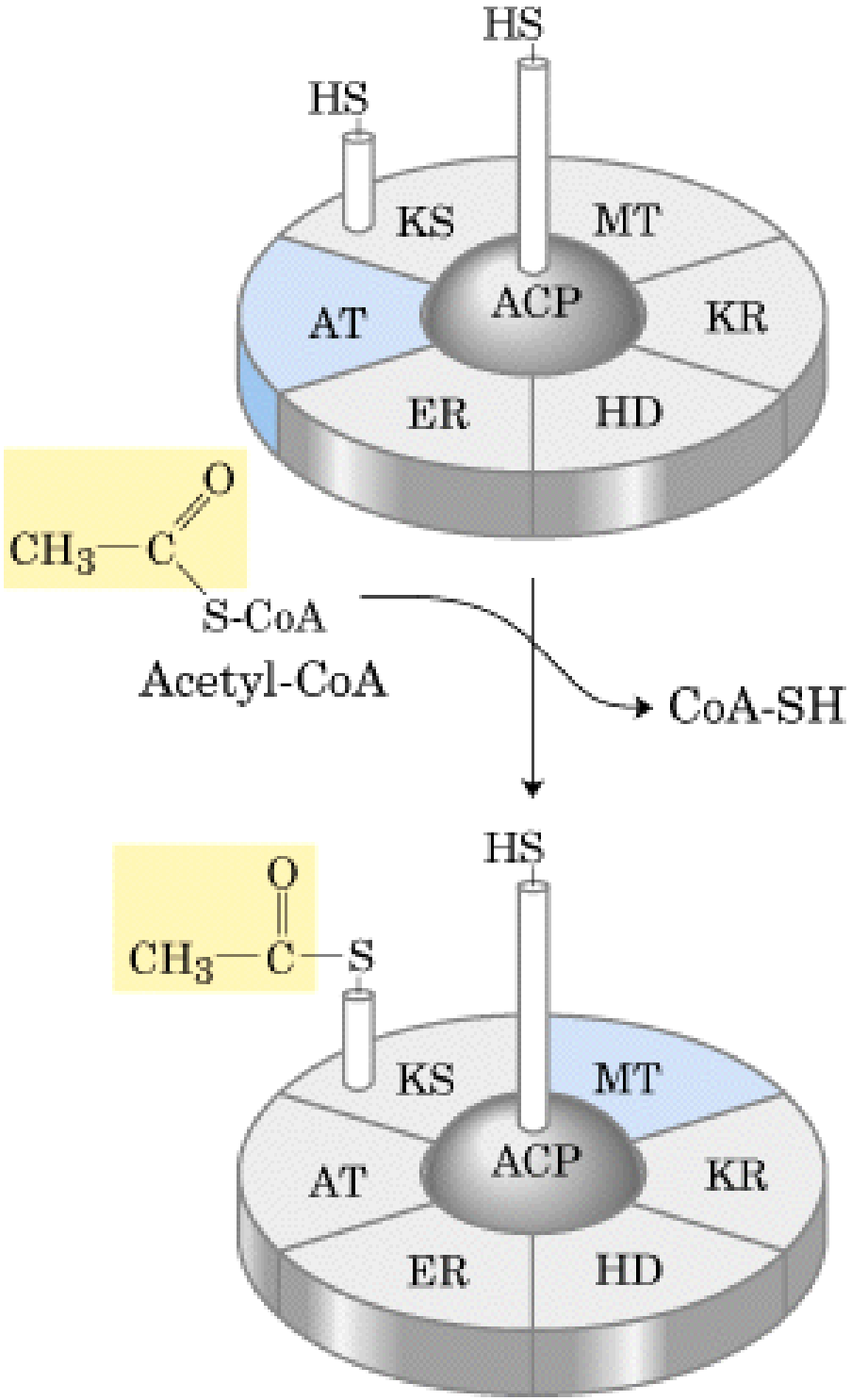
The acyl carrier protein (ACP) is very similar to CoA (thus can be regarded as "macro CoA")

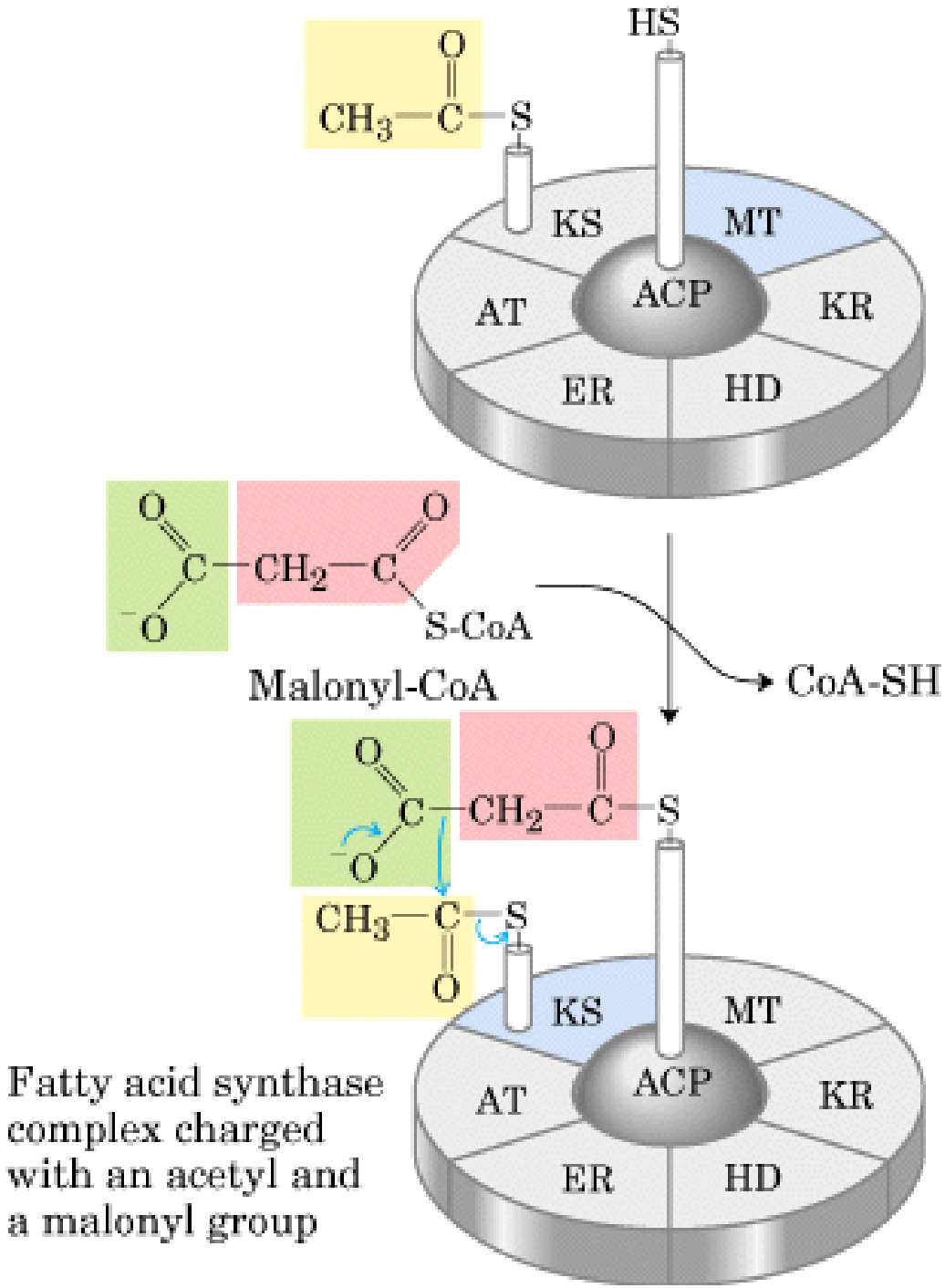


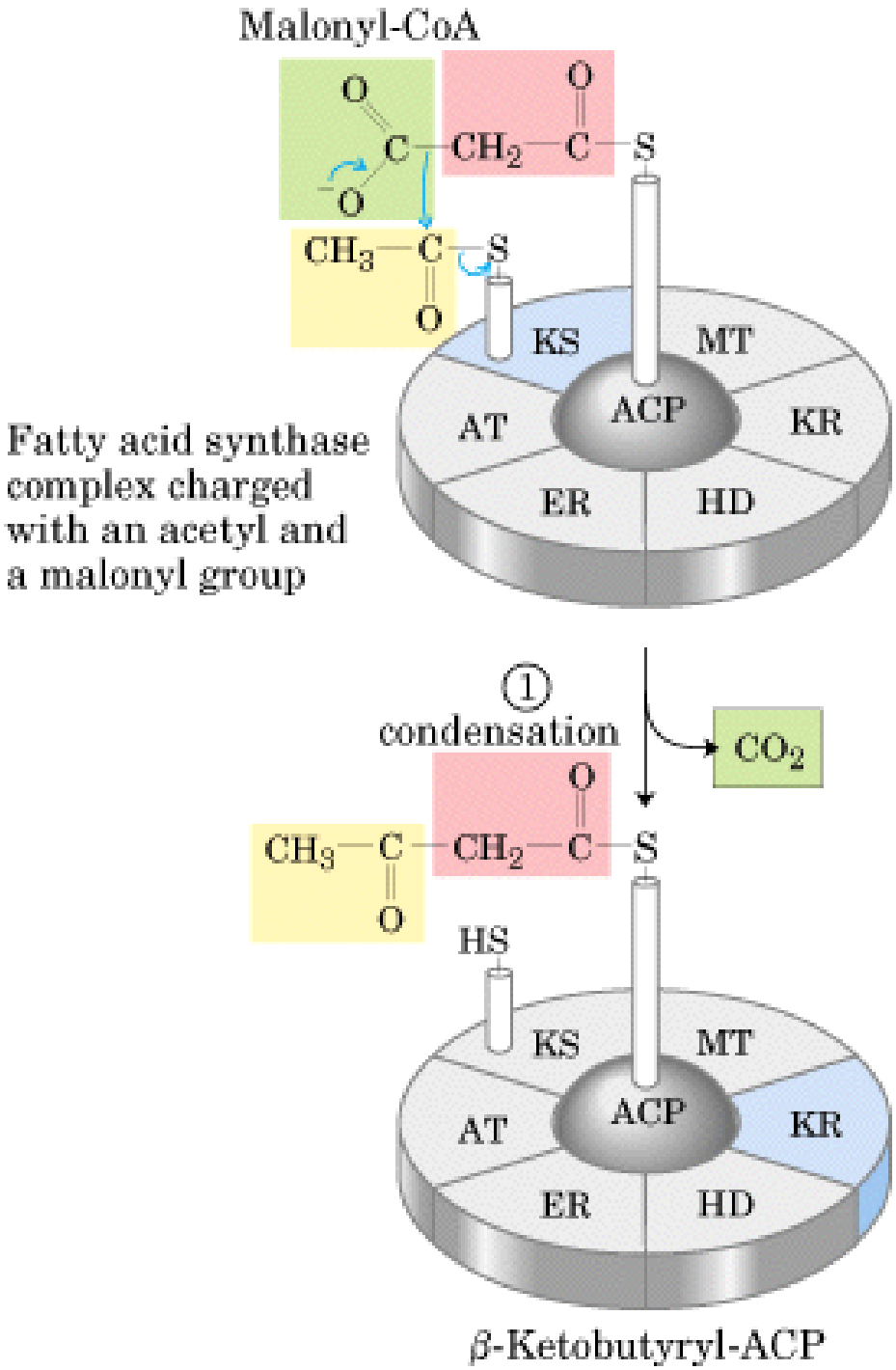
Phosphopantetheine group of coenzyme A

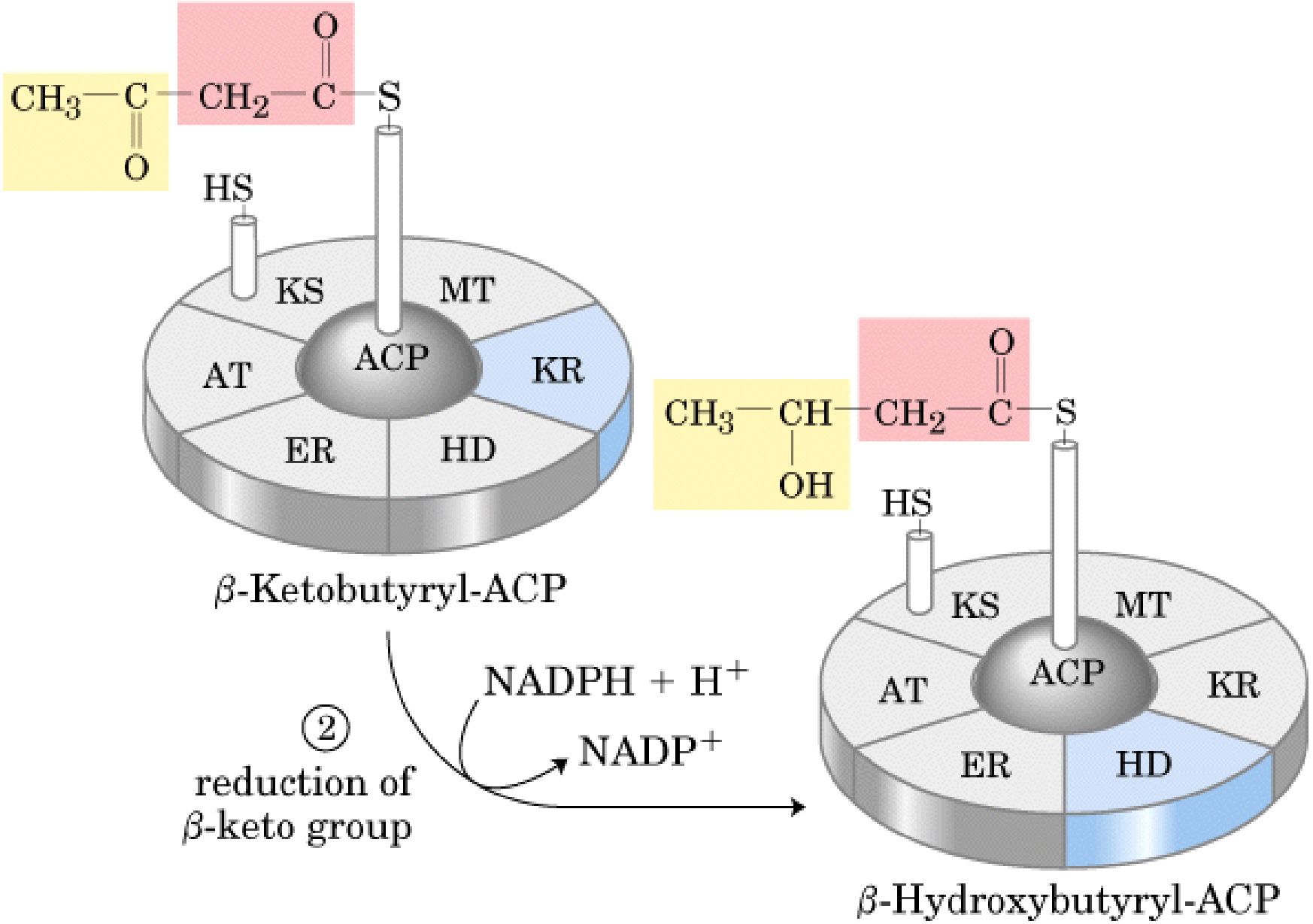


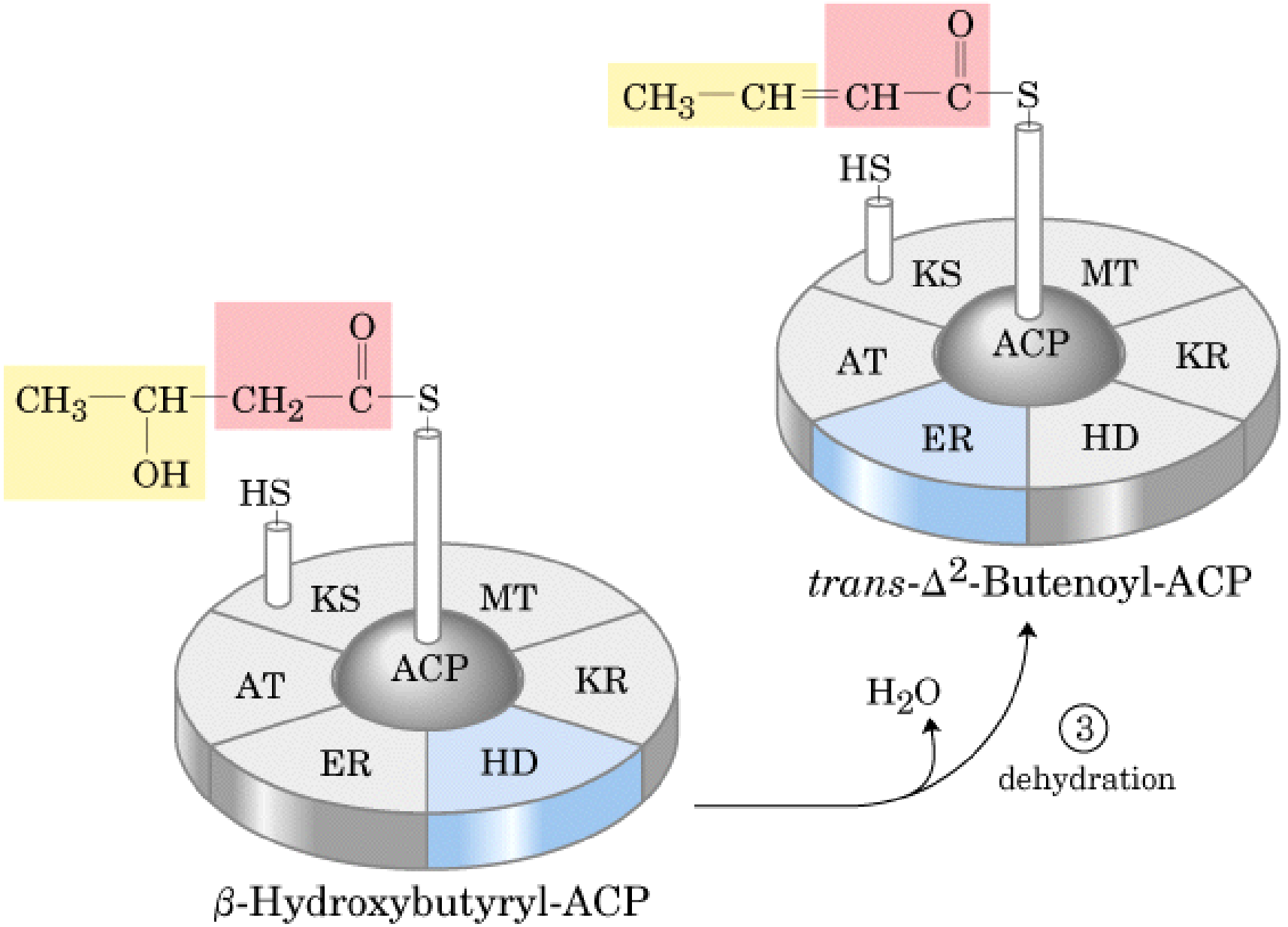
Phosphopantetheine prosthetic group of ACP

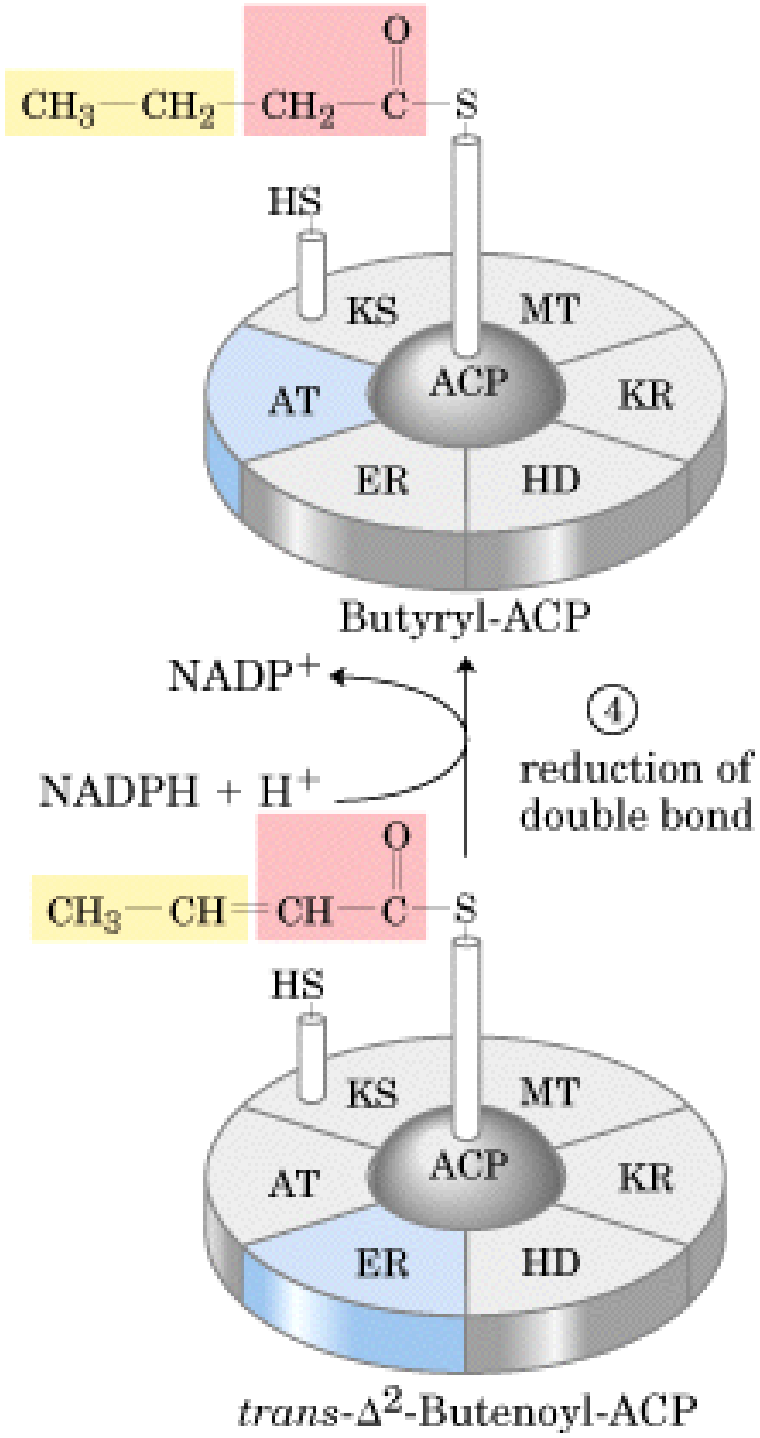

HS
HS
KS
MT
AT
ACP
KR
ER
HD
CH3—C—S-CoA
O
Acetyl-CoA
CoA-SH
HS
CH3—C—S
O
KS
MT
AT
ACP
KR
ER
HD

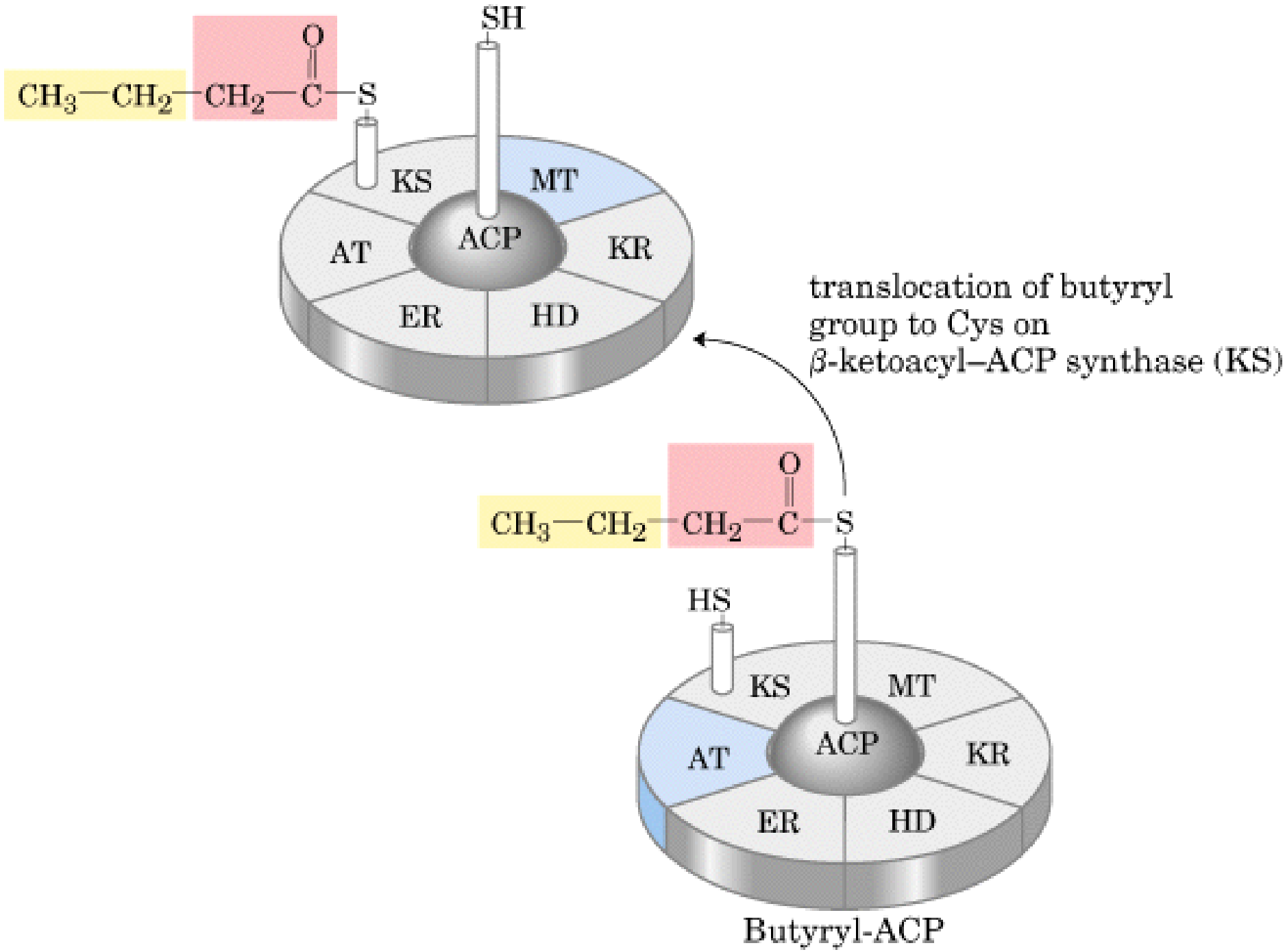

HS
CH3—C—S
KS
MT
AT
ACP
KR
ER
HD
S-CoA
Malonyl-CoA
CoA-SH
CH2
CO2
Fatty acid synthase complex charged with an acetyl and a malonyl group

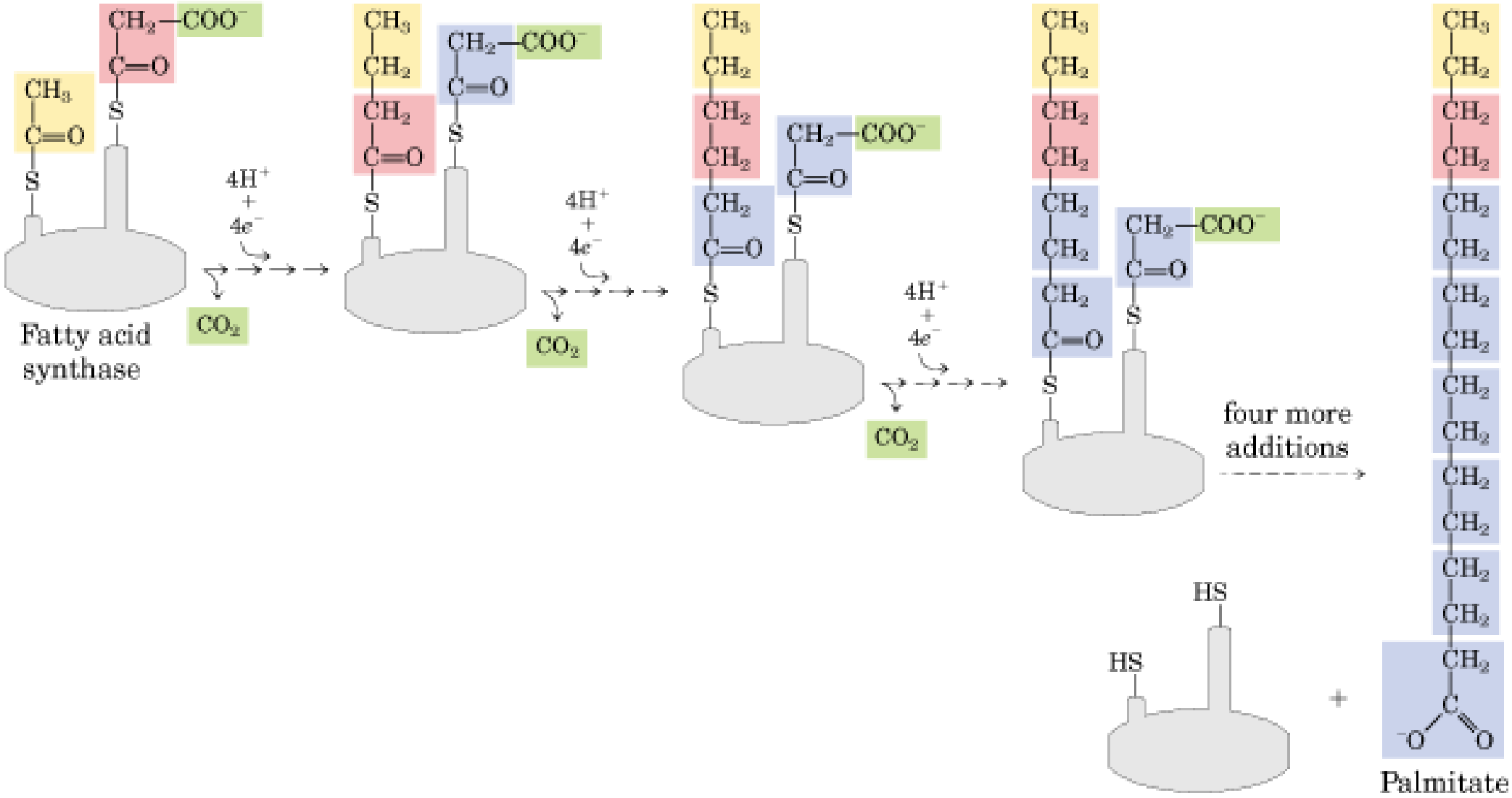

Malonyl-CoA
Fatty acid synthase complex charged with an acetyl and a malonyl group
KS
MT
AT
ACP
KR
ER
HD
1
condensation
CO2
HS
β-Ketobutyryl-ACP


CH3—C—CH2—C—S
O
O
HS
KS
MT
AT
ACP
KR
ER
HD
β-Ketobutyryl-ACP
NADPH + H+
NADP+
② reduction of β-keto group
CH3—CH—CH2—C—S
OH
β-Hydroxybutyryl-ACP


CH3—CH—CH2—C—S
O
OH
HS
KS
MT
AT
ACP
KR
ER
HD
β-Hydroxybutyryl-ACP
CH3—CH=CH—C—S
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trans-Δ2-Butenoyl-ACP
H2O
3
dehydration

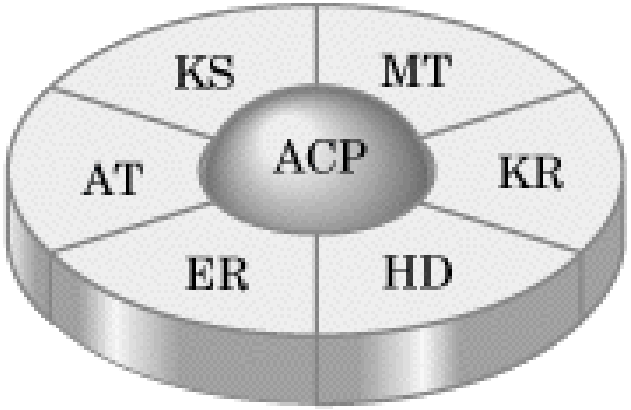

CH3—CH2—CH2—C—S
O
HS
KS
MT
AT
ACP
KR
ER
HD
Butyryl-ACP
NADP+
NADPH + H+
4
reduction of double bond
CH3—CH=CH—C—S
O
HS
KS
MT
AT
ACP
KR
ER
HD
trans-Δ2-Butenoyl-ACP


SH
O
CH3—CH2—CH2—C—S
KS
MT
AT
ACP
KR
ER
HD
translocation of butyryl group to Cys on β-ketoacyl–ACP synthase (KS)
O
CH3—CH2—CH2—C—S
HS
KS
MT
AT
ACP
KR
ER
HD
Butyryl-ACP


Fatty acid synthase
4H+ + 4e−
CO2
four more additions
HS
HS
Palmitate

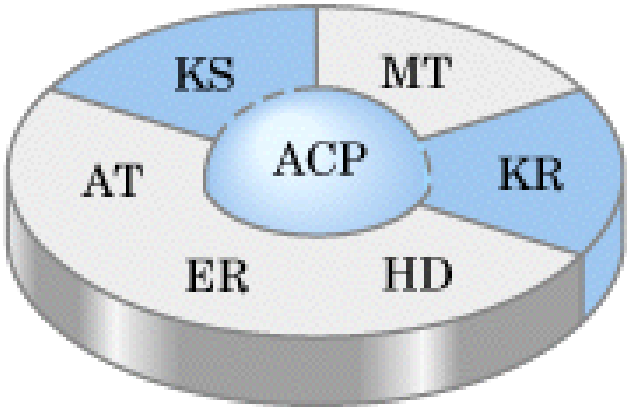
Bacteria, Plants

Seven activities in seven separate polypeptides



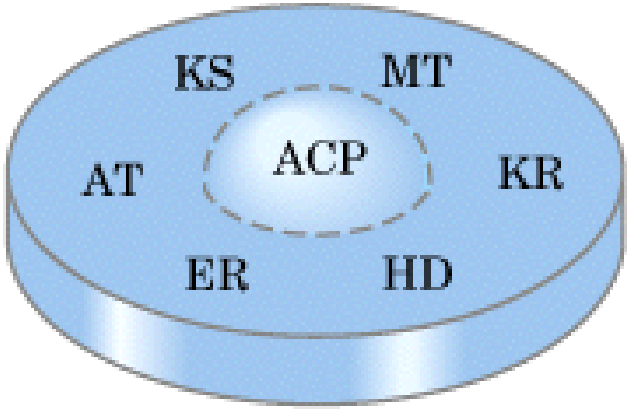
Yeast

Seven activities in two separate polypeptides



Vertebrates

Seven activities in one large polypeptide



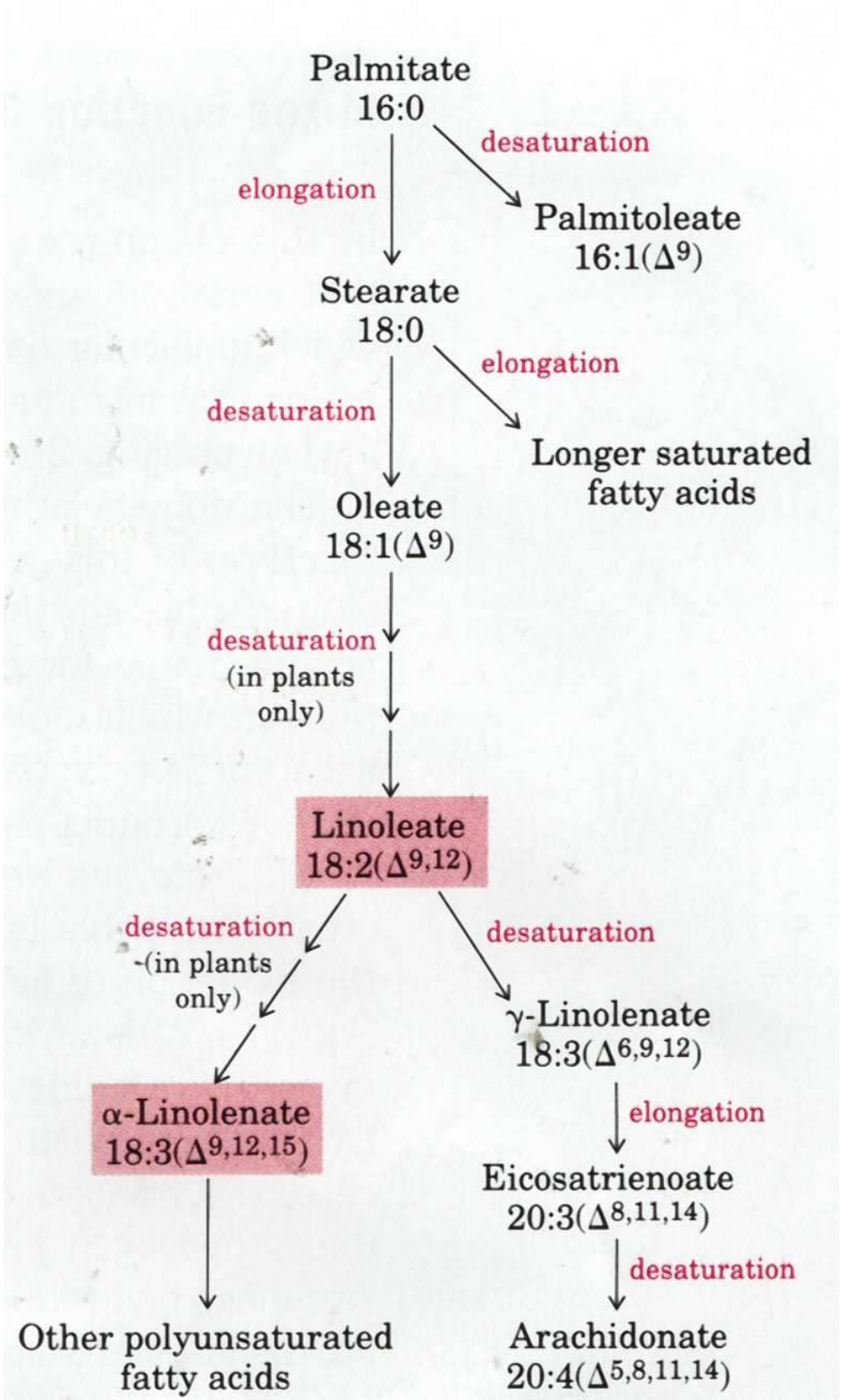
5. Palmitate can be further elongated and desaturated in smooth ER

- Palmitoyl-CoA can be further elongated by the fatty acid elongation system present mainly in the **smooth endoplasmic reticulum （ER）**, with two-carbon units also donated by **malonyl-CoA**.
- Stearoyl-CoA (硬脂酰）can be desaturated between C-9 and C-10 to produce oleate（油酸）, 18:1(Δ^9).



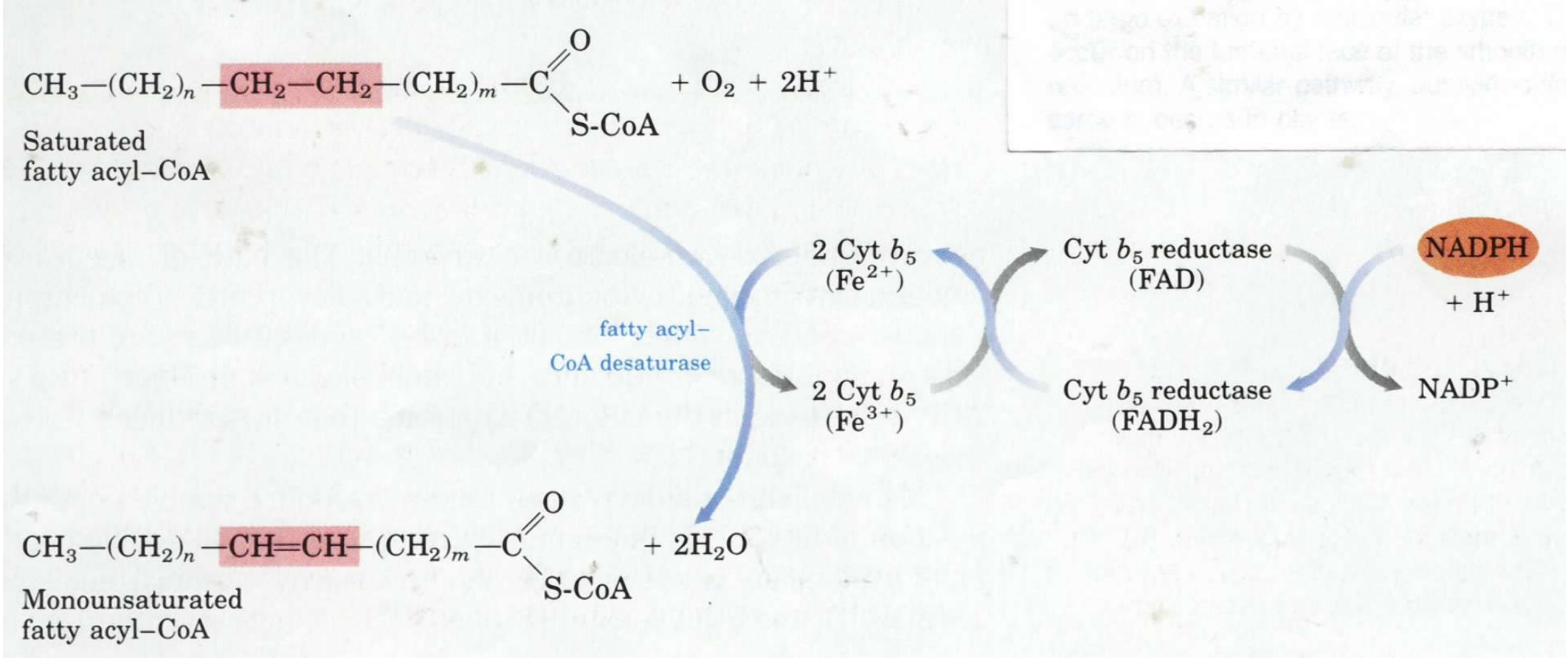
- The double bonds are introduced by the catalysis of fatty acyl-CoA **desaturase**, where both the fatty acyl group and NADPH are oxidized by O_2.
- **Linoleate (亚油酸) and linolenate （亚麻酸）** cannot be synthesized by mammals and are therefore termed essential fatty acids as they have to be ingested in the diet.





Palmitate is the Precursor for the biosynthesis of other fatty acids





Fatty acyl-CoA is desaturated (oxidized) by O_2 and NADPH. (the electrons from both fatty acyl-CoA and NADPH)

6. Essential fatty acids

- The polyunsaturated fatty acids **linoleate and linolenate** cannot be synthesized by mammals and are therefore termed essential fatty acids as they have to be ingested in the diet.



7. Newly synthesized fatty acids have mainly two alternative fates in cells

- Fate I: be incorporated into triacylglycerols as a form to store metabolic energy in long terms.
- Fate II: be incorporated into membrane phospholipids (during rapid growth).

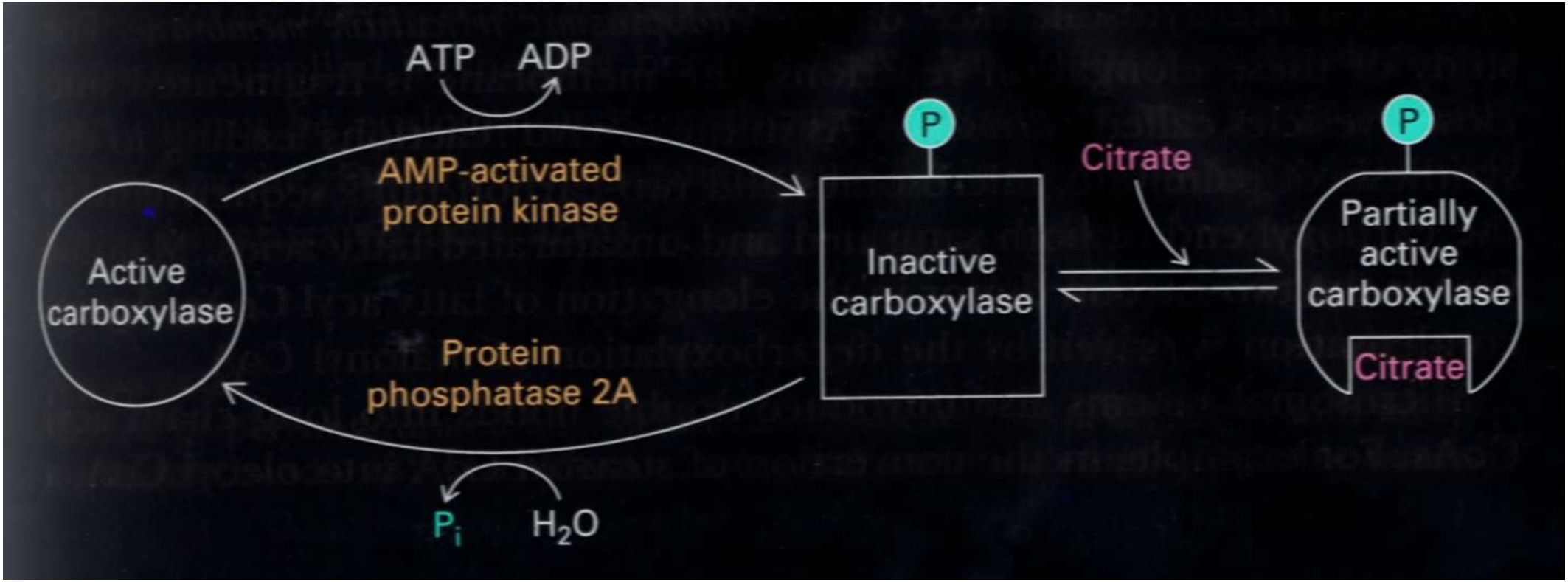


8. The rate of fatty acid biosynthesis is controlled by acetyl-CoA carboxylase

Acetyl-CoA carboxylase is regulated by allosteric effectors and reversible phosphorylation.



Citrate partially activate the phosphorylated acetyl-CoA carboxylase



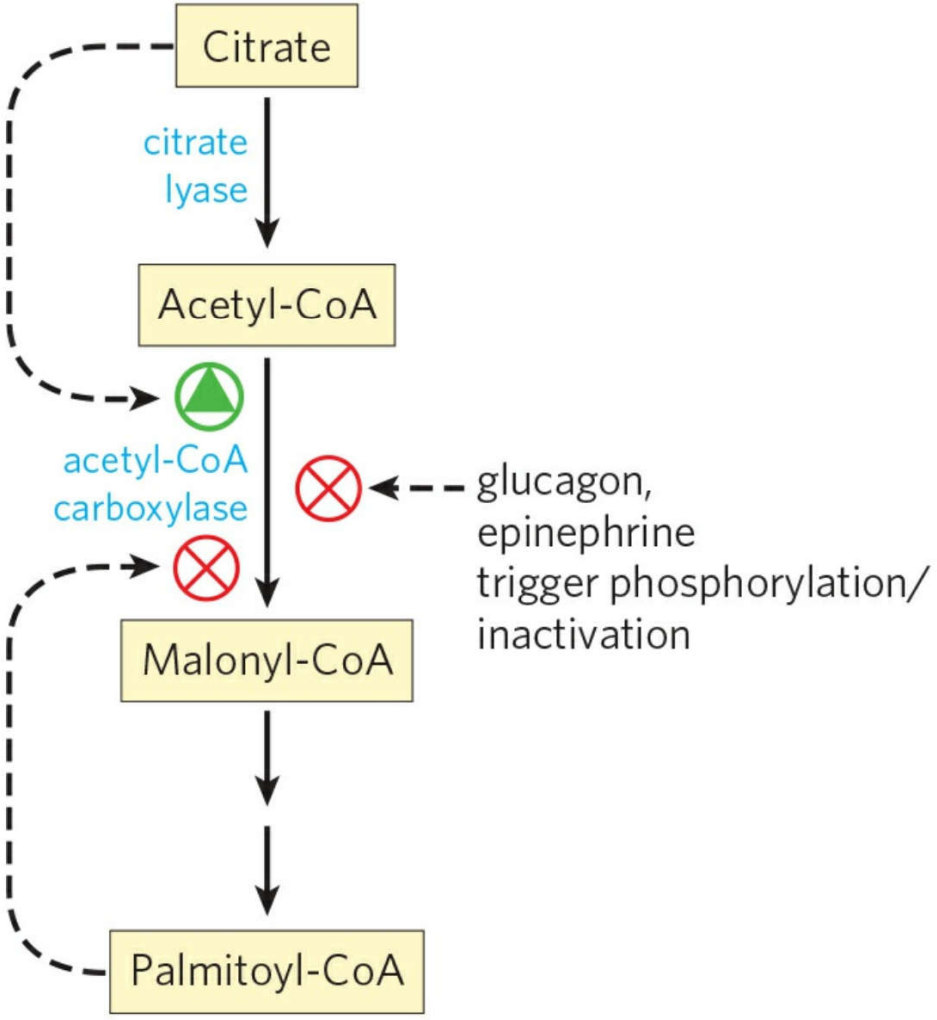
Biochemistry



- Acetyl-CoA carboxylase, catalyzing the committing and rate-limiting step of fatty acid synthesis, is allosterically **inhibited by palmitoyl-CoA and phosphorylation, activated by citrate.**
- **Glucagon** and **epinephrine** inhibit protein phosphatase 2A, thus keeping acetyl CoA carboxylase in the inactive form and blocking fatty acid synthesis. **Insulin stimulates acetyl CoA carboxylase.**



Regulation of fatty acid synthesis

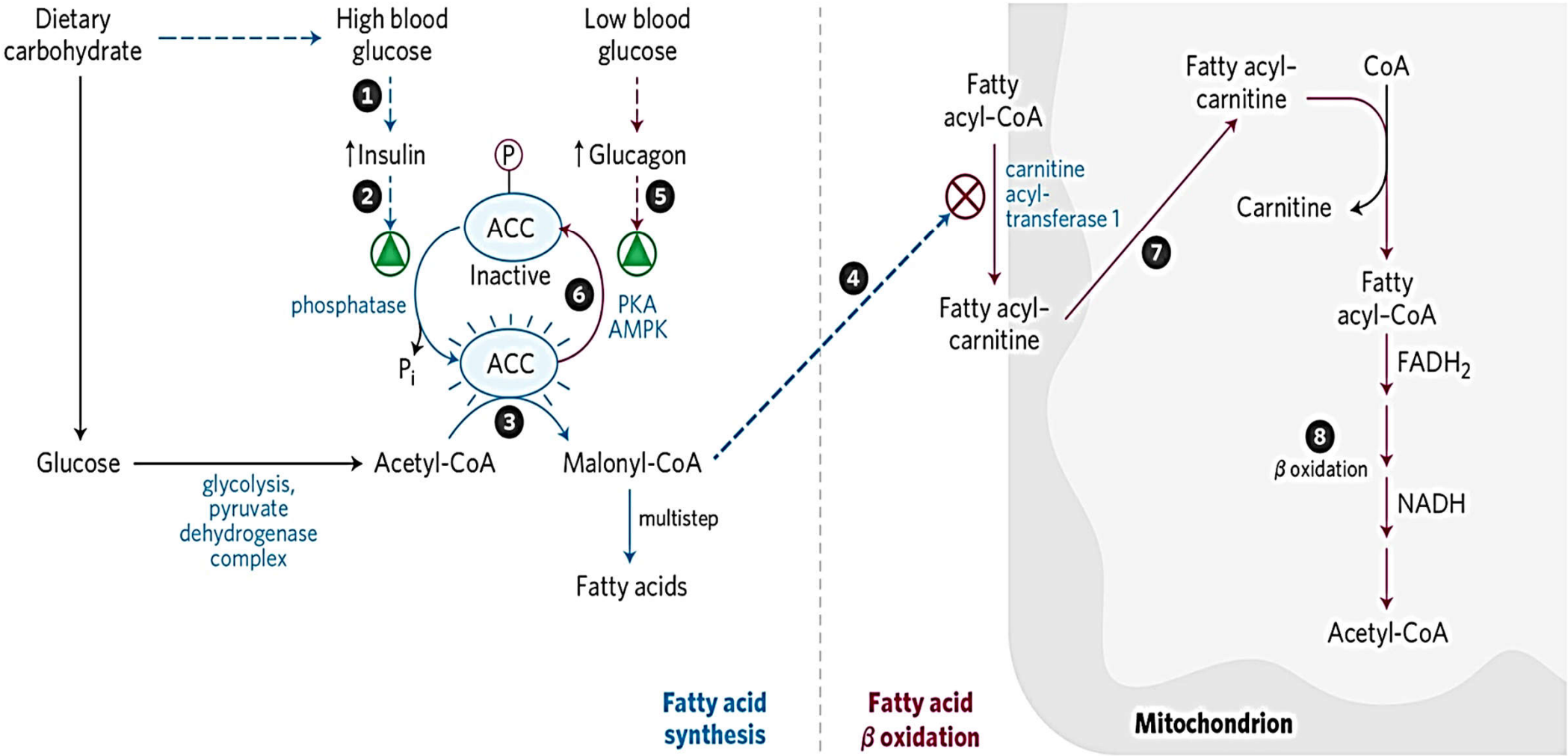


(a)



(b)

Coordinated regulation of fatty acid synthesis and breakdown



Summary

- Fatty acid biosynthesis takes a different pathway from the reverse of its degradation and takes place in different cellular compartments.
- The aceytl-CoA units are transported out of mitochondrial matrix as citrate.



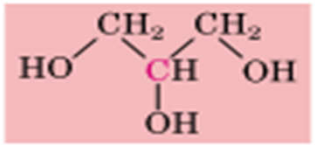
- Acetyl-CoA carboxylase catalyzes the rate-limiting step of fatty acid synthesis and is highly regulated by allosteric and covalent modifications.
- Palmitate, the usual final product of fatty acid synthesis, can be further elongated and desaturated in sER.



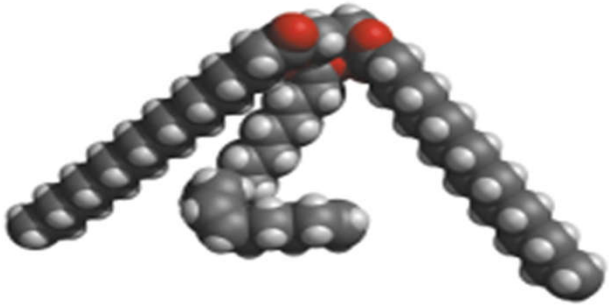
Metabolism of Triacylglycerols K4 catabolism and anabolism

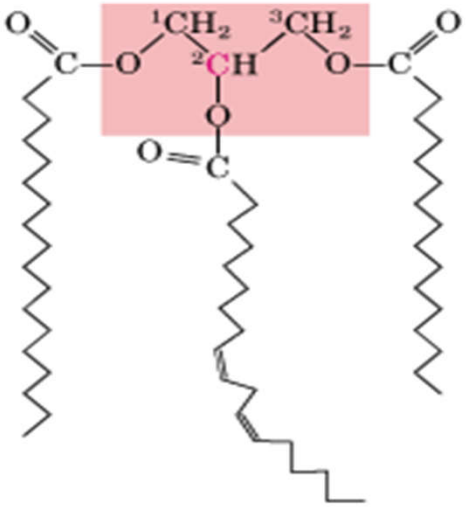


1. Breakdown of triacylglycerols



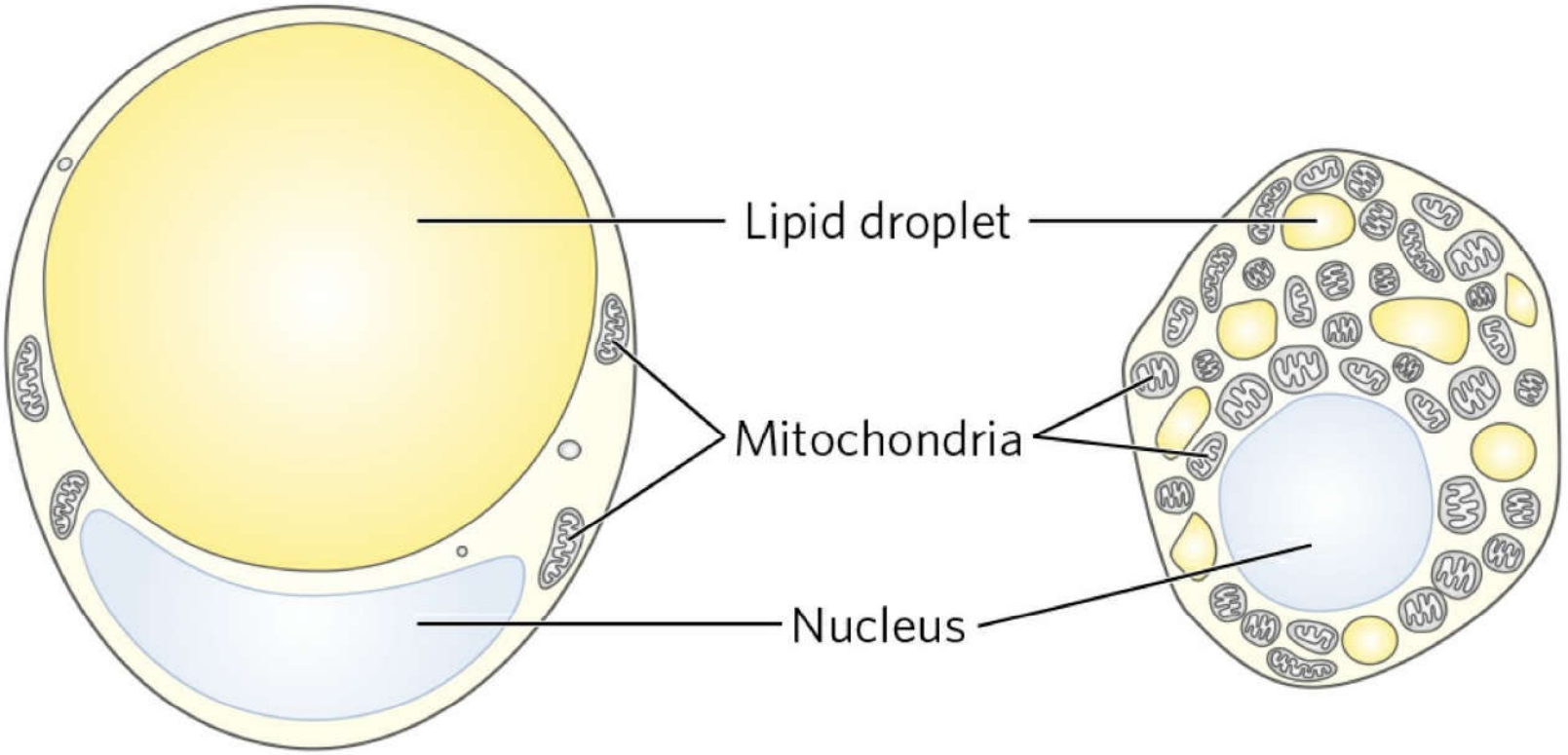
Glycerol





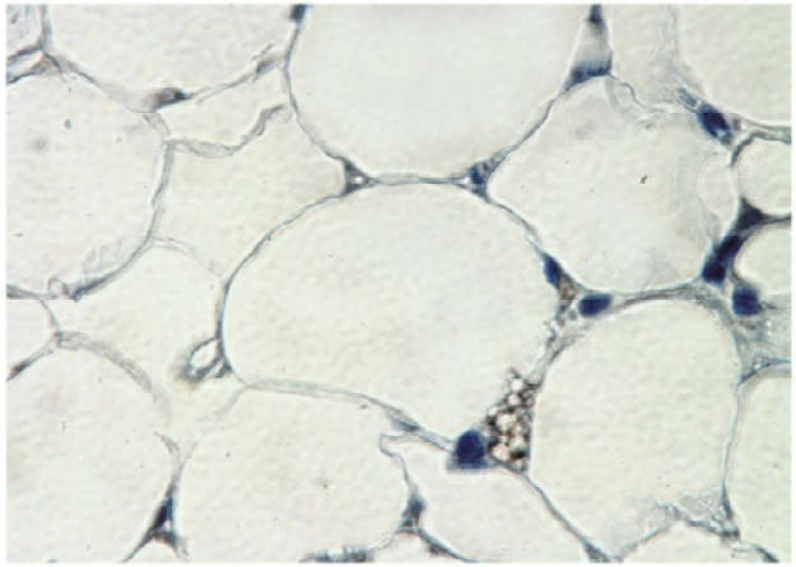
1-Stearoyl, 2-linoleoyl, 3-palmitoyl glycerol,
a mixed triacylglycerol



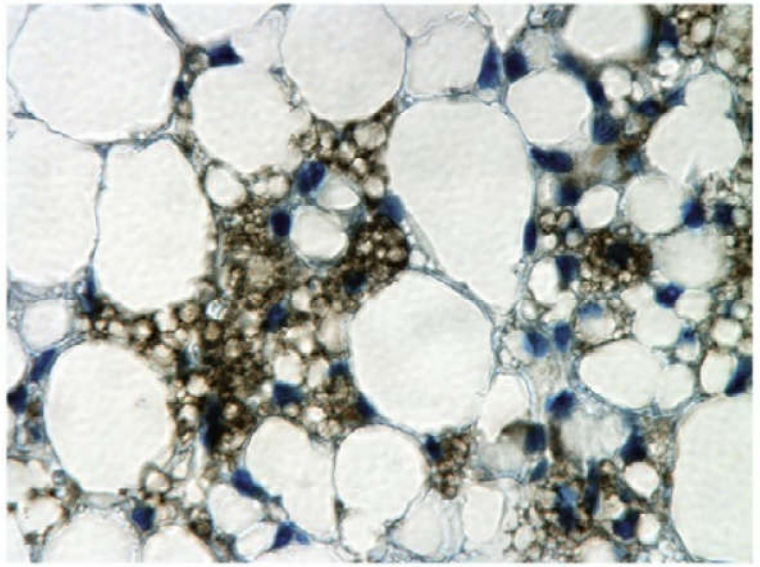


(a) White adipocyte

(b) Brown adipocyte

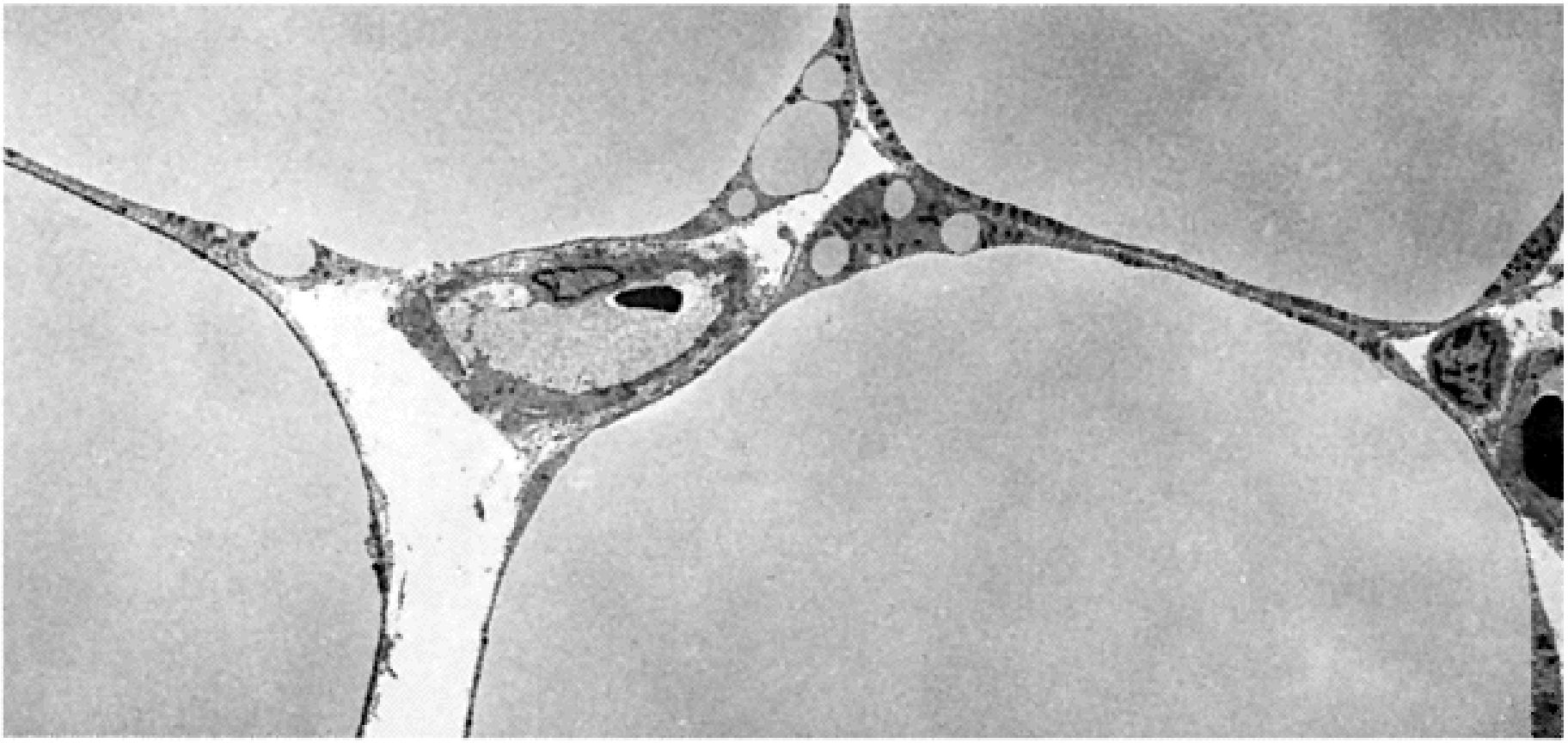


(c)



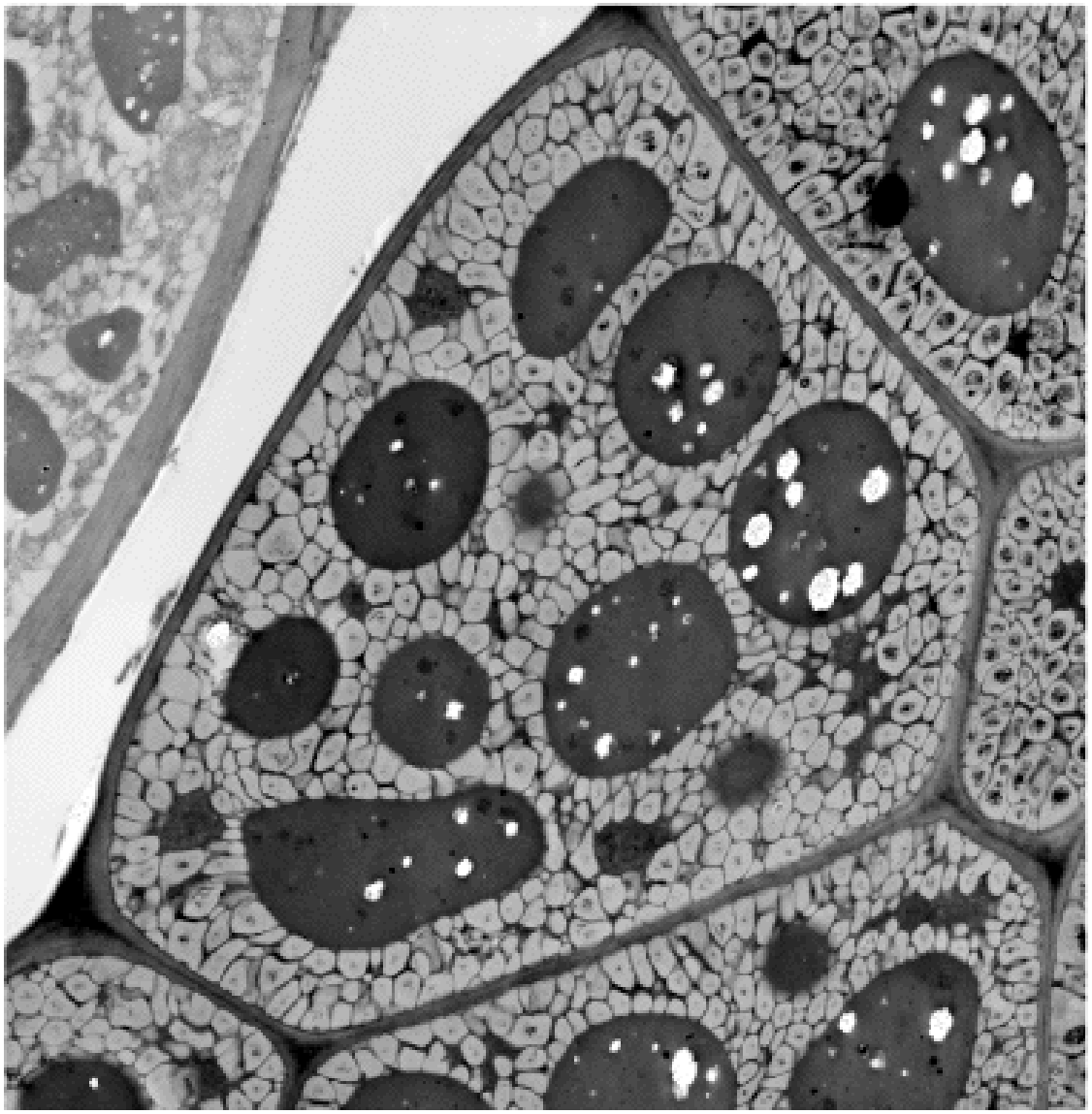
(d)





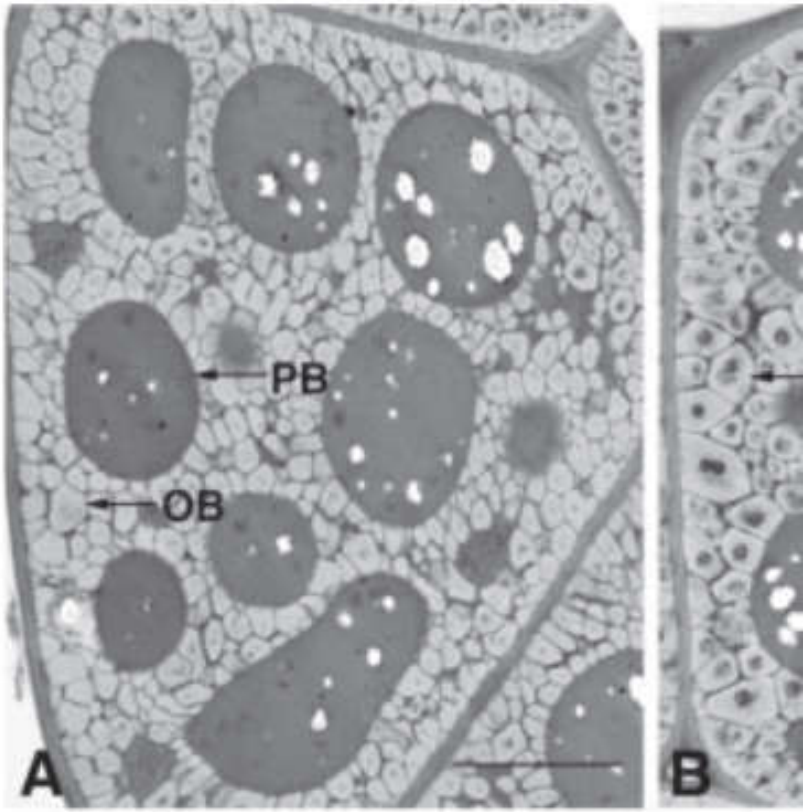
8 μm

(a)



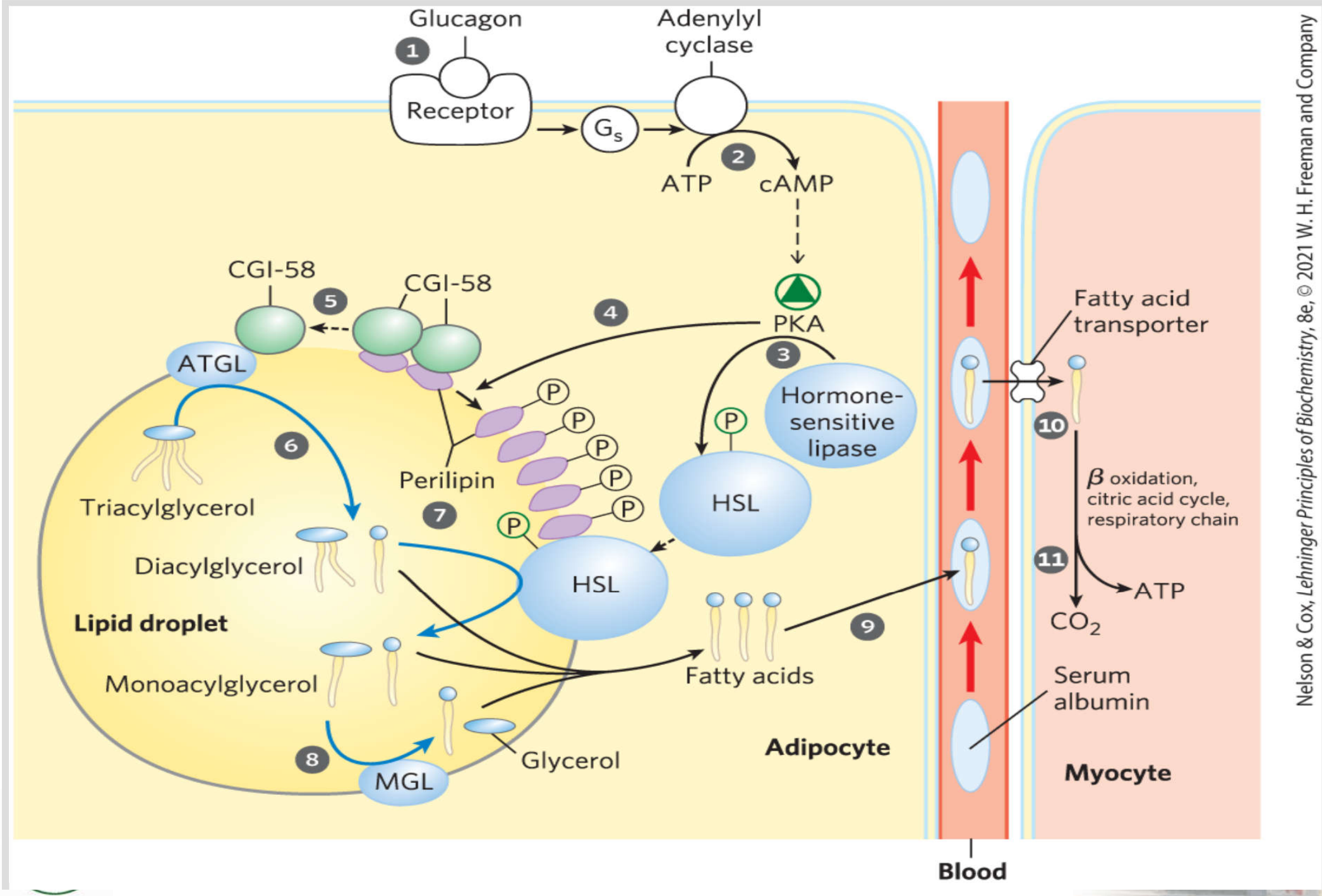
(b)

3 μm





Mobilization of triacylglycerols stored in adipose tissue



Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2021 W. H. Freeman and Company

1. Breakdown of triacylglycerols

The fatty acids in triacylglycerols are released from the glycerol backbone by the action of **lipases**.

The free fatty acids can then degraded by β–oxidatin to produce energy.

The glycerol is converted into dihydroxyacetone phosphate which enters glycolysis.



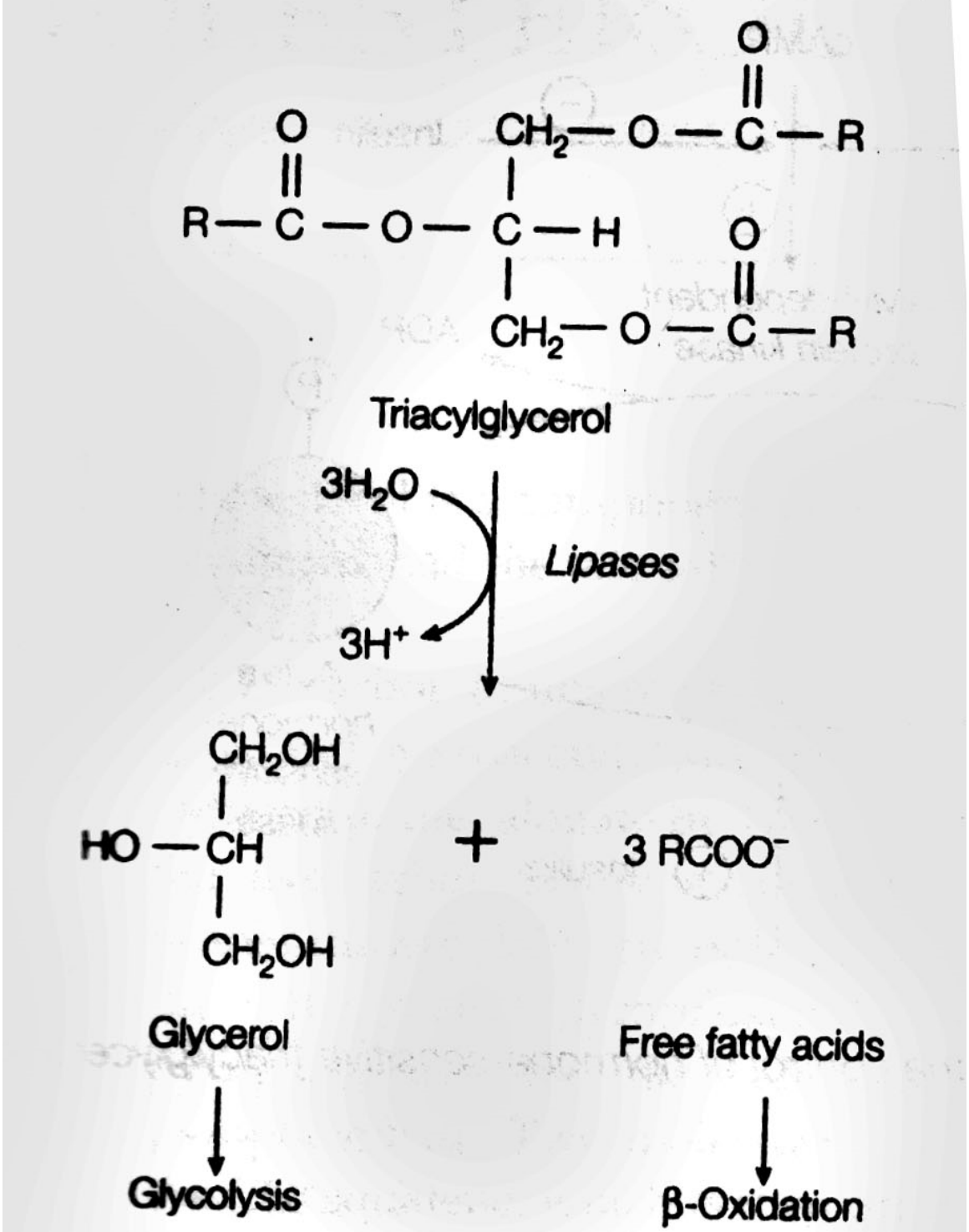
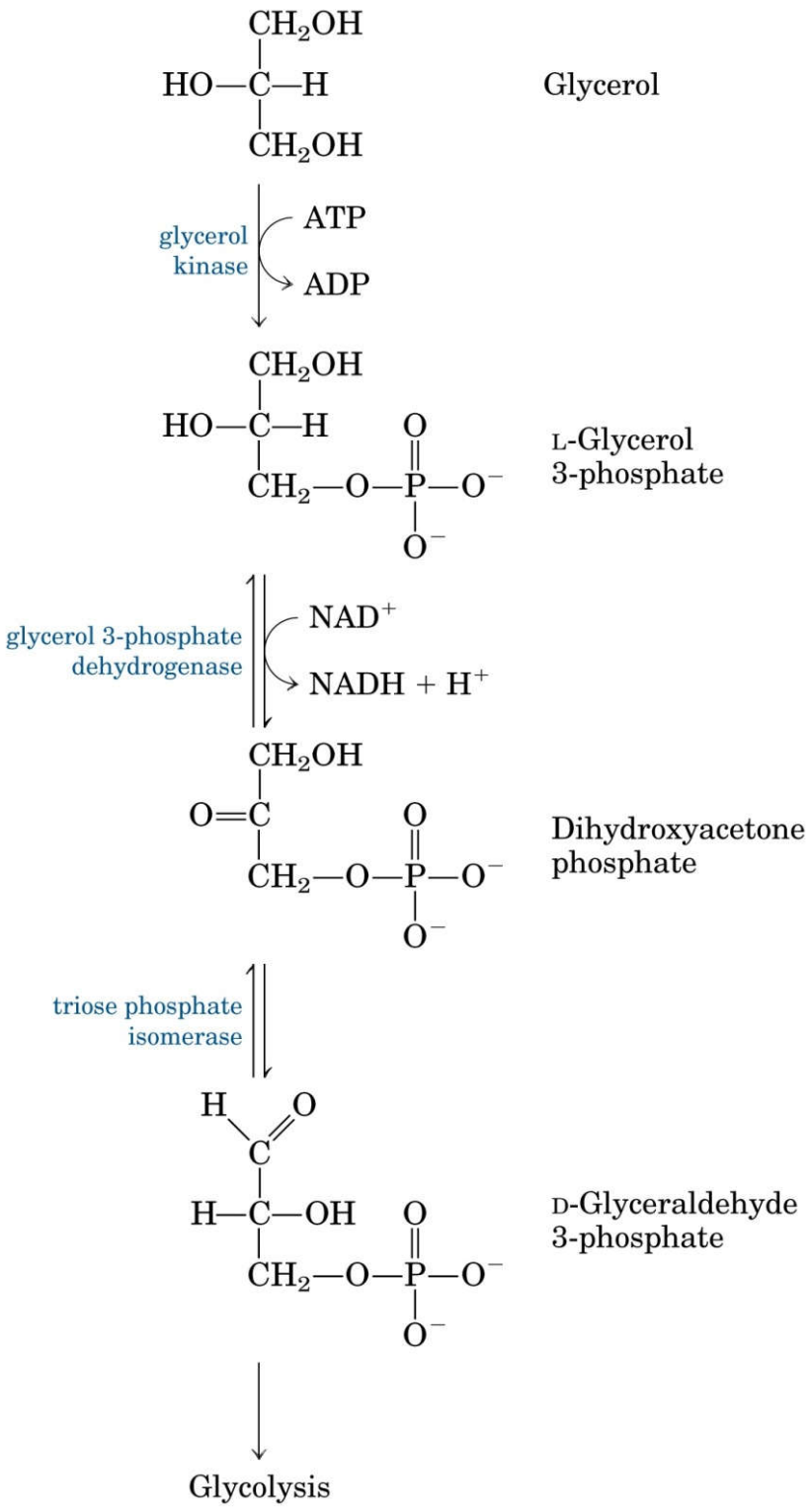
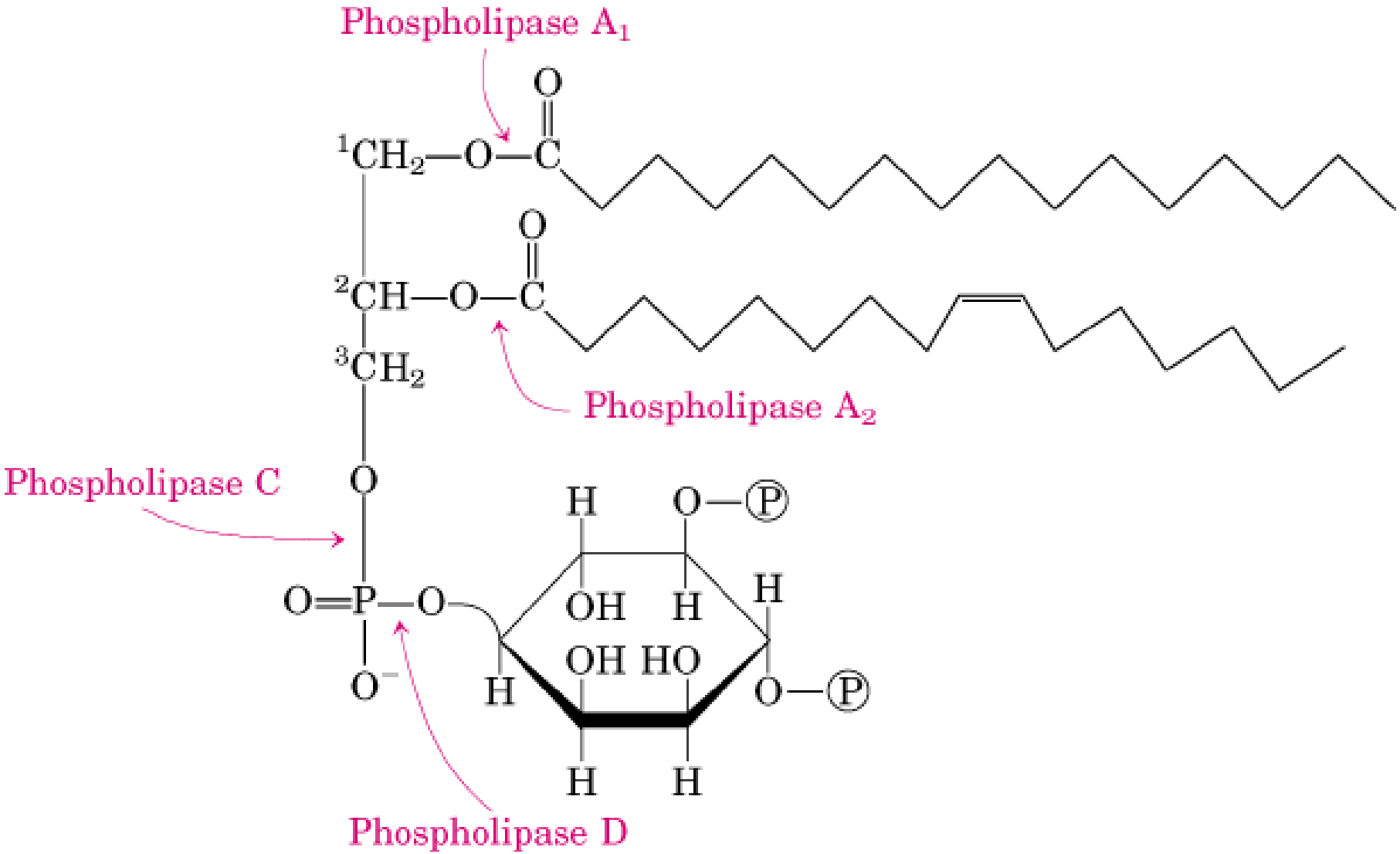


Fig. 3. Breakdown of triacylglycerols.

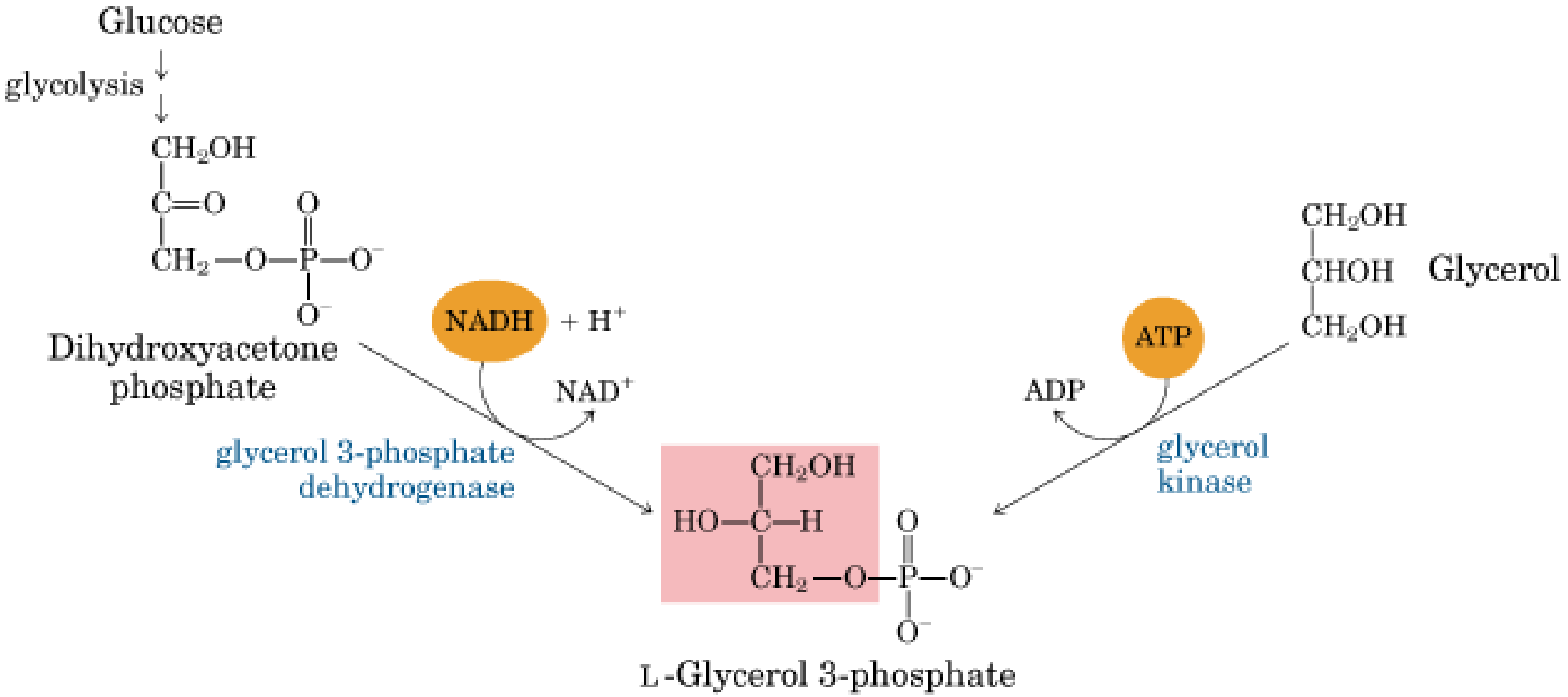

CH2OH
HO—C—H
CH2OH
Glycerol
glycerol kinase
ATP
ADP
CH2OH
HO—C—H
CH2—O—P—O⁻
O⁻
L-Glycerol 3-phosphate
glycerol 3-phosphate dehydrogenase
NAD+
NADH + H+
CH2OH
O=C
CH2—O—P—O⁻
O⁻
Dihydroxyacetone phosphate
triose phosphate isomerase
H
O
C
H—C—OH
CH2—O—P—O⁻
O⁻
D-Glyceraldehyde 3-phosphate
Glycolysis

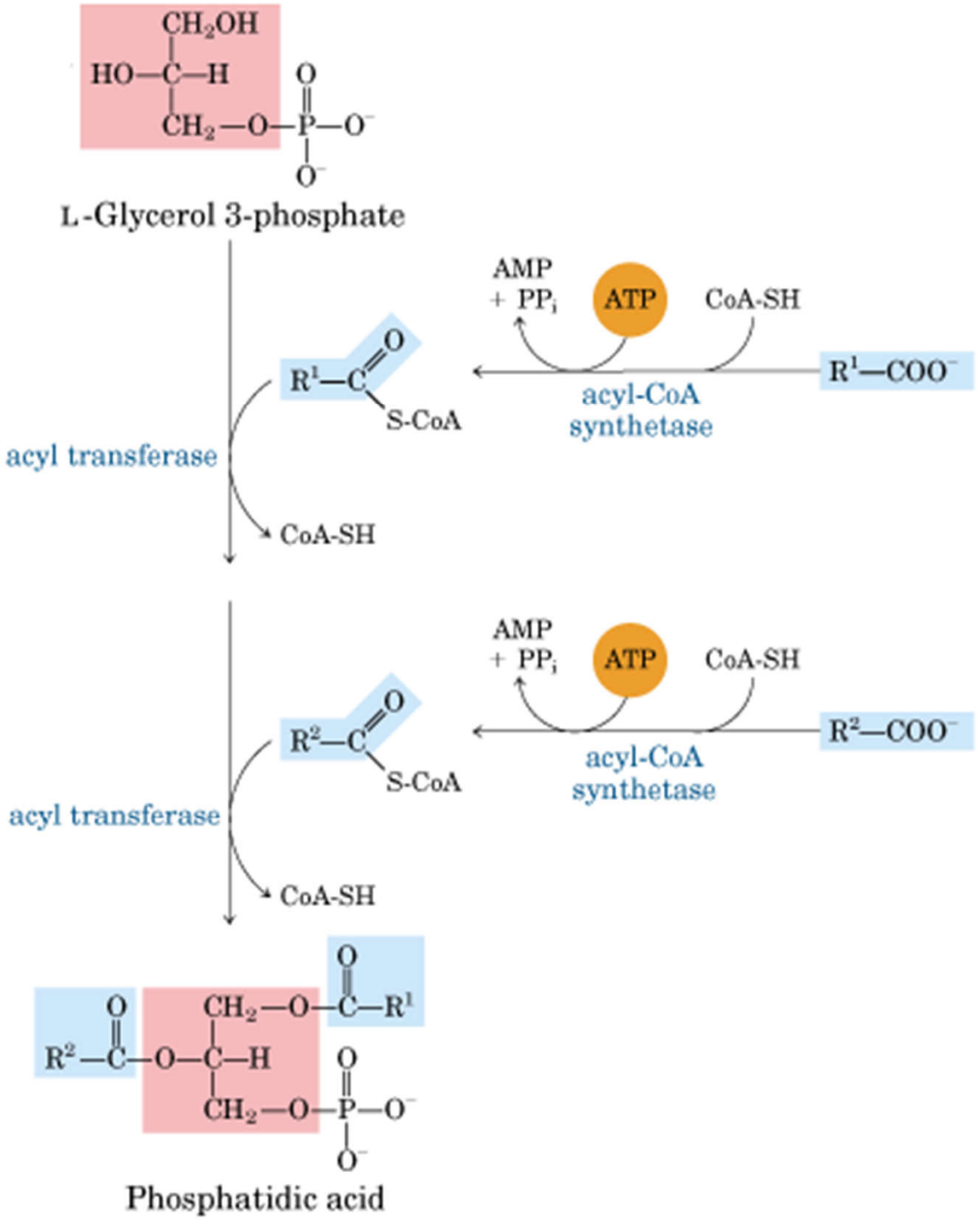

Phospholipase A1
Phospholipase A2
Phospholipase C
Phospholipase D

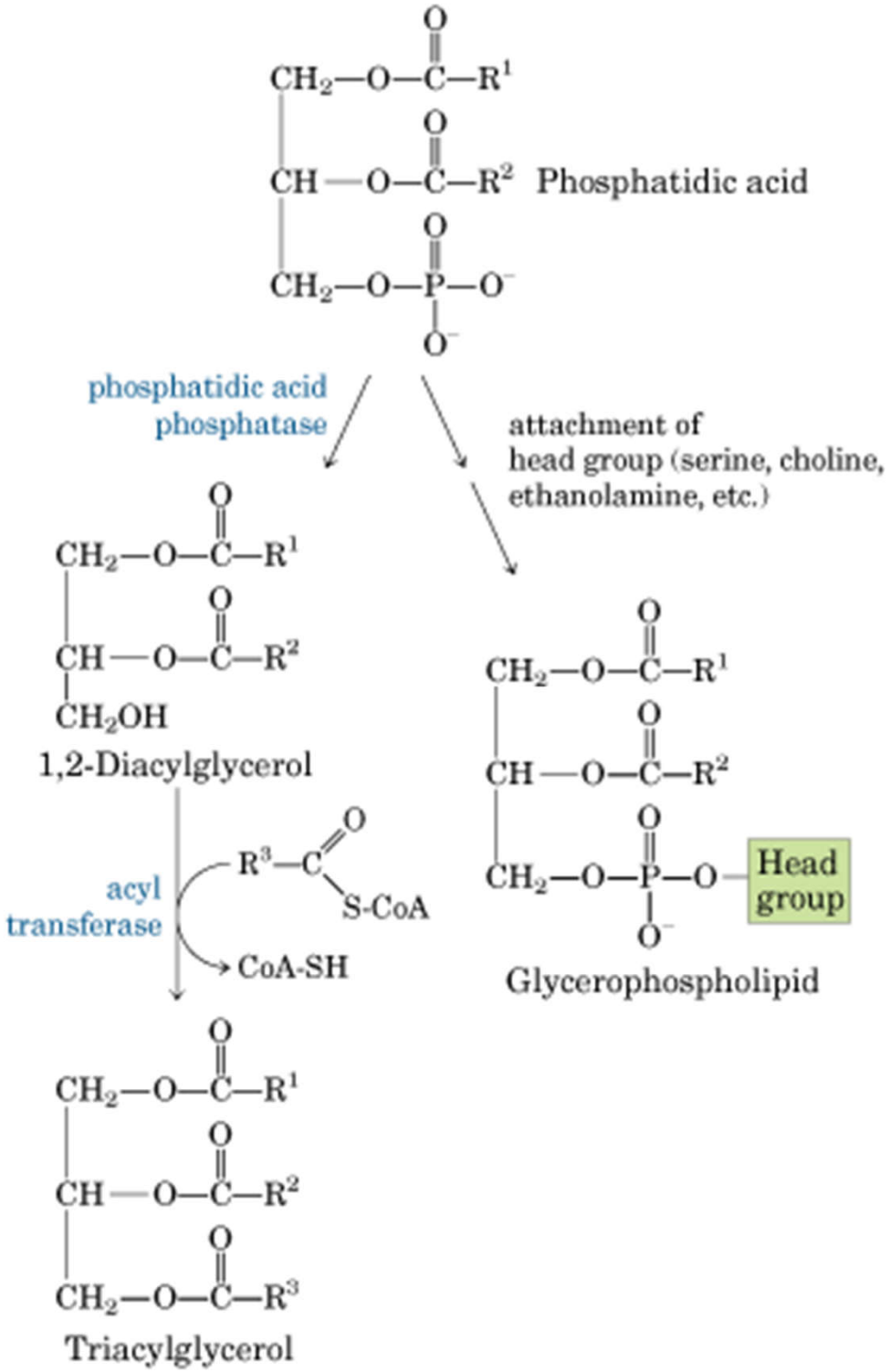
2. Triacylglycerols are synthesized from glycerol 3-phosphate and acyl CoAs

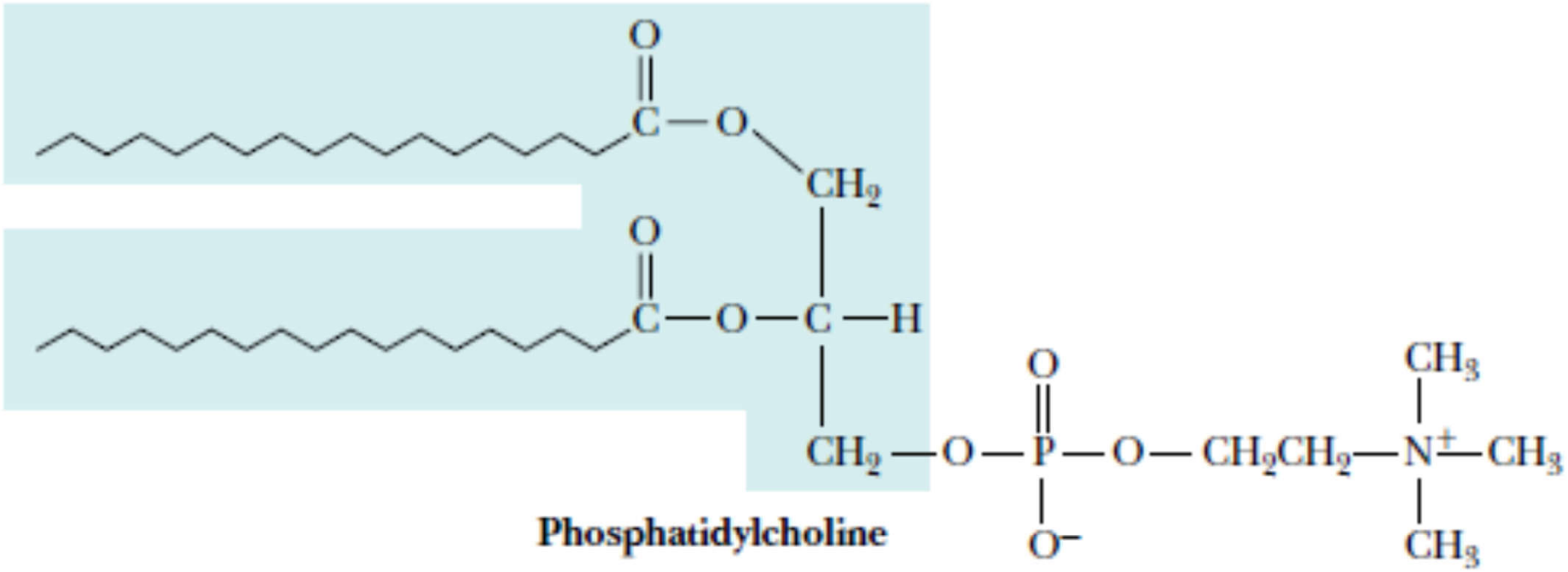
- **L-glycerol 3-phosphate** is derived from either glycerol or dihydroxyacetone phosphate.
- **Acyl CoAs** are added on to glycerol 3-phosphate to form phosphatidic acid.
- **A phosphatidic acid** is converted to a triacylglycerol via a dephosphorylation reaction (catalyzed by **phosphatidic acid phosphatase**) and a acyl transferring reaction.



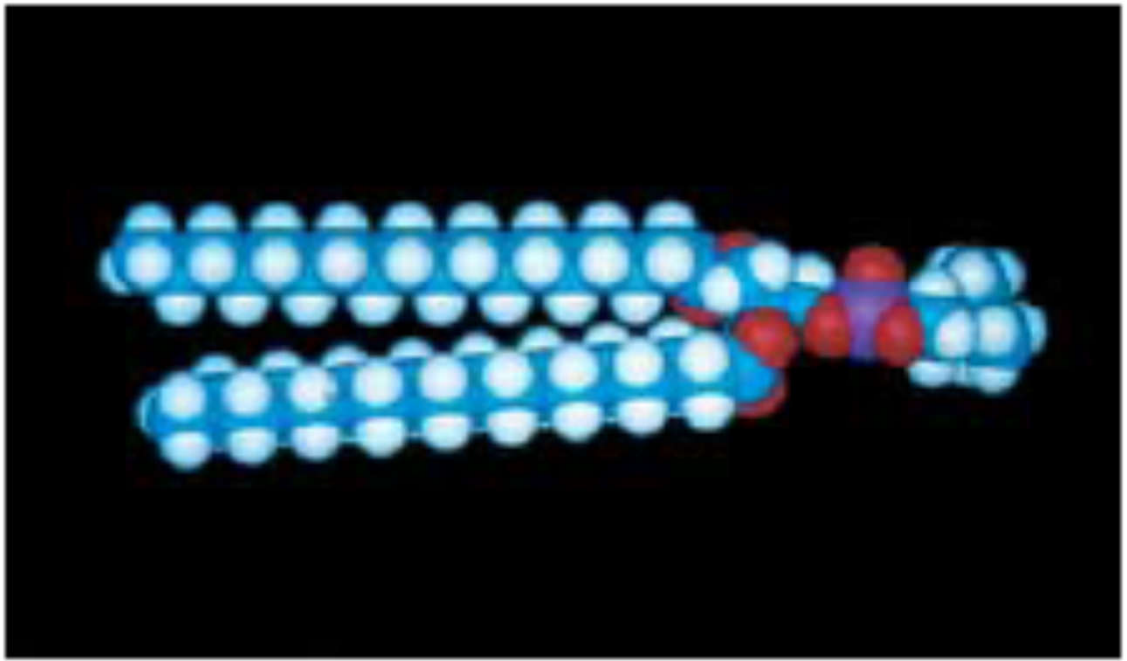

Glucose
glycolysis
Dihydroxyacetone phosphate
NADH + H+
NAD+
glycerol 3-phosphate dehydrogenase
ATP
ADP
glycerol kinase
Glycerol
L-Glycerol 3-phosphate


L-Glycerol 3-phosphate
AMP
+ PPi
ATP
CoA-SH
acyl-CoA
synthetase
acyl transferase
CoA-SH
AMP
+ PPi
ATP
CoA-SH
acyl-CoA
synthetase
acyl transferase
CoA-SH
Phosphatidic acid


Phosphatidic acid
phosphatidic acid phosphatase
attachment of head group (serine, choline, ethanolamine, etc.)
1,2-Diacylglycerol
acyl transferase
S-CoA
CoA-SH
Head group
Glycerophospholipid
Triacylglycerol



Phosphatidylcholine





END

Thank you

Biochemistry

